Towards Semantically-Aware Robot Navigation: From Geometric Maps to Vision-Text Models

Praveen Kumar

Ph.D. Scholar (20204272)

Control and Automation

Department of Electrical Engineering

Thesis Supervisor(s):
Dr. Tushar Sandhan, and Dr. K.R. Guruprasad

IIT Kanpur

31-July-2026

List of Publications

1. P. Kumar, P. Gupta and T. Sandhan, “NeuralPathLite: Fast and Robust Diffusion-Based Path Planning for Autonomous Navigation,” in IEEE Transactions on Automation Science and Engineering, vol. 23, pp. 12471-12485, 2026, doi: 10.1109/TASE.2026.3710089. (Published)

2. P. Kumar and T. Sandhan, “Vision-Aided Online A* Path Planning for Efficient and Safe Navigation of Service Robots,” International Journal of Intelligent Robotics and Applications (Springer), 2026, doi: 10.1007/s41315-026-00540-z. **(Published)**

3. P. Kumar, K. R. Guruprasad, and T. Sandhan, “Multi-Task Visual Perception Network with LLM Conditioning for Autonomous Navigation,” in Proceedings of the 2026 International Joint Conference on Neural Networks (IJCNN), presented at the 2026 IEEE World Congress on Computational Intelligence (WCCI), Maastricht, The Netherlands, June 21–26, 2026. **(Published)**

4. P. Kumar and T. Sandhan, “Path-Tracking Control of the 4WIS4WID Electric Vehicle by Direct Inverse Control using Artificial Neural Network,” 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-8, doi: 10.1109/ICCCNT56998.2023.10306595. **(Published)**

5. P. Kumar, R. M. Singhee, K. R. Guruprasad and T. Sandhan, “VTMMap Driven Vision-Language Navigation: From Visual Relations into Textual Paths,” in IEEE Transactions on Automation Science and Engineering (Submitted).

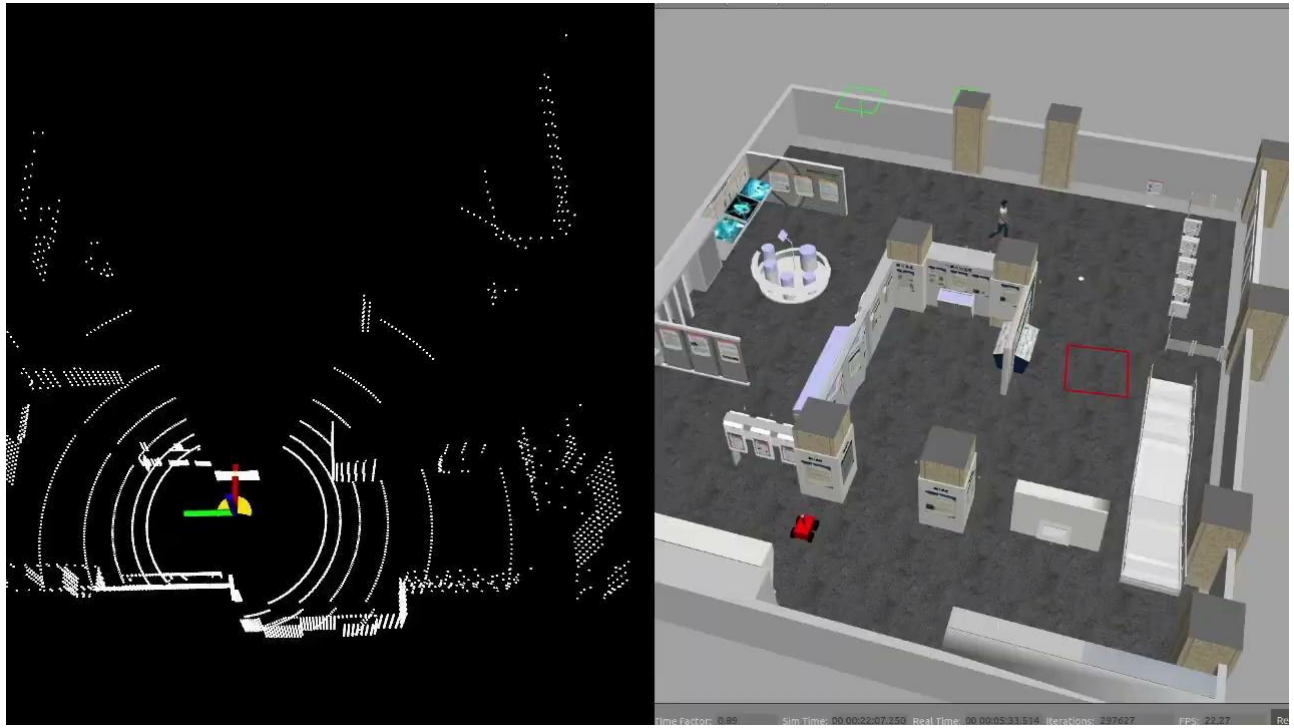
6. P. Kumar, R. M. Singhee, K. R. Guruprasad and T. Sandhan, “Optimizing Human-in-the-Loop Vision-Language Navigation Through Hardware-Software Co-Design,” IEEE Robotics and Automation Practice (Submitted).

This presentation is organized based on the findings reported in these publications.

Outline

1. Introduction & Research Motivation
    a. Systemic Bottlenecks of Metric SLAM & LiDAR Hardware
    b. Semantic Blindness
    c. Path Planning Algorithm limitations
    d. Cartesian Coordinate Barrier
    e. Thesis Statement, Challenges & Research Questions (RQ1–RQ4)
2. Core Scientific Contributions and their Results
    a. Vision-Aided Context-Aware A* Planning & “Beware Lists”
    b. NeuralPathLite: Fast Diffusion-Based Path Planner
    c. Mapless Interactive Vision-Language Navigation with Drift Resets
    d. VTMMap: Geometry-Free Vision-to-Text World Models & Text Paths
3. List of Publications
4. Conclusion & Future Possibilities

Systemic Bottlenecks of Metric SLAM & LiDAR Hardware



- High Computational and Hardware Overhead.
- Cumulative Odometry Drift and Map Degradation in Open-Loop Traversal.
- Semantic Disconnect in Human-Centric Environments.

Systemic Bottleneck: Open-Loop Odometry Drift & Map Degradation

1. Step-Wise Motion Model in SE(2)

$$\mathbf{x}_t = \mathbf{x}_{t-1} \oplus (\mathbf{u}_t + \boldsymbol{\epsilon}_t), \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t)$$

2. Long-Horizon Unbounded Error Growth

$$\mathbf{x}_T = \mathbf{x}_0 \oplus \left(\bigoplus_{t=1}^{T} (\mathbf{u}_t + \boldsymbol{\epsilon}_t) \right) \implies \mathbf{P}_T = \sum_{t=1}^{T} \mathbf{J}_t \boldsymbol{\Sigma}_t \mathbf{J}_t^T \propto \mathcal{O}(T)$$

3. Factor Graph Optimization & Open-Loop Breakdown

$$\mathbf{X}^* = \arg\min_{\mathbf{X}} \left(\sum_{t=1}^{T} \|\mathbf{x}_t - f(\mathbf{x}_{t-1}, \mathbf{u}_t)\|^2_{\boldsymbol{\Sigma}_t} + \sum_{(i,j) \in \mathcal{E}_{\text{loop}}} \|\mathbf{x}_i - g(\mathbf{x}_j, \mathbf{z}_{ij})\|^2_{\boldsymbol{\Omega}_{ij}} \right)$$

$$\text{Open-Loop Condition: } \mathcal{E}_{\text{loop}} = \emptyset \implies \lim_{T \to \infty} \|\mathbf{p}_{\text{true}}(T) - \mathbf{p}_{\text{map}}(T)\| = \infty$$

- **Odometry Noise Accumulation** — S. Thrun, "Probabilistic robotics," Communications of the ACM, vol. 45, no. 3, pp. 52-57, 2002.
  - Pose error covariance grows as O(T) over long horizons.
- **Open-Loop Loop Closure Failure** — C. Cadena et al., "Past, Present, and Future of Simultaneous Localization and Mapping: Toward the Robust-Perception Age," IEEE Transactions on Robotics, vol. 32, no. 6, pp. 1309-1332, 2016.
  - Without loop closure (E_loop = empty), localization error diverges asymptotically.

Applications



Landmines in a battlefield-like environment



Industrial Chemical Spill Hazard



Warehouse Environment: Color-Coded Safety and Navigation Zones



Indoor Car Parking



Reserved Traffic Lane Markings



Color-Based Lane Identification

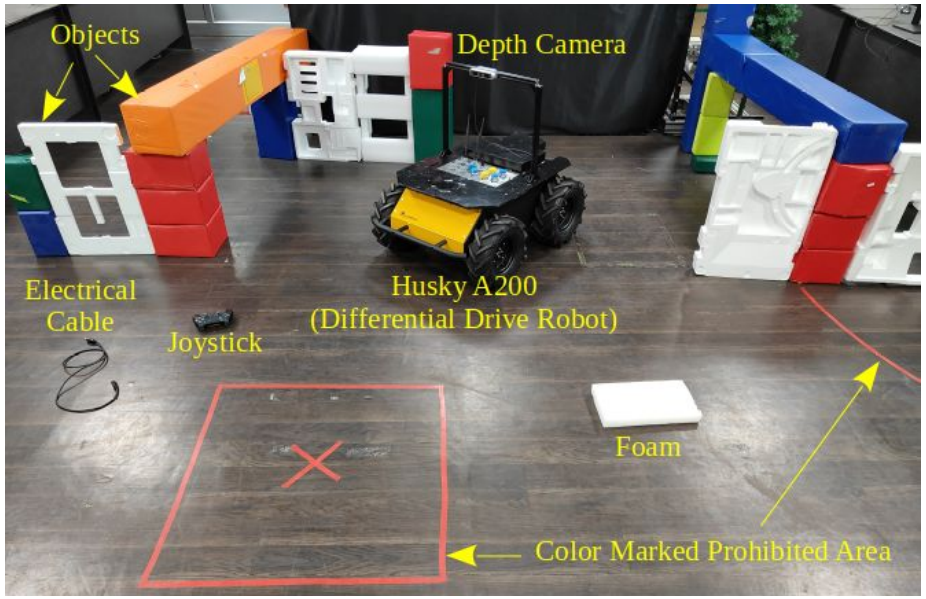
Continue…



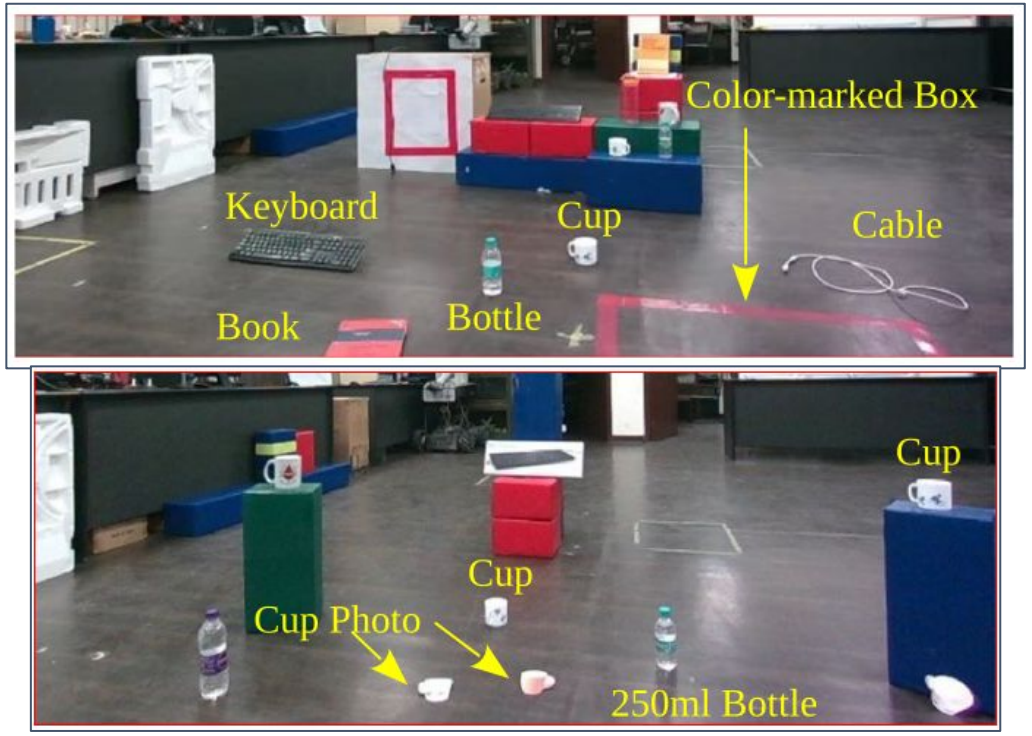
Restricted Access Area



Flooded Road



Real-world Setup



Motivating Examples for Semantic-Aware Autonomous Navigation

Path Planning Algorithm limitations





Can we build a single AI path planner that achieves fast real-time generation, scales effortlessly regardless of map size, resolution, or visual representation, and remains robust against out-of-distribution environments without path planning failures?

Cartesian Coordinate Barrier

User



Robot



Environment: Example of House



Synthetic Image Generated by Gemini Pro.

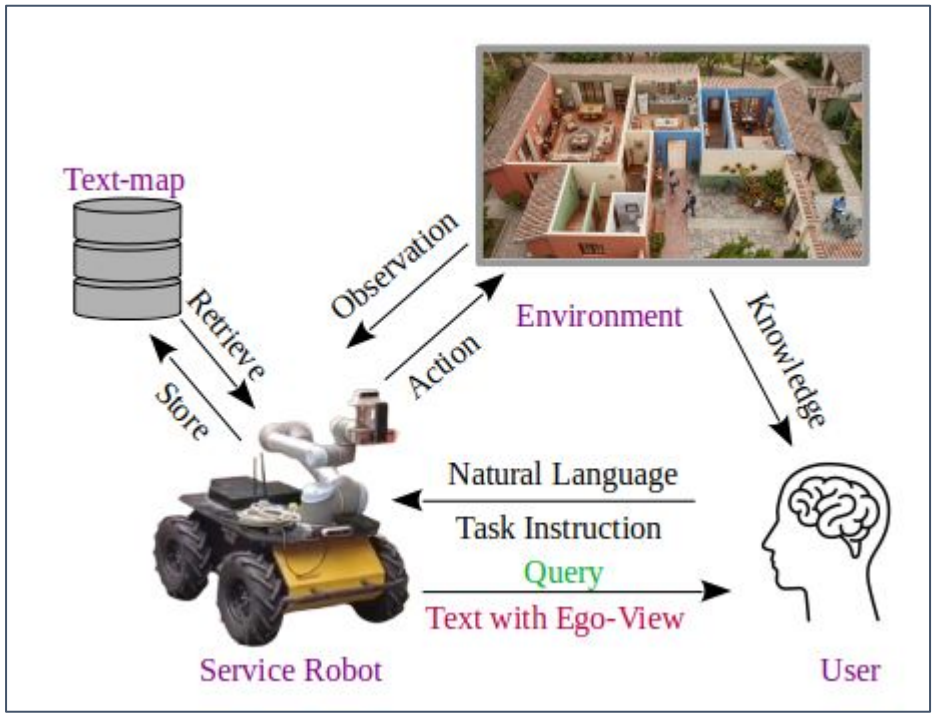
❓ **How can I determine the 2D coordinates of my destination?**

❓ **Can I simply say, "Go to the kitchen," instead of providing 2D coordinates?**

❓ **Can the robot remember my house's previously explored places so that it understands my instructions and assists me like an intelligent personal assistant?**

Thesis Statement, Challenges & Research Questions (RQ1–RQ4)

"Autonomous mobile robot navigation in human-centric environments can be rendered significantly safer, drift-mitigation, computationally frugal, and naturally commandable by transitioning from rigid metric-geometric SLAM maps toward semantically-aware visual costmaps, generative neural path planning, and visual-text memory models."



Thesis Statement

Challenges

1. Contextual Traversability.
2. Generative Path Planning and Fleet Scalability.
3. Interactive Human-Robot Navigation and Drift Mitigation.
4. Vision-Text Memory Mapping and Natural Language Control.

Research Questions (RQ1–RQ4)

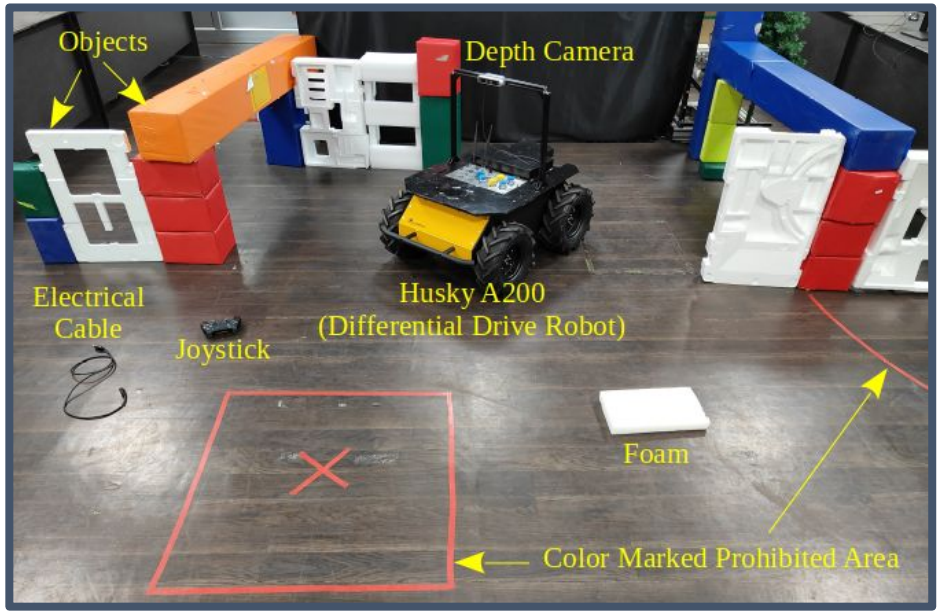
1. (Semantic Perception): How can egocentric RGB-D visual cues and semantic segmentation be projected online to adjust costmaps for safe local path planning around a user-defined “beware list” of surface items, ensuring the robot avoids driving over or overriding semantically sensitive objects?

2. (Generative Path Planning): Can lightweight image-conditioned diffusion models achieve real-time, computationally efficient DDPM-based path generation in high-resolution, visually complex environments, and how can a Secure Copy Protocol (SCP) ensure scalable execution across multi-robot fleets?

3. (Drift-Free VLN): Can multi-task visual perception networks combined with Large Language Models (LLMs) enable robots to interpret human language commands, execute vision-grounded action plans, and eliminate long-horizon odometry drift through target-anchored visual landmark resets?

4. (Vision-Text Memory & Text-Space Navigation): Is it possible to eliminate Cartesian coordinate maps (x, y, θ) by representing environment memory as a Vision-Text Memory Map (VTMMap), and how can text-space path planning combined with interactive user communication progressively enhance a service robot’s commandability and operational efficiency?

a. Vision-Aided Context-Aware A* Planning & "Beware Lists"



Real-world Experimental Setup

(Semantic Perception): How can egocentric RGB-D visual cues and semantic segmentation be projected online to adjust costmaps for safe local path planning around a user-defined "beware list" of surface items, ensuring the robot avoids driving over or overriding semantically sensitive objects?

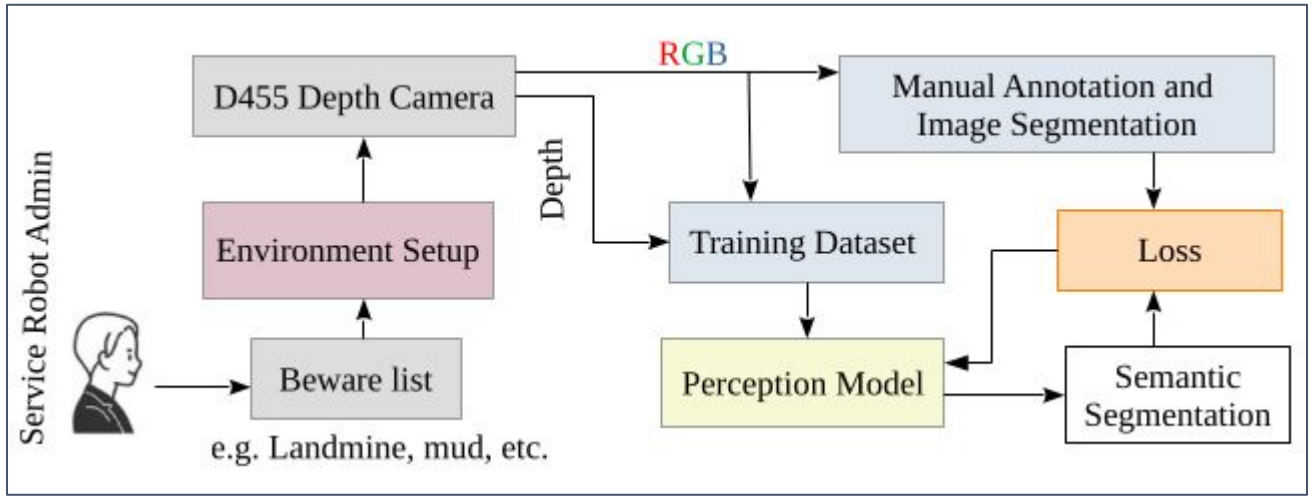
Literature Survey & SOTA Comparison

Navigation Pipeline	Succ. (NH) [22/22]	Succ. (H) [20/20]	Planning Time (ms)	Path Length (m)	Path Smoothness (Rad)
RTAB-Map + A^* + DWA	22/22	0/20	24	12.1	0.84
OpenVSLAM + A^* + DWA	22/22	0/20	26	12.3	0.87
Kimera + A^* + DWA	22/22	0/20	27	12.2	0.86
ROS Navigation Stack (A^* + DWA)	22/22	0/20	25	12.4	0.82
OpenVSLAM + RRT* + TEB	22/22	0/20	98	13.3	0.61
RTAB-Map + FAR Planner	22/22	0/20	20	11.8	0.74
OpenVSLAM + ViT-A^*	22/22	0/20	110	12.6	0.79
Kimera + DiPPeR + TEB	21/22	0/20	132	12.8	0.55
Proposed Vision-Aided Online A^* (Ours)	**22/22 (100%)**	**18/20 (90%)**	**48**	**13.6**	**0.91**

- Sumikura, S., Shibuya, M., & Sakurada, K. (2019, October). OpenVSLAM: A versatile visual SLAM framework. In *Proceedings of the 27th ACM international conference on multimedia* (pp. 2292-2295).

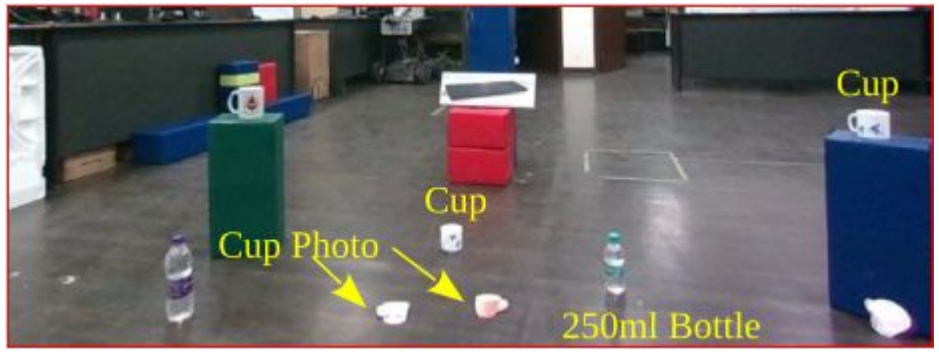
Vision-Aided Context-Aware A* Planning & "Beware Lists"

Beware List: All yellow-labeled objects are treated as avoidance targets, except the photo.



Perception module training pipeline





Environment Setup with Beware Items

Workspace Partitioning:

$$\mathcal{O} = \mathcal{O}_{\text{rigid}} \cup \mathcal{O}_{\text{free}} \cup \mathcal{B}$$

Divides workspace into rigid walls ($\mathcal{O}_{rigid}$), bare floor ($\mathcal{O}_{free}$), and Beware items ($\mathcal{B}$).

Continue…

Beware Item Property:

$$\forall b \in \mathcal{B}, \quad \text{Traversable}_{\text{phys}}(b) = \text{True}, \quad \text{SemanticRisk}(b) > 0$$

Physically traversable by wheels, but carrying high semantic risk (phones, papers, chemical spills).

Semantic Costmap Formulation:

$$C(\mathbf{p}) = C_{\text{geo}}(\mathbf{p}) + \sum_{b \in \mathcal{B}} w_b \cdot \mathbb{I}\left(\mathbf{p} \in \Omega(b)\right)$$

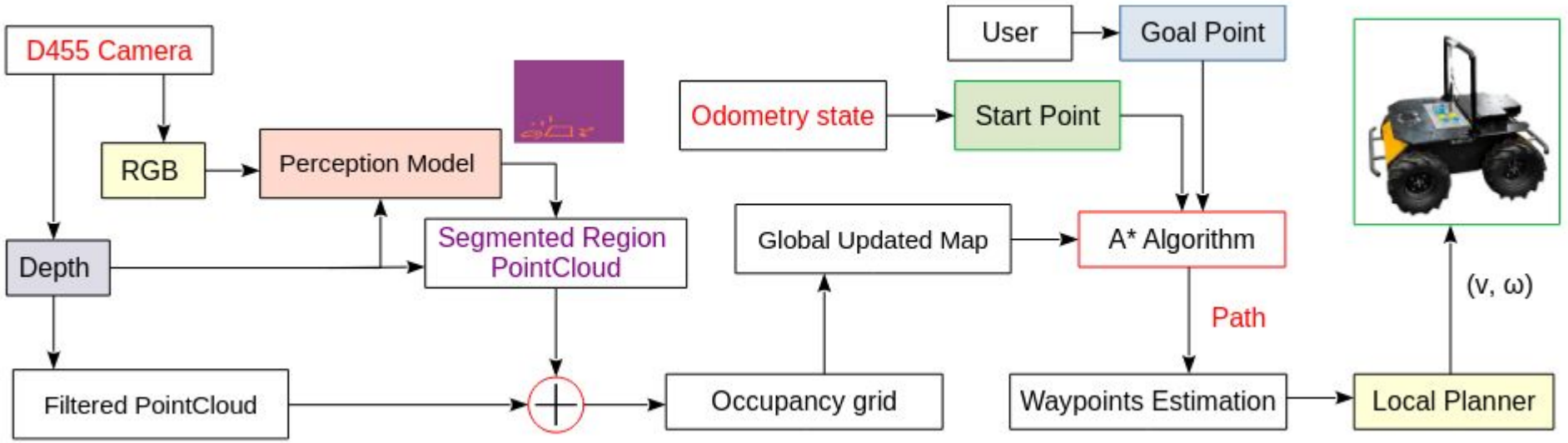
Paints penalty w_b *onto cell* $\mathbf{p}$ *if inside Beware footprint* $\Omega(b)$. $\mathbb{I}(\cdot)$ *is an indicator function (*1 *if* $\mathbf{p} \in \Omega(b)$, 0 *otherwise).*

Zero-Override Safety Constraint:

$$\sigma^*(s) \cap \left(\bigcup_{b \in \mathcal{B}} \Omega(b) \right) = \emptyset, \quad \forall s \in [0, 1]$$

Ensures zero spatial overlap ($\emptyset$) *between planned path* $\sigma^*(s)$ *and Beware item areas.*

Continue…



Navigation System Overview

Continue…

Table 1: Quantitative comparison of semantic segmentation models on the custom obstacle detection dataset. All models are fine-tuned from pretrained segmentation weights. Best results are in bold, second best are underlined.

	RGB Segmentation		**RGB-D Segmentation**			
Metric	DeepLabV3-R50 [17]	SegFormer-B2 [13]	ESANet-R34 [18]	MIPANet [16]	RedNet-R50 [3]	DFormerv2-S [19]
Architecture	CNN	Transformer	CNN	CNN	CNN	Transformer (GSA)
Pretrained On	COCO	ADE20K	NYUv2	NYUv2	NYU-like	NYUv2
mIoU (%)	79.69	**93.66**	71.95	74.13	91.90	<u>92.52</u>
Obstacle IoU (%)	59.77	**87.43**	44.44	48.68	83.94	<u>85.17</u>
F1 / Dice	0.748	**0.933**	0.615	0.655	0.913	<u>0.920</u>
Precision	0.769	0.938	0.691	**0.946**	0.901	<u>0.919</u>
Recall	0.729	<u>0.928</u>	0.555	0.501	**0.925**	0.921
Pixel Acc (%)	99.62	**99.90**	99.46	99.59	99.86	<u>99.87</u>
Params (M)	39.63	27.35	46.88	271.89	81.94	**26.67**
FPS	**40.0**	34.5	27.5	18.5	<u>38.1</u>	10.4
Inference (ms)	**25.0**	29.0	36.4	54.0	<u>26.2</u>	96.2
Real-time	Yes	Yes	No	No	Yes	No

Continue…

3D Point Generation of Beware Items:

- Semantic Masking & Depth Gating

$$M_b(u,v) = \mathbb{I}\Big(S(u,v) = b\Big), \quad b \in \mathcal{B}$$

$$D_b(u,v) = D(u,v) \cdot M_b(u,v)$$

Isolates depth D_b strictly for surface items in Beware List $\mathcal{B}$.

- Pinhole 3D Camera Projection (K^{-1})

$$\begin{bmatrix} X_b \\ Y_b \\ Z_b \end{bmatrix} = D_b(u,v) \cdot \mathbf{K}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

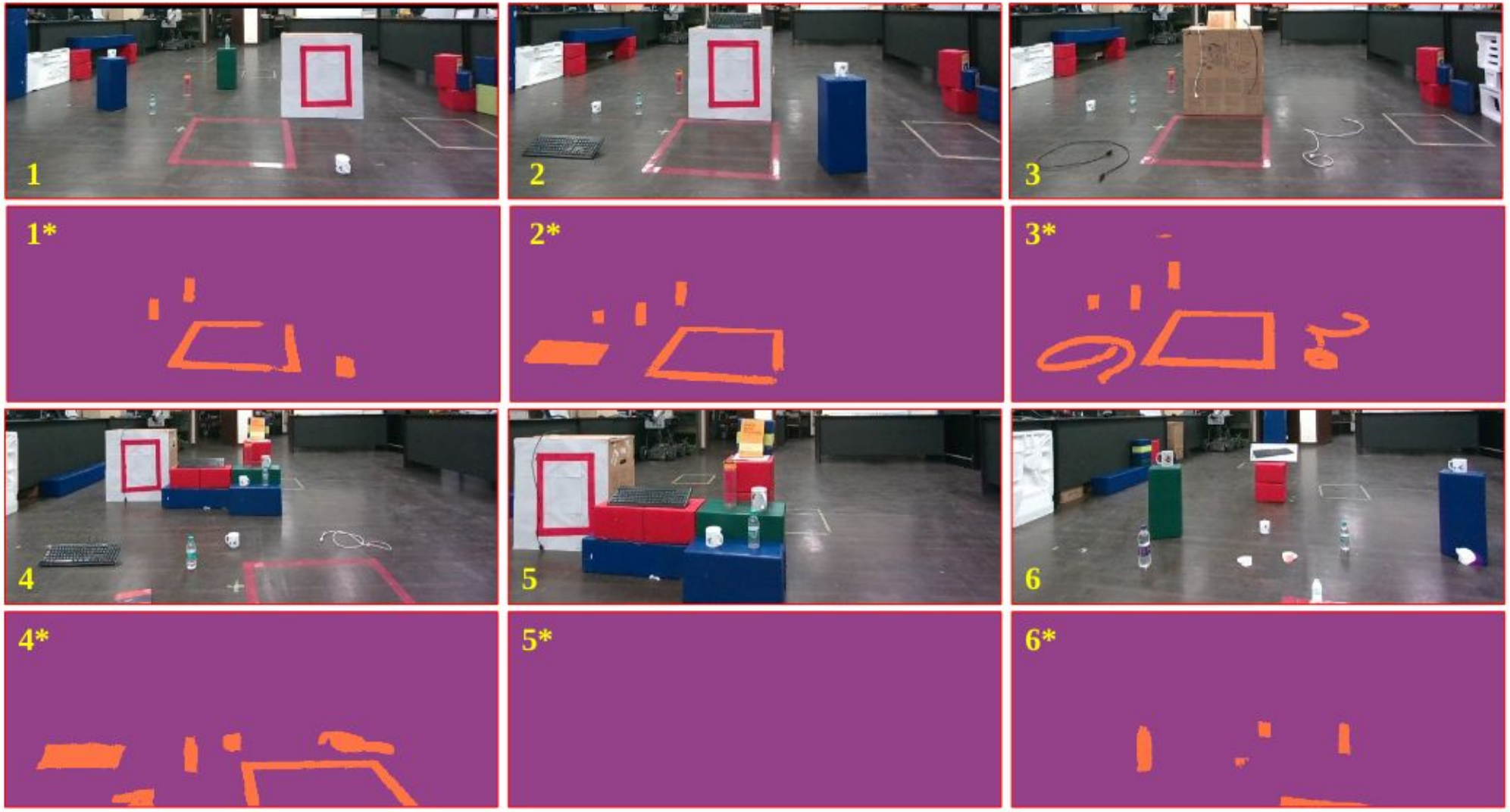
Back-projects 2D pixel coordinates (u,v) to 3D point cloud $\mathcal{P}_{3D}(b)$.

- Costmap Projection & Obstacle Avoidance

$$C(\mathbf{p}) = C_{\text{geo}}(\mathbf{p}) + \sum_{b \in \mathcal{B}} w_b \cdot \mathbb{I}\left(\mathbf{p} \in \text{Proj}(\mathcal{P}_{3D}(b))\right)$$

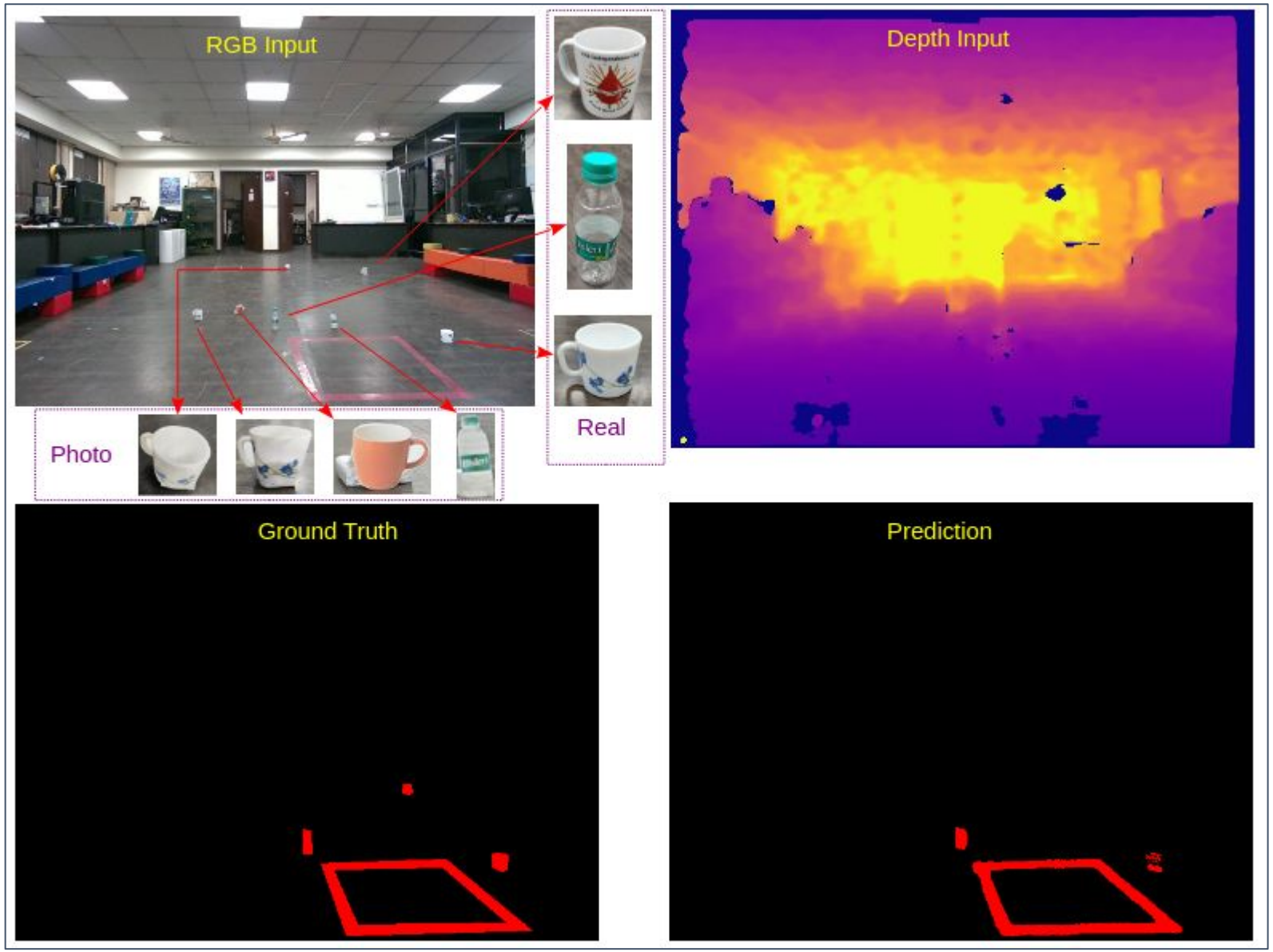
Injects 3D points into online costmap $C(\mathbf{p})$ to enforce zero-override planning.

Results



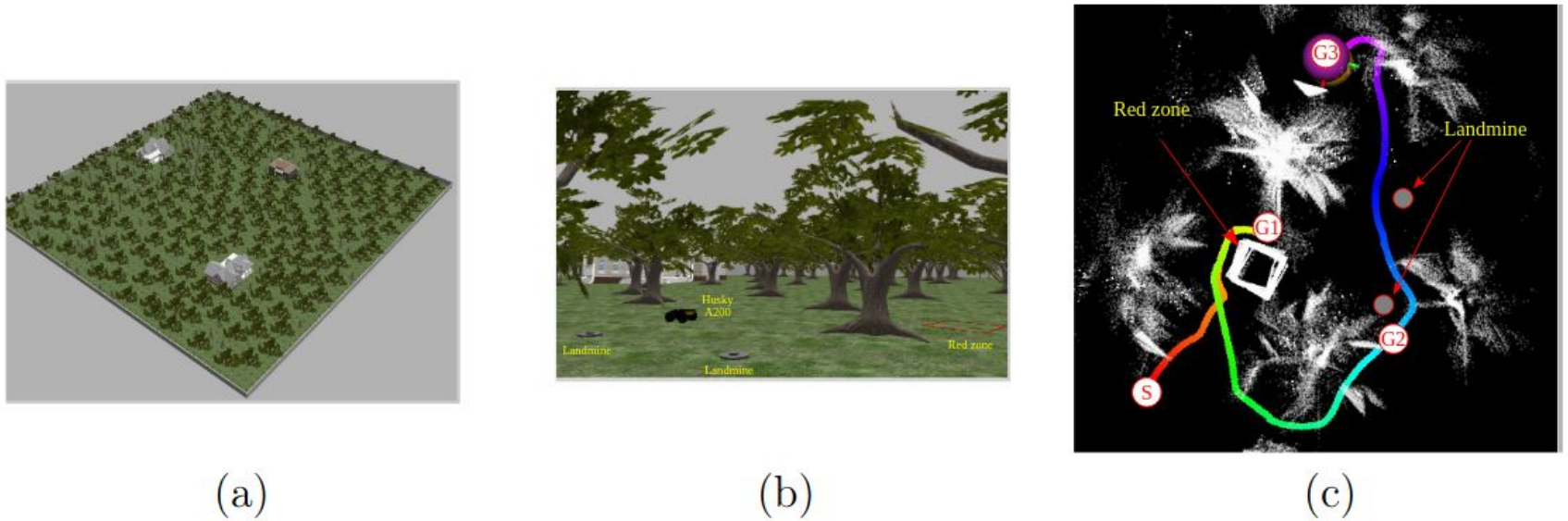
RGB-D Semantic Segmentation Results: Frames 1^* to 6^* show the corresponding semantic segmentation outputs generated by our trained RedNet model. Quantitatively, the model achieved an mIoU of 0.923, pixel accuracy of 0.992, background IoU of 0.99, and obstacle IoU of 0.86.

Results



Qualitative comparison between RedNet-R50 predictions and ground truth on the custom obstacle dataset.

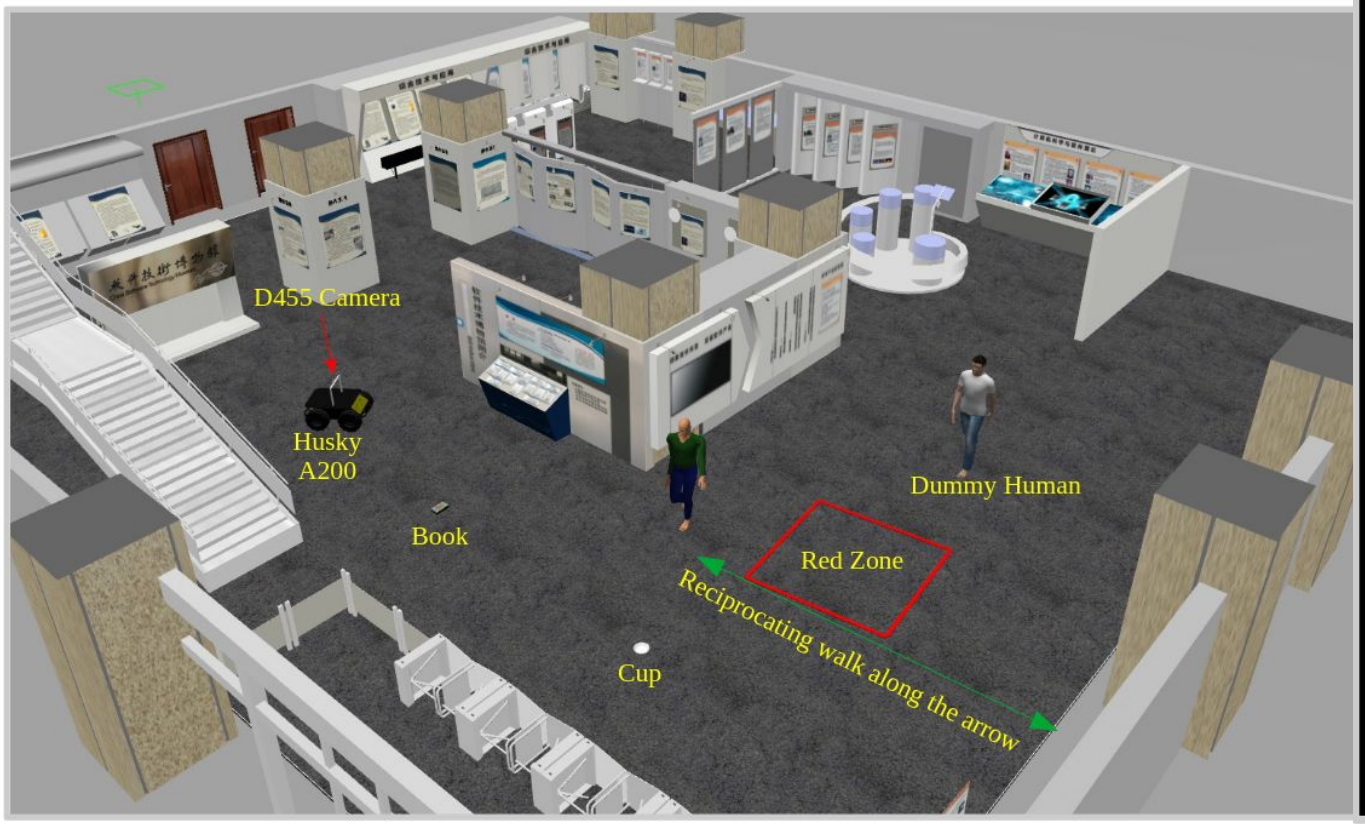
Results



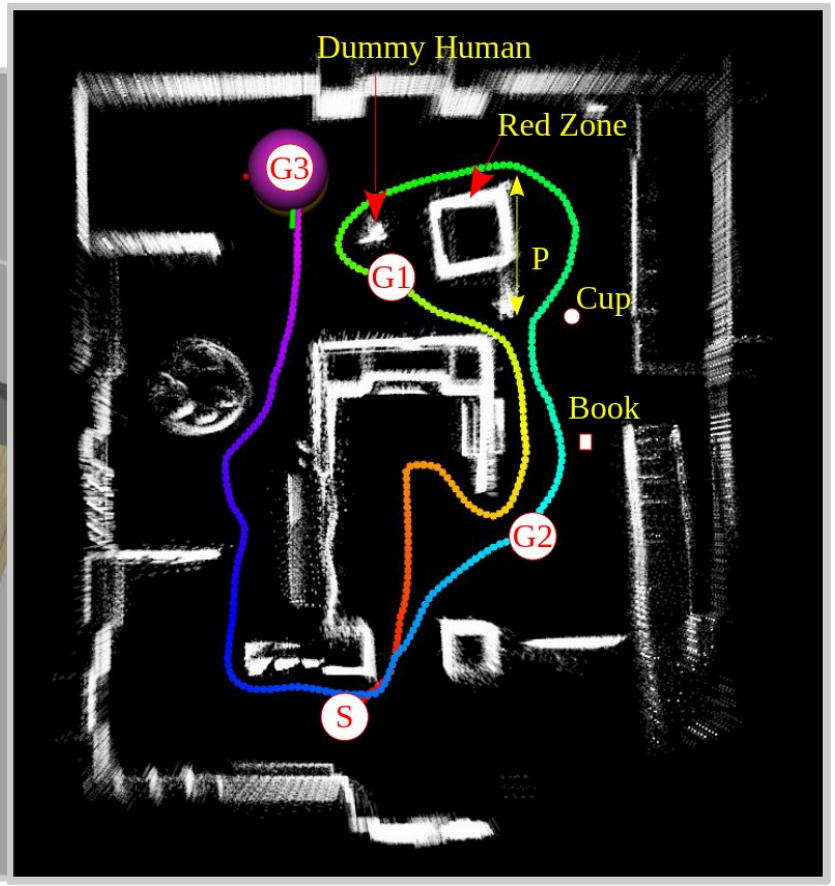
(a) (b) (c)

Simulation in a Forest Environment with Semantic Hazards. (a) Top down view of the Gazebo forest environment (150m x 150m), featuring scattered buildings, numerous trees, and designated hazardous zones (landmines and a red zone). (b) A small portion of the forest environment near the Husky A200 robot area with the red zone, and buried landmines. (c) Overhead visualization of the robot's exploration and navigation trajectory (multi color line) from Start (S) through intermediate goals (G1, G2) to the final destination (G3)

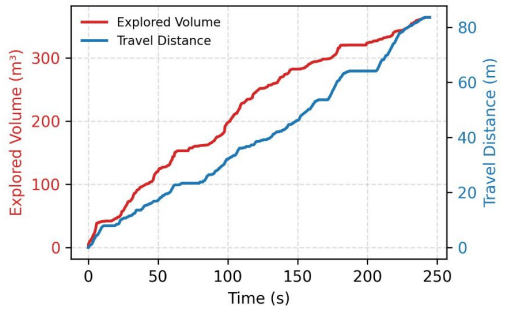
Results



Gazebo Simulation Environment Setup.

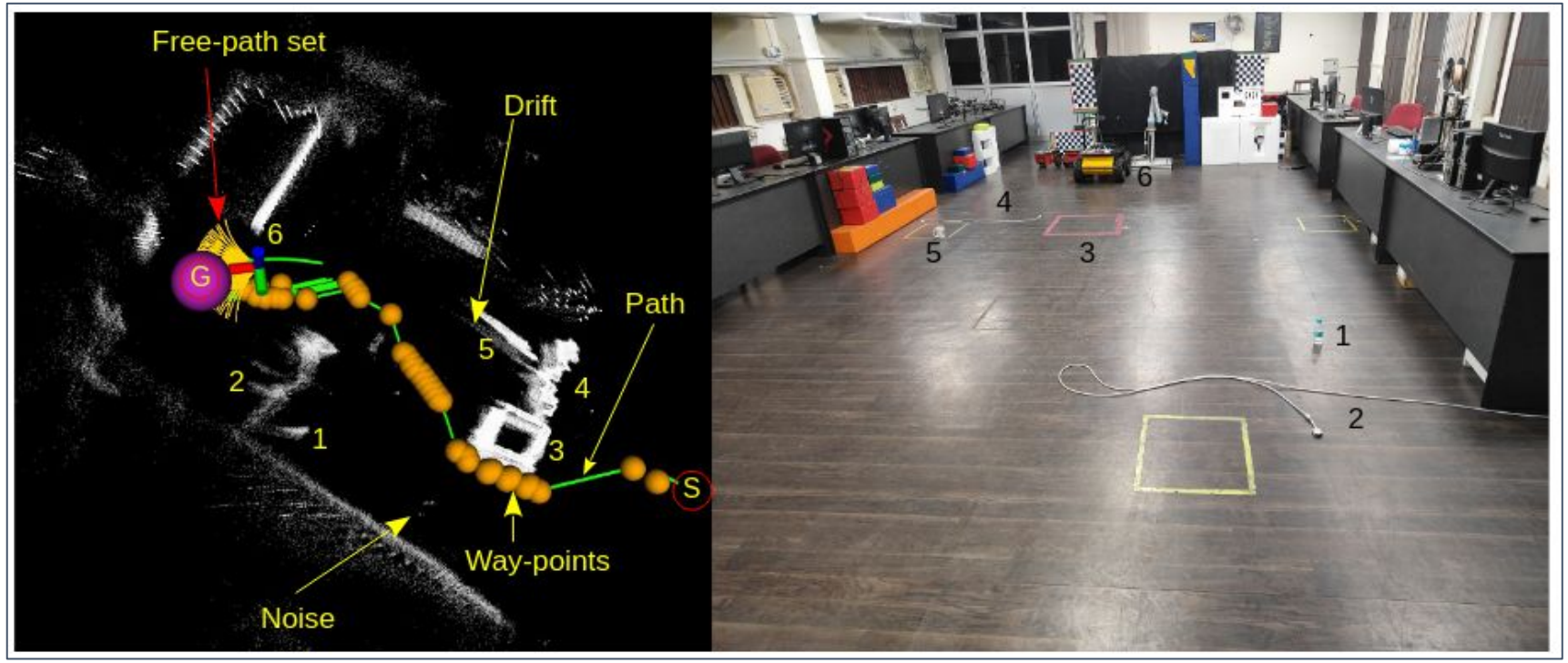


Top-down View of a Multi-Goal Simulation Trajectory.



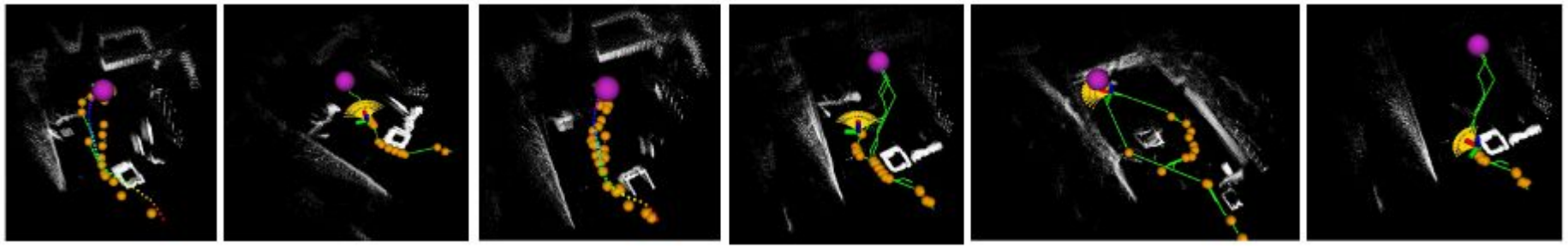
The results highlight the capability of the proposed algorithm to navigate efficiently while avoiding geometric and non-geometric obstacles.

Results



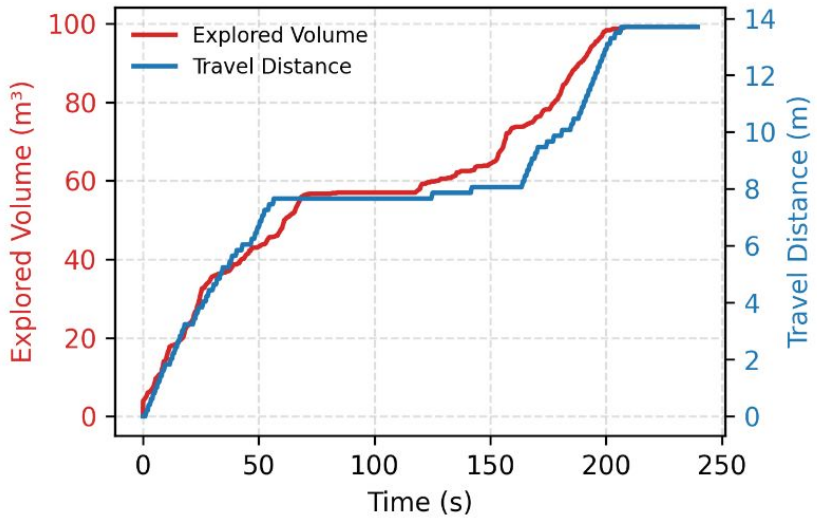
Traversed path and point cloud.

Real-world Setup.

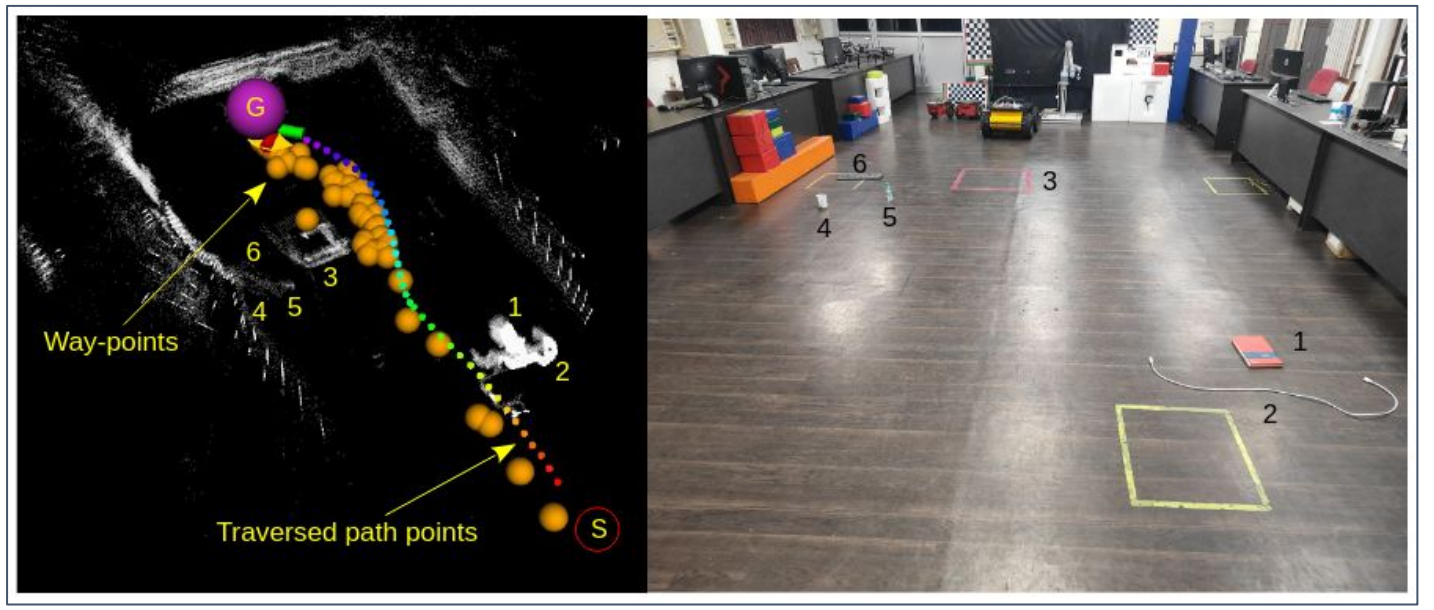


Point cloud views of the real-world environment illustrating successful path planning and traversal across multiple goal points, with our algorithm avoiding both obstacles and user-specified restricted areas.

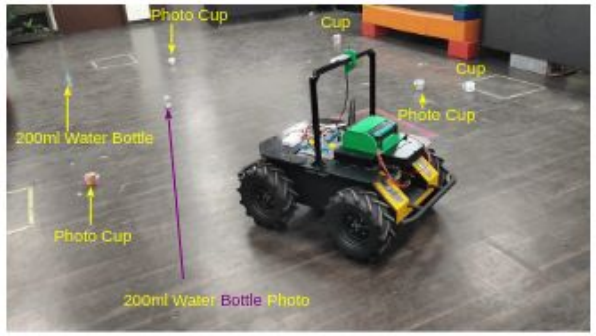
Results



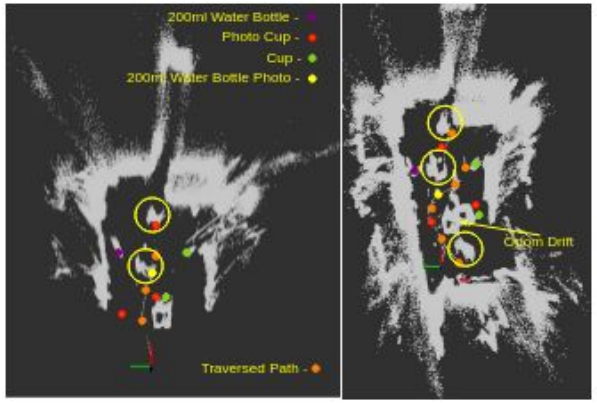
The graphs illustrate the algorithm's efficiency in navigation and exploration within a complex environment.



Real-World Navigation Performance



(a)



(b)

(a) Experimental environment

(b) Top-view point cloud representation with semantic segmentation results and the traversed path.

Results

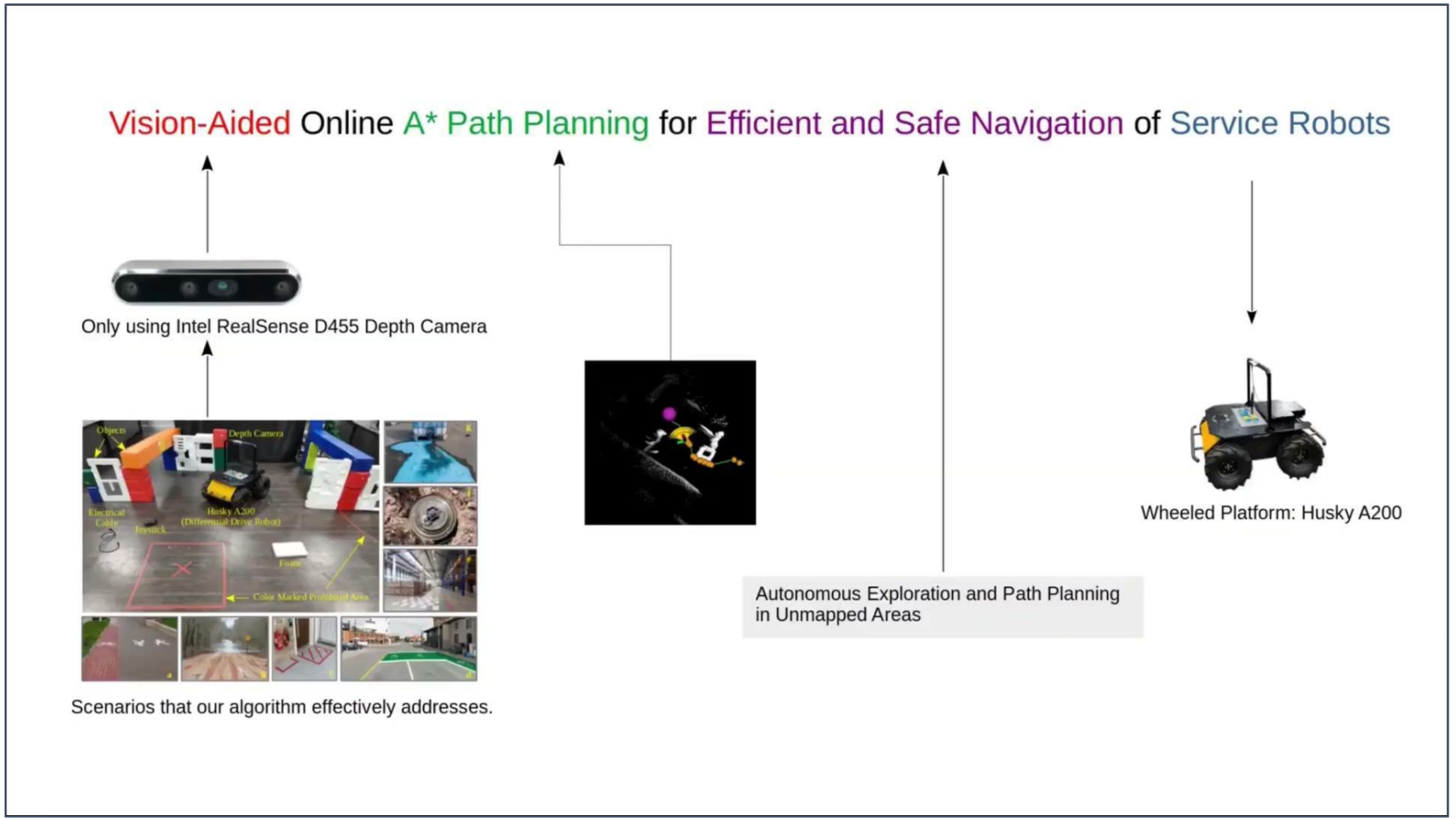
TABLE 2: Comparison of RGB-D navigation pipelines over 42 trials. NH: trials without hazards (22), H: trials with hazards (20). Time and Len denote average planning time (ms) and path length (m), respectively. Smooth indicates trajectory smoothness measured as average heading change.

Navigation Pipeline	Succ.(NH)	Succ.(H)	Time	Len	Smooth
RTAB-Map [26] + A* + DWA	22/22	0/20	24	12.1	0.84
OpenVSLAM [27] + A* + DWA	22/22	0/20	26	12.3	0.87
Kimera [28] + A* + DWA	22/22	0/20	27	12.2	0.86
ROS Navigation Stack [29] (A* + DWA)	22/22	0/20	25	12.4	0.82
OpenVSLAM [27] + RRT* [7] + TEB [30]	22/22	0/20	98	13.3	0.61
RTAB-Map [26] + FAR Planner [8]	22/22	0/20	**20**	**11.8**	0.74
OpenVSLAM [27] + ViT-A* [31]	22/22	0/20	110	12.6	0.79
Kimera [28] + DiPPeR [1] + TEB [30]	21/22	0/20	132	12.8	**0.55**
Proposed Method	**22/22**	**18/20**	48	13.6	0.91

F. Yang, C. Cao, H. Zhu, J. Oh and J. Zhang, "FAR Planner: Fast, Attemptable Route Planner using Dynamic Visibility Update," *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Kyoto, Japan, 2022, pp. 9-16, doi: 10.1109/IROS47612.2022.9981574.

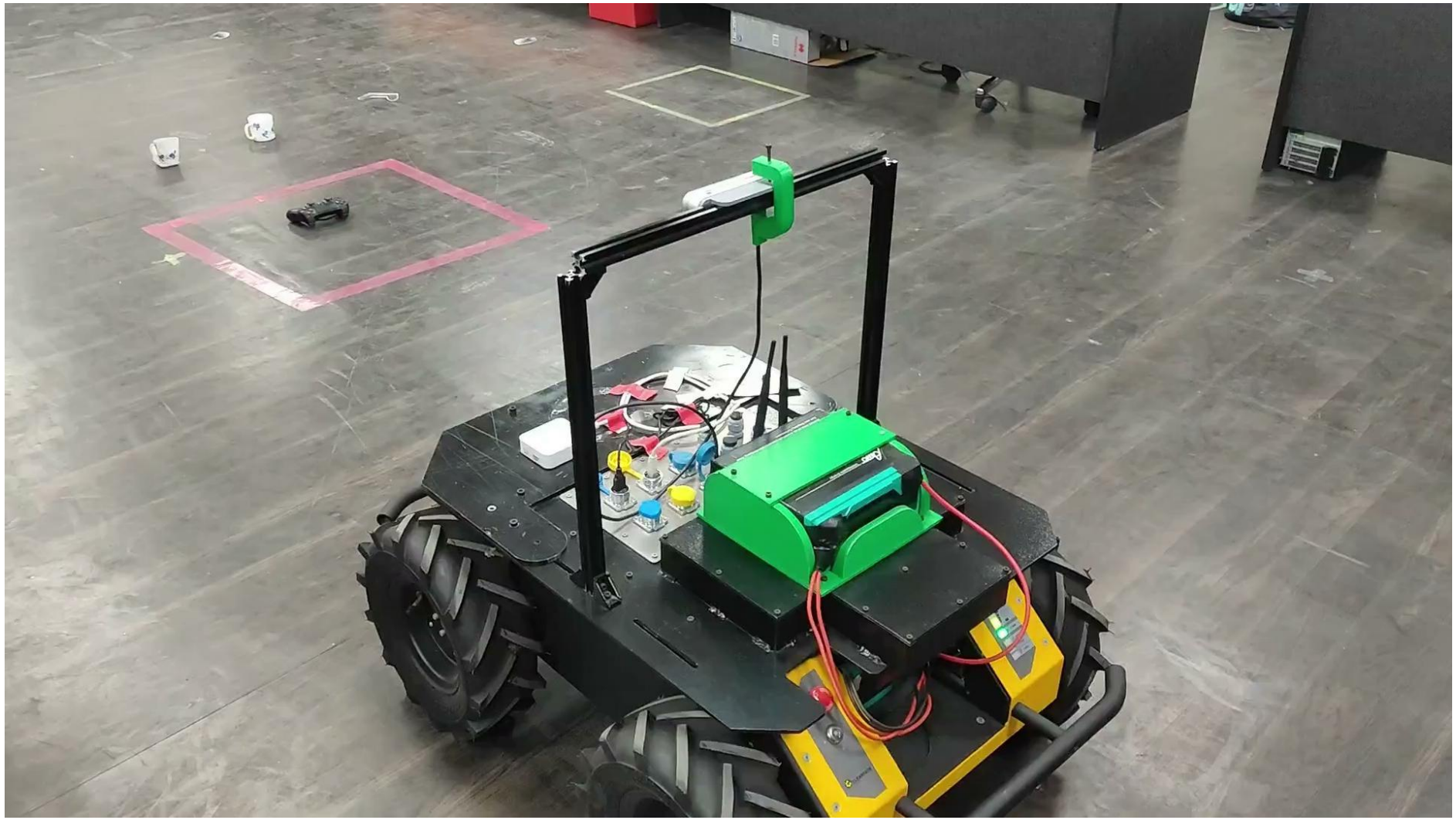
Source Code and Demonstration Videos

Source Code Link



- P. Kumar and T. Sandhan, "Vision-Aided Online A* Path Planning for Efficient and Safe Navigation of Service Robots," International Journal of Intelligent Robotics and Applications (Springer), 2026, doi: 10.1007/s41315-026-00540-z. **(Published)**

Demonstration Videos



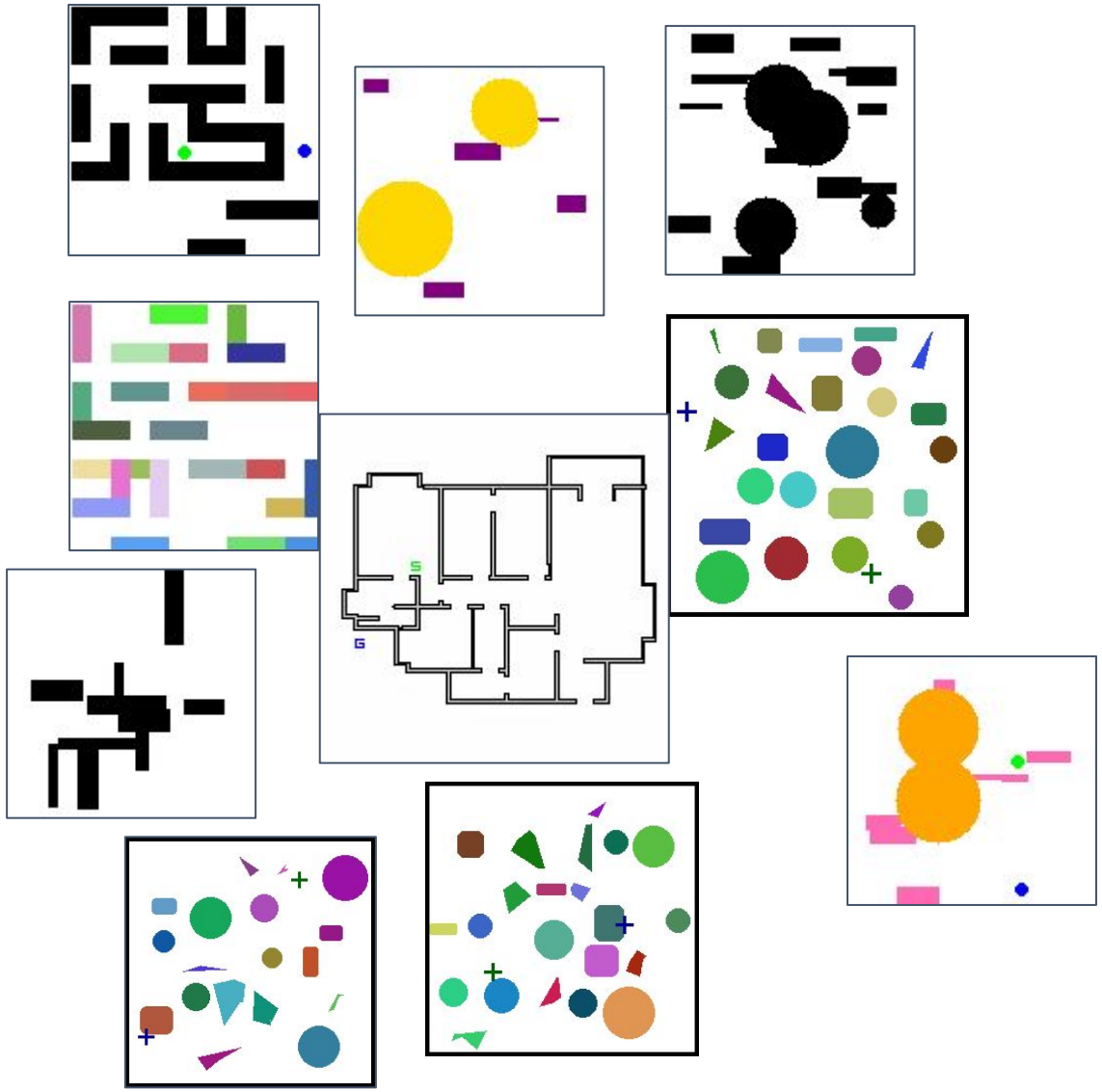
b. NeuralPathLite: Fast Diffusion-Based Path Planner

(Generative Path Planning): Can lightweight image-conditioned diffusion models achieve real-time, computationally efficient DDPM-based path generation in high-resolution, visually complex environments, and how can a Secure Copy Protocol (SCP) ensure scalable execution across multi-robot fleets?

NeuralPathLite: Fast Diffusion-Based Path Planner

NeuralPathLite: Existing generative path planners trade off execution speed for trajectory accuracy. NeuralPathLite bridges this fundamental gap by decoupling start-goal positional semantics from obstacle geometry.

- **Real-Time Generative Planning:** Lightweight DDPM achieving **0.16 s latency** (7.4× speedup over SOTA planners).
- **Semantic Visual Anchoring:** Disentangles start-goal semantics from visual obstacles $\Longrightarrow$ **96.12% OOD success** (prevents path planning failures).
- **Centralized Swarm Offloading:** Server-Client Protocol (SCP) offloading computation for edge fleets (Clearpath Husky A200).



Literature Survey & SOTA Comparison

Paradigm	Key Methods	Core Strengths	Critical Bottlenecks / Failure Modes
Classical Search	A^*, Dijkstra	Resolution optimal	Exponential computational latency $\mathcal{O}(N \log N)$ in dense maps
Sampling-Based	RRT*, PRM	Fast space exploration	Rough trajectory paths, non-deterministic execution variance
Existing Neural/DDPM	ViT-A^*, DiPPeR	Neural heuristic speed	Heavy compute backbones, pathfinding failures in OOD maps
NeuralPathLite (Ours)	**IEEE T-ASE 2026**	**0.16s latency, 96.12% OOD**	**Lightweight UNet + Disentangled Semantic Visual Anchoring**

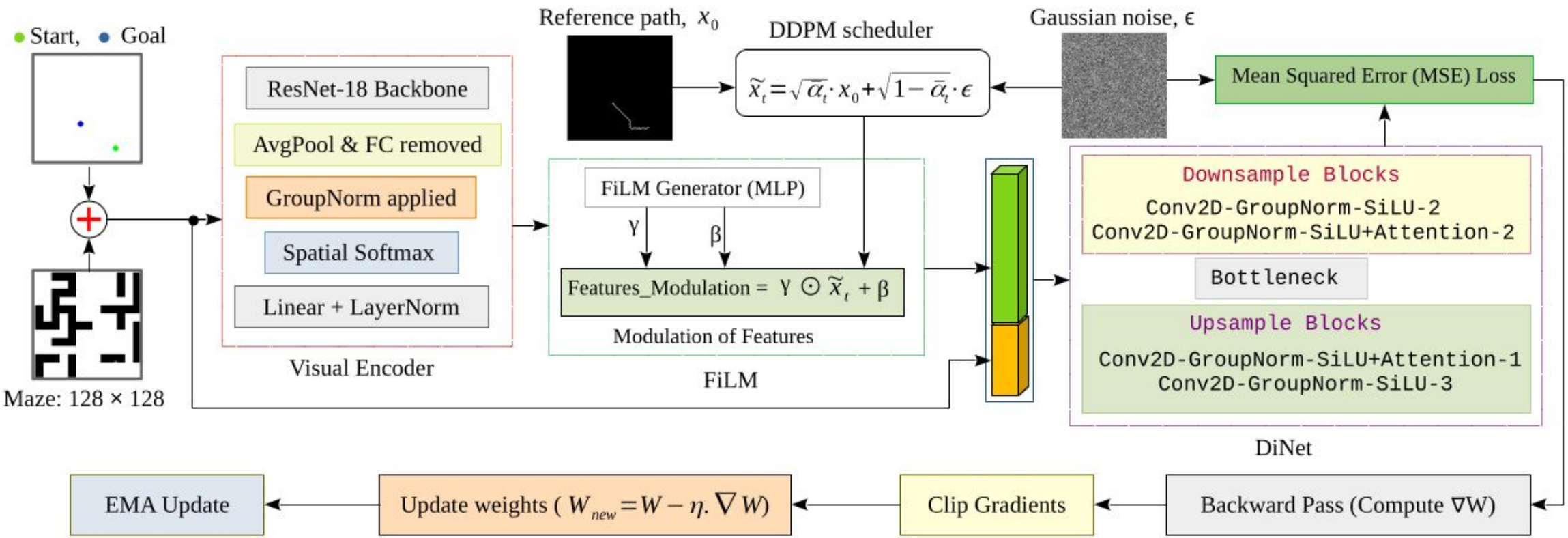
J. Liu, M. Stamatopoulou and D. Kanoulas, "DiPPeR: Diffusion-based 2D Path Planner applied on Legged Robots," *2024 IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024, pp. 9264-9270, doi: 10.1109/ICRA57147.2024.10610013.

Continue…

MPNet vs NeuralPathLite.

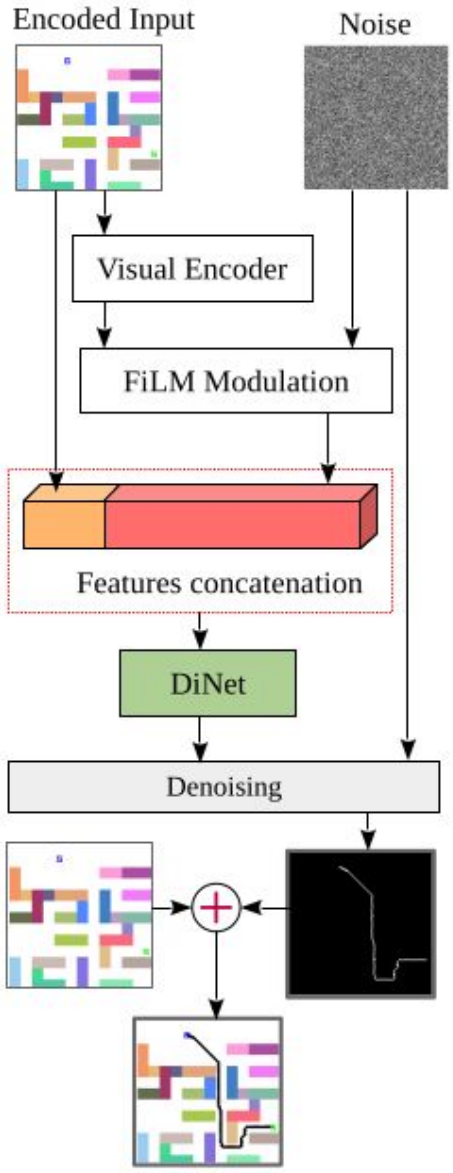
Aspect	**MPNet (TRO, 2021)**	**NeuralPathLite**
Input	Point clouds / voxels	2D maps with visual start–goal cues
Goal	State configurations	Visual markers
Planning	Iterative bidirectional planning	Diffusion-based generation
Search	Hybrid replanning possible	No explicit search (fixed steps)
Inference	∼1 s (reported mean)	∼0.169 s
Completeness	Via hybrid planners	No formal guarantees
Environment	2D–7D spaces	High-clutter maze environments

Methodology



Training pipeline of NeuralPathLite

Continue…



Inference pipeline of NeuralPathLite.

1. Optimal Path Formulation & Forward Diffusion

$$\mathcal{P}^* = \arg\min_{\mathcal{P}\in\Omega} c(\mathcal{P}), \qquad x_t = \sqrt{\bar{\alpha}_t}\cdot x_0 + \sqrt{1-\bar{\alpha}_t}\cdot\epsilon, \quad \epsilon\sim\mathcal{N}(\mathbf{0},\mathbf{I})$$

$$\text{Square Cosine Schedule:} \quad \bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad f(t) = \cos^2\left(\frac{\frac{t}{T}+s}{1+s}\cdot\frac{\pi}{2}\right)$$

2. Semantic Visual Anchoring (SVA) & FiLM Modulation

$$\text{ResNet-18 Spatial Softmax:} \quad z = f_{\text{proj}}\Big(\text{SpatialSoftmax}(f(\tilde{I}))\Big) \in \mathbb{R}^d$$

$$\text{FiLM Path Modulation:} \quad \tilde{p}_{\text{mod}} = \gamma\cdot x_t + \beta$$

3. Reverse Denoising Step (DiNet) & Training Loss

$$x_{t-1} = \lambda\left(x_t - \eta\,\theta(x_t, M, t)\right) + \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$$

$$\mathcal{L}_{\text{MSE}} = \|\theta(\tilde{p}_{\text{mod}}, t) - \epsilon\|_2^2$$

NeuralPathLite Results

TABLE 1: INFERENCE PERFORMANCE ON THE SET OF MAZES FIG. 5.
RD INDICATES REVERSE DIFFUSION, AND SR IS THE SUCCESS RATE

RD Steps (t)	(a)		(b)		(c)		(d)		(e)		(f)	
	Time (s)	SR (%)	Time (s)	SR (%)	Time (s)	SR (%)	Time (s)	SR (%)	Time (s)	SR (%)	Time (s)	SR (%)
5	0.16	66	0.17	92	0.17	90	0.18	90	0.16	89	0.18	91
10	0.34	79	0.32	94	0.34	95	0.34	92	0.32	90	0.32	94
20	**0.64**	**84**	**0.72**	**97**	**0.68**	**96**	**0.64**	**95**	**0.65**	**92**	**0.68**	**98**

TABLE 2: INFERENCE SUCCESS RATE OF OUR MODEL AND DiPPeR [40].

Dataset	Success Rate (%) [40]	Success Rate (%) **Ours**
Validation, Fig. 5(b)	81	**97**
Out-of-Distribution, Fig. 5(c)–(f)	83	**95.25**
Average	82	**96.12**

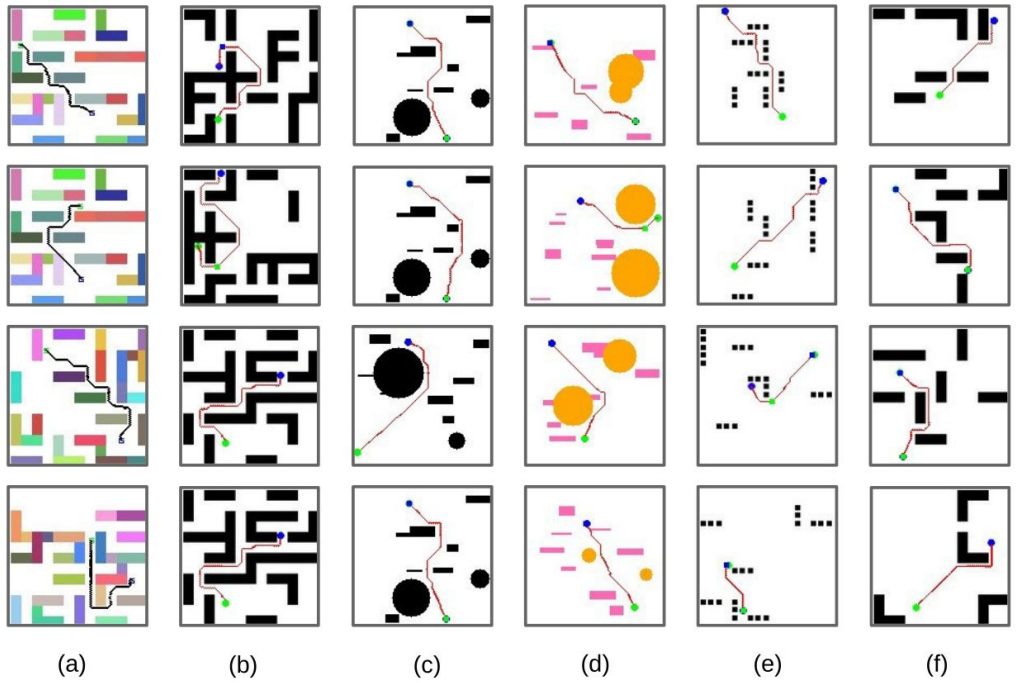


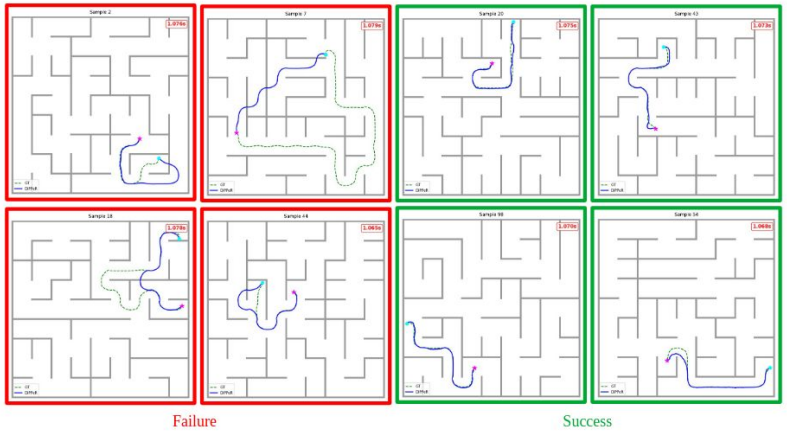
Fig.5: The generated paths successfully connect the given start and goal points across a variety of navigation scenarios, demonstrating strong generalization.

Continue…

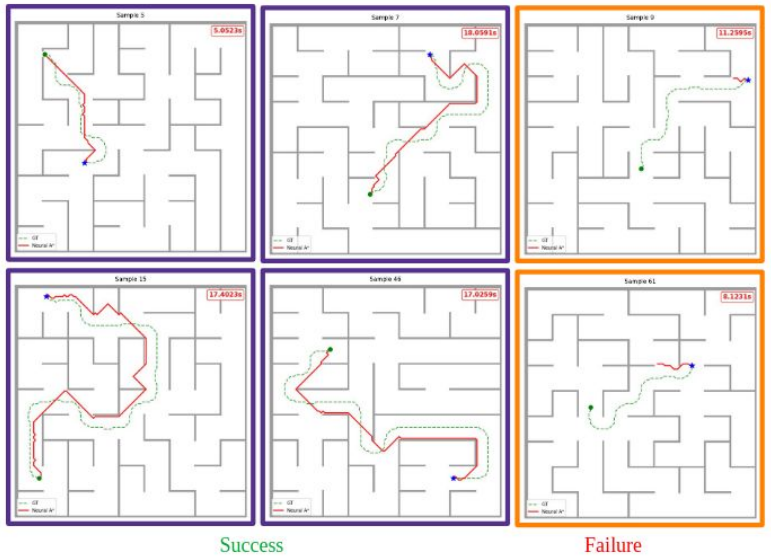
TABLE III

SUCCESS RATE COMPARISON ON SINGLE-PATH MAZE BENCHMARK (DIPPER DATASET)

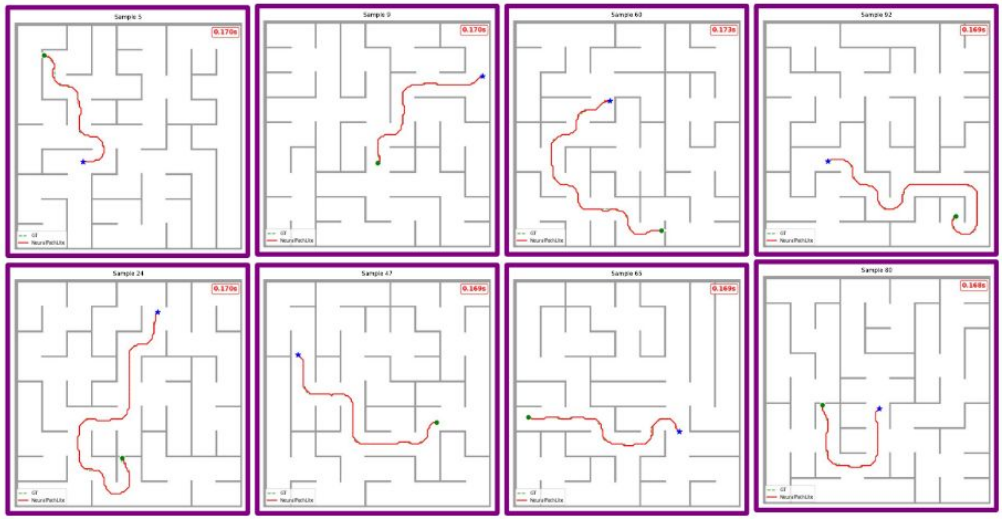
Planner	Type	Success Rate (%) ↑
Neural A* [9]	Hybrid	70
DiPPeR [40]	Learning-based	23
NeuralPathLite (Ours)	**Learning-based**	**92**



Qualitative Analysis of DiPPeR on Single-Path Mazes



Qualitative Analysis of Neural A* on Single-Path Mazes.



Performance of NeuralPathLite in Intricate Maze Environments.

Comparison with SOTA

TABLE 3: QUANTITATIVE COMPARISON OF PATH PLANNERS ACROSS THE MRPB BENCHMARK ENVIRONMENTS ILLUSTRATED IN Next Slide.

Environment	Planner	Inference Time (s) ↓	Path Length ↓	Explored Nodes ↓
(a) room02 360 × 360	Standard A*	0.58 ± 0.43	397.7 ± 235.9 st	46,831.8 ± 34,869.3
	Naive A*	4.63 ± 4.23	397.7 ± 235.9 st	46,900.2 ± 34,911.6
	Neural A* [6]	10.99 ± 7.46	748.60 ± 578.79 px	8,826.3 ± 5,560.4
	DiPPeR [7]	1.2520 ± 0.0034	315.64 ± 237.67 px	0 ± 0
	NeuralPathLite (Ours)	**0.1695 ± 0.0009**	**241.9 ± 184.9 px**	**0 ± 0**
(b) office01 280 × 280	Standard A*	0.19 ± 0.22	268.4 ± 164.4 st	17,151.2 ± 20,525.7
	Naive A*	1.12 ± 1.48	268.4 ± 164.4 st	17,173.9 ± 20,537.5
	Neural A* [6]	8.97 ± 4.80	349.66 ± 233.92 px	6,883.4 ± 3,654.5
	DiPPeR [7]	1.2523 ± 0.0055	295.75 ± 159.31 px	0 ± 0
	NeuralPathLite (Ours)	**0.1694 ± 0.0007**	**186.3 ± 97.4 px**	**0 ± 0**
(c) office02 600 × 600	Standard A*	0.35 ± 0.22	360.3 ± 134.2 st	26,603.2 ± 16,162.4
	Naive A*	3.32 ± 2.85	360.3 ± 134.2 st	26,646.2 ± 16,184.9
	Neural A* [6]	6.50 ± 2.94	1490.21 ± 879.92 px	4,980.0 ± 2,229.3
	DiPPeR [7]	1.2502 ± 0.0027	351.67 ± 134.08 px	0 ± 0
	NeuralPathLite (Ours)	**0.1685 ± 0.0015**	**128.0 ± 65.1 px**	**0 ± 0**
(d) mall 760 × 760	Standard A*	1.58 ± 0.60	701.9 ± 139.6 st	117,964.7 ± 42,162.1
	Naive A*	25.15 ± 15.16	701.9 ± 139.6 st	118,010.8 ± 42,194.7
	Neural A* [6]	11.05 ± 4.68	1821.40 ± 934.63 px	8,426.5 ± 3,551.7
	DiPPeR [7]	1.2540 ± 0.0025	576.21 ± 198.40 px	0 ± 0
	NeuralPathLite (Ours)	**0.1694 ± 0.0007**	**262.6 ± 180.6 px**	**0 ± 0**

Continue…

Qualitative Comparison of Path Planning Algorithms. Representative navigation examples from the MRPB benchmark.

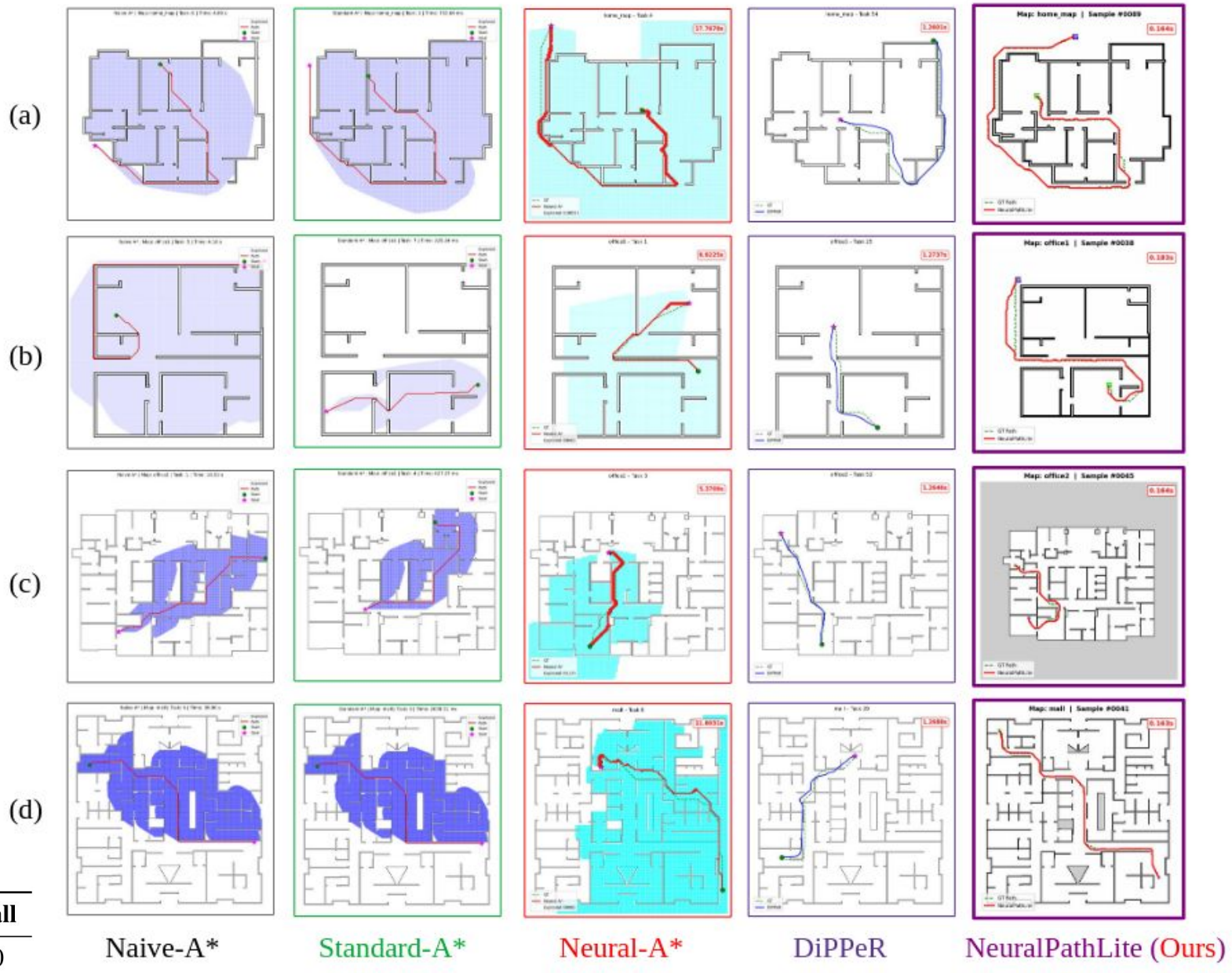


TABLE 4: SUCCESS RATE (%) COMPARISON ON MRPB DATASET ENVIRONMENTS

Method	Room02	Office01	Office02	Mall
DiPPeR [40]	58	56	21	20
NeuralPathLite (Ours)	**84**	**75**	**76**	**70**

Continue…

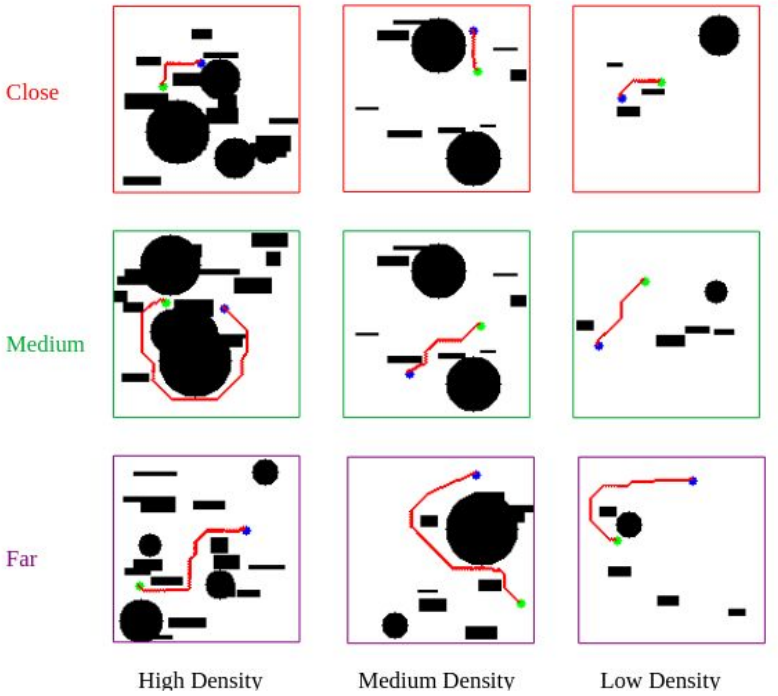
COMPARATIVE EVALUATION ON MRPB BENCHMARKS ACROSS DIFFERENT ENVIRONMENTS.

Planner	**Room02** (360 × 360)			**Office01** (280 × 280)			**Office02** (600 × 600)			**Mall** (760 × 760)		
	Time (s)	**Len.**	**Sm.**	**Time (s)**	**Len.**	**Sm.**	**Time (s)**	**Len.**	**Sm.**	**Time (s)**	**Len.**	**Sm.**
RRT*	25.05±0.03	1.14±0.56	0.05±0.02	25.05±0.01	0.98±0.45	0.03±0.02	25.04±0.04	0.27±0.14	0.02±0.01	25.03±0.03	0.70±0.24	0.05±0.03
RRT	0.49±0.58	1.22±0.60	0.17±0.09	0.13±0.10	1.02±0.47	0.07±0.03	0.41±0.35	0.30±0.15	0.07±0.04	1.64±1.21	0.74±0.27	0.19±0.12
RRT-Con.	0.33±0.26	1.23±0.60	0.16±0.07	0.17±0.16	1.03±0.47	0.10±0.04	0.38±0.27	0.30±0.15	0.08±0.04	1.06±0.76	0.74±0.26	0.16±0.09
PRM	25.14±0.07	1.15±0.57	0.21±0.06	25.12±0.03	0.98±0.46	0.12±0.06	25.09±0.05	0.27±0.14	0.16±0.05	25.09±0.03	0.70±0.25	0.22±0.07
PRM*	25.15±0.05	1.15±0.57	0.08±0.03	25.16±0.05	0.98±0.45	0.05±0.02	25.08±0.05	0.27±0.14	0.07±0.03	25.10±0.03	0.70±0.25	0.10±0.03
Inf. RRT*	25.04±0.03	1.14±0.57	0.05±0.02	25.06±0.01	0.98±0.45	0.03±0.01	25.02±0.02	0.27±0.14	0.03±0.01	25.04±0.04	0.69±0.24	0.05±0.03
A*	0.43±0.27	**0.77±0.30**	0.19±0.19	0.19±0.16	**0.70±0.31**	0.24±0.14	0.35±0.11	**0.20±0.10**	0.37±0.21	1.54±0.57	**0.52±0.18**	0.40±0.24
NeuralPathLite (Ours)	**0.16±0.02**	0.89±0.46	**0.14±0.04**	**0.16±0.01**	0.71±0.30	**0.15±0.08**	**0.16±0.01**	0.18±0.09	**0.19±0.13**	**0.17±0.00**	0.65±0.31	**0.12±0.03**

Ablation & Robustness

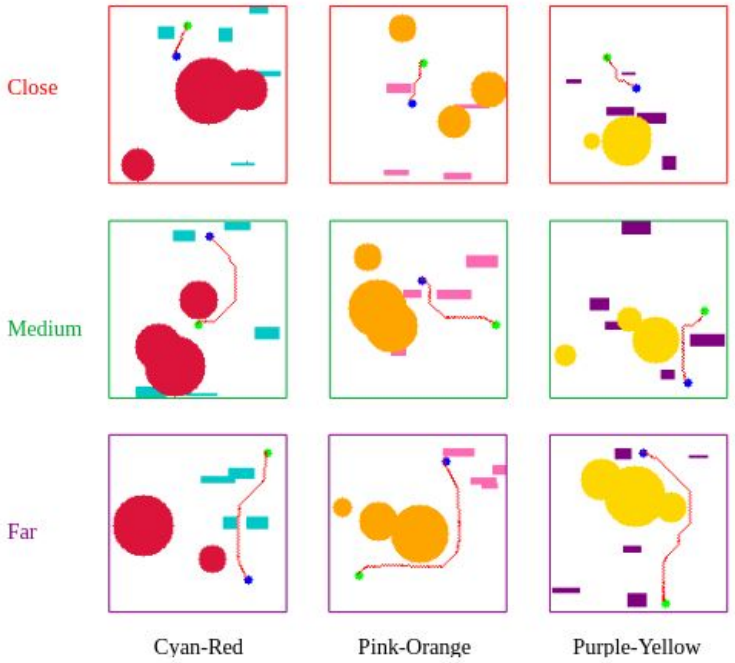
TABLE 4: QUANTITATIVE EVALUATION OF OOD GENERALIZATION ACROSS VARYING MAZE TOPOLOGIES , OBSTACLE SCALES, AND COLOR DISTRIBUTIONS .

OOD Factor	Maze Type (Fig. ref)	Diff.	Close (%)	Med. (%)	Far (%)	Avg. (%)
Shape	Mixed BW (Fig. 5(c))	Low	100	96	86	94.00
		Med	100	100	96	98.67
		High	97	92	82	90.33
Color	Pink+Orange (Fig. 5(d))	–	97	93	75	88.33
	Purple+Yellow	–	94	86	71	83.67
Scale	Tiny Squares (Fig. 5(e))	Low	97	88	80	88.33
		Med	97	95	83	91.67
		High	93	91	84	89.33
Layout	Room Walls (Fig. 5(f))	Few	100	100	96	98.67
		Mod.	100	99	94	97.67
		Dense	99	92	82	91.00
Overall Average			97.64	93.82	84.45	91.97

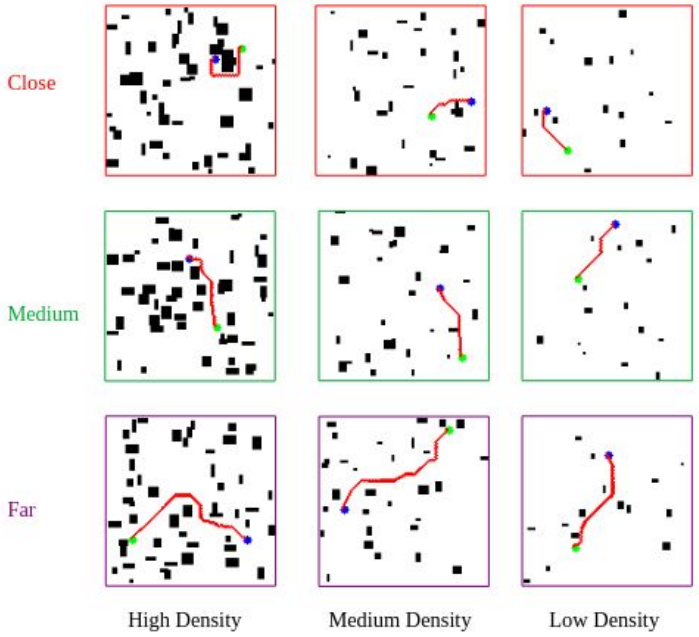


Robustness of NeuralPathLite across Varying Path Lengths and Clutter Densities.

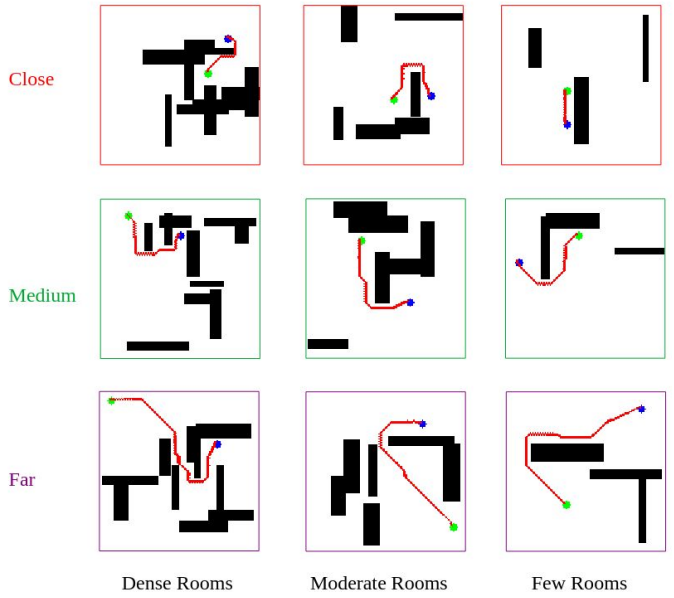
Continue…



Color and Distance Robustness of NeuralPathLite.



Generalization Performance of NeuralPathLite across Unseen Environments.



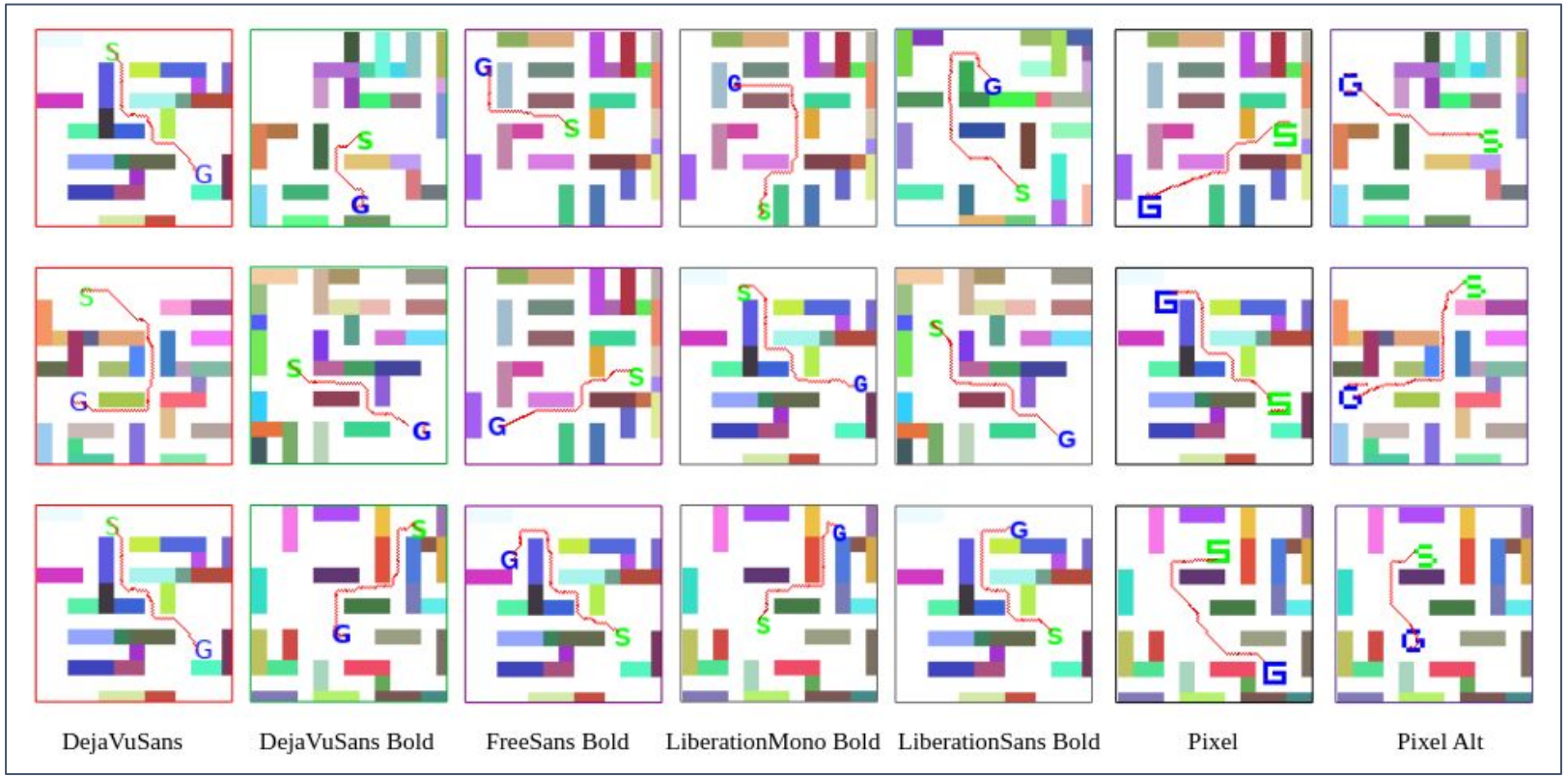
Qualitative Generalization Analysis of NeuralPathLite

Continue…

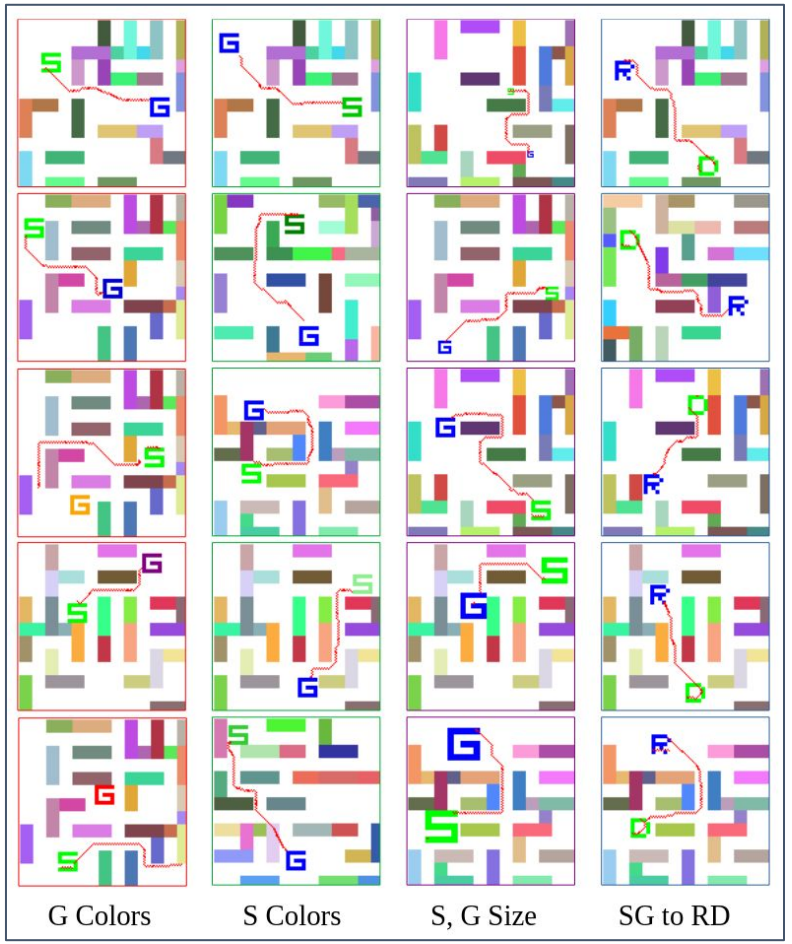
ROBUSTNESS ANALYSIS OF SEMANTIC MARKERS UNDER VISUAL VARIATIONS . THE BASELINE CUSTOM “PIXEL” MARKERS OCCUPY AN EFFECTIVE SPATIAL FOOTPRINT OF 15 × 15 PIXELS ON THE ENVIRONMENT MAP . STANDARD TRUETYPE FONTS ARE RENDERED AT 16 PT TO CLOSELY MATCH THIS SPATIAL FOOTPRINT.

Test Category	Varied Parameter	Samples	Success (%)
Font Style Variation	*(Constant: S=Green, G=Blue, Size ≈ 15 × 15 px)*		
	Pixel (Baseline, 15 × 15 px)	200	84
	Pixel (Rounded Corners, 15 × 15 px)	200	77
	DejaVuSans-Bold (16pt)	200	90
	LiberationMono-Bold (16pt)	200	78
	FreeSansBold (16pt)	200	91
	DejaVuSans (16pt)	200	86
	LiberationSans-Bold (16pt)	200	90
Marker Size Variation	*(Constant: Font=Pixel, S=Green, G=Blue)*		
	Size (10 × 10 px)	200	77
	Size (15 × 15 px) - Baseline	200	85
	Size (20 × 20 px)	196	70
	Size (25 × 25 px)	152	62
Marker Color Variation	*(Constant: Font=Pixel, Size=15 × 15 px)*		
Start (S) Color	Green (0, 255, 0) - Baseline	200	80
	Dark Green (0, 200, 0)	200	89
	Lime (50, 205, 50)	200	83
	Forest (0, 128, 0)	200	82
	Light Green (144, 238, 144)	200	24
Goal (G) Color	Blue (0, 0, 255) - Baseline	200	81
	Red (255, 0, 0)	200	02
	Orange (255, 165, 0)	200	02
	Dark Blue (0, 0, 200)	200	85
	Purple (128, 0, 128)	200	03
Semantic Token Variation	*(Constant: Font=Pixel, Size=15 × 15 px, Colors=Green/Blue)*		
	$S \rightarrow D, G \rightarrow R$	200	80

Continue…



Qualitative Ablation Study on Font Style Robustness.



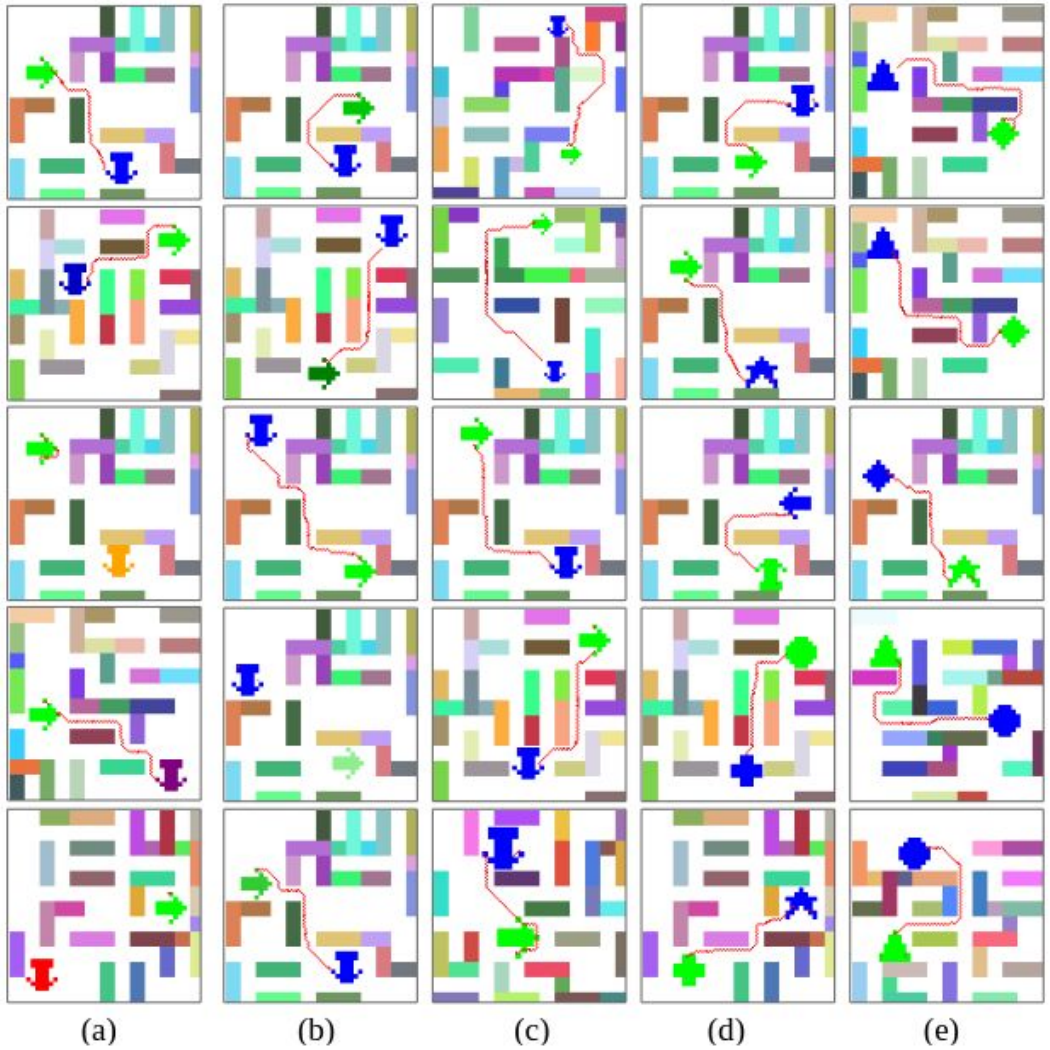
Qualitative Ablation on Visual Marker Robustness.

Continue…

ABLATION STUDY ON VISUAL MARKER ROBUSTNESS. FOR THE SIZE AND COLOR ABLATIONS, THE CONTROL CONFIGURATION EVALUATES OUT - OF -DISTRIBUTION GENERALIZATION USING UNSEEN ARROW SYMBOLS AT SCALE 3 (21 × 21 PX).

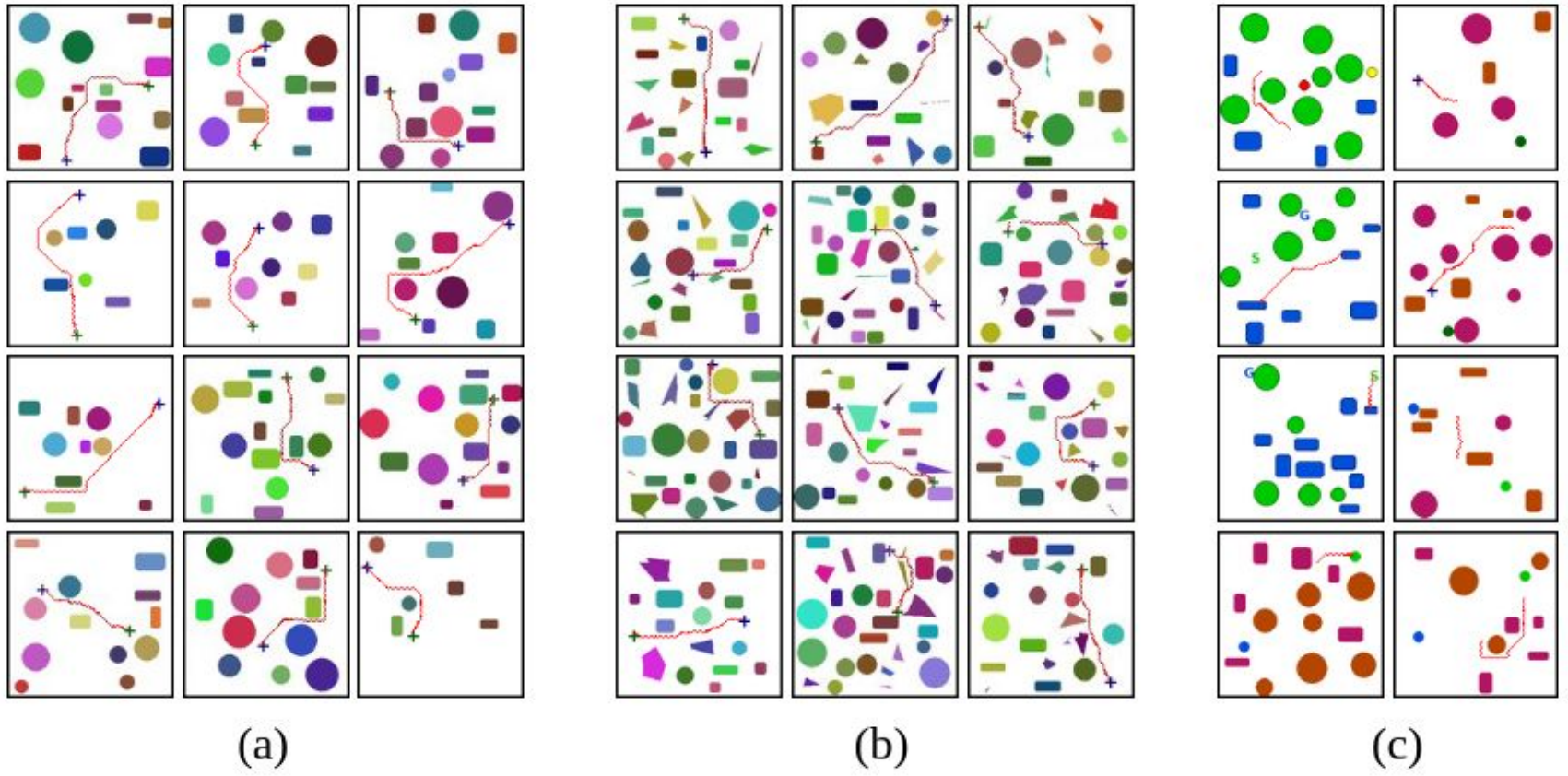
Test Category	Varied Parameter	Samples	Success (%)
Control Parameters: *Morphology: →→↓ (Unseen) | Scale: 3 (21 × 21 px) | Colors: Green/Blue*			
Symbol Style Variation	*(Constant: S=Green, G=Blue, Size=Scale 3 (21 × 21 px))*		
	→→↓	195	81.53
	↑→←	199	85.00
	⋆ → ♦	200	49.00
	▲ → ●	195	50.00
	+ → ⋆	197	44.16
	♦ → ▲	199	72.86
	→→ ⋆	198	64.14
	● → +	192	89.64
Marker Size Variation	*(Constant: Symbol=→→↓, S=Green, G=Blue)*		
	Scale 2 (14 × 14 px)	191	83.25
	Scale 3 (21 × 21 px) - Control	198	73.74
	Scale 4 (28 × 28 px)	65	74.53
Marker Color Variation	*(Constant: Symbol=→→↓, Size=Scale 3 (21 × 21 px))*		
Start (S) Color	Green (0, 255, 0) - Control	196	80.10
	Dark Green (0, 200, 0)	180	77.16
	Lime (50, 205, 50)	189	82.65
	Forest (0, 128, 0)	186	78.40
	Light Green (144, 238, 144)	184	09.42
Goal (G) Color	Blue (0, 0, 255) - Control	196	79.60
	Red (255, 0, 0)	188	00.00
	Orange (255, 165, 0)	186	00.00
	Dark Blue (0, 0, 200)	193	74.10
	Purple (128, 0, 128)	194	64.43

Continue…



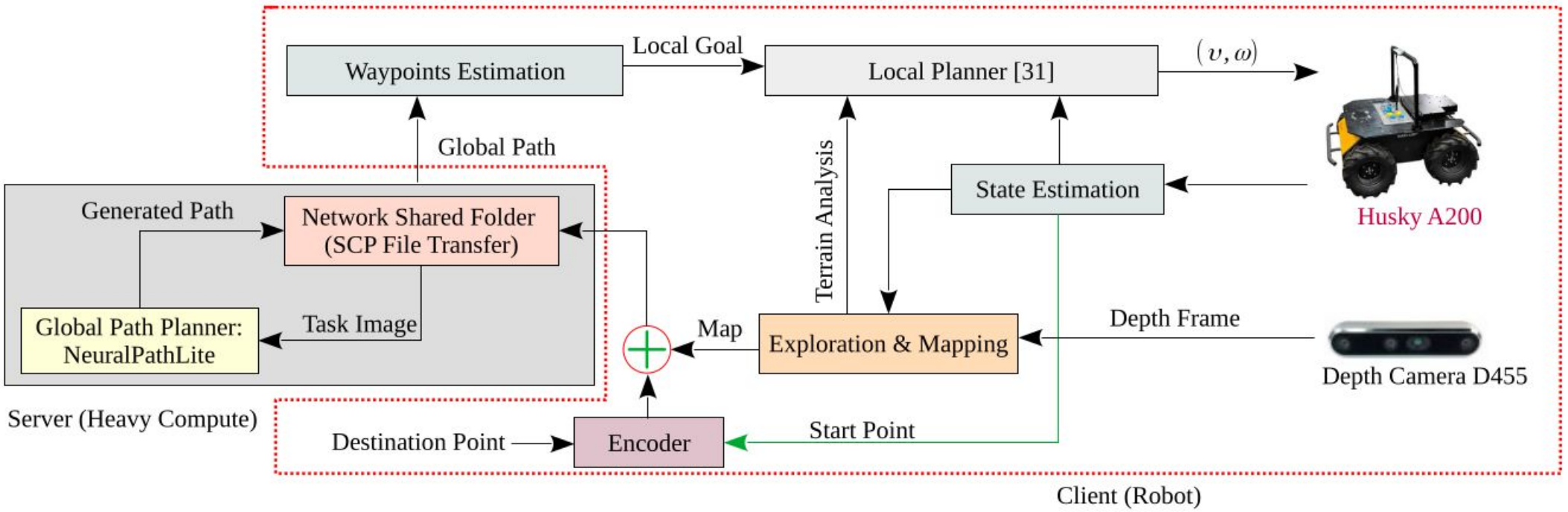
Qualitative Ablation Study on Visual Marker Robustness: Navigation results demonstrate the generalization capability of NeuralPathLite across diverse out-of-distribution (OOD) marker configurations.

Continue…



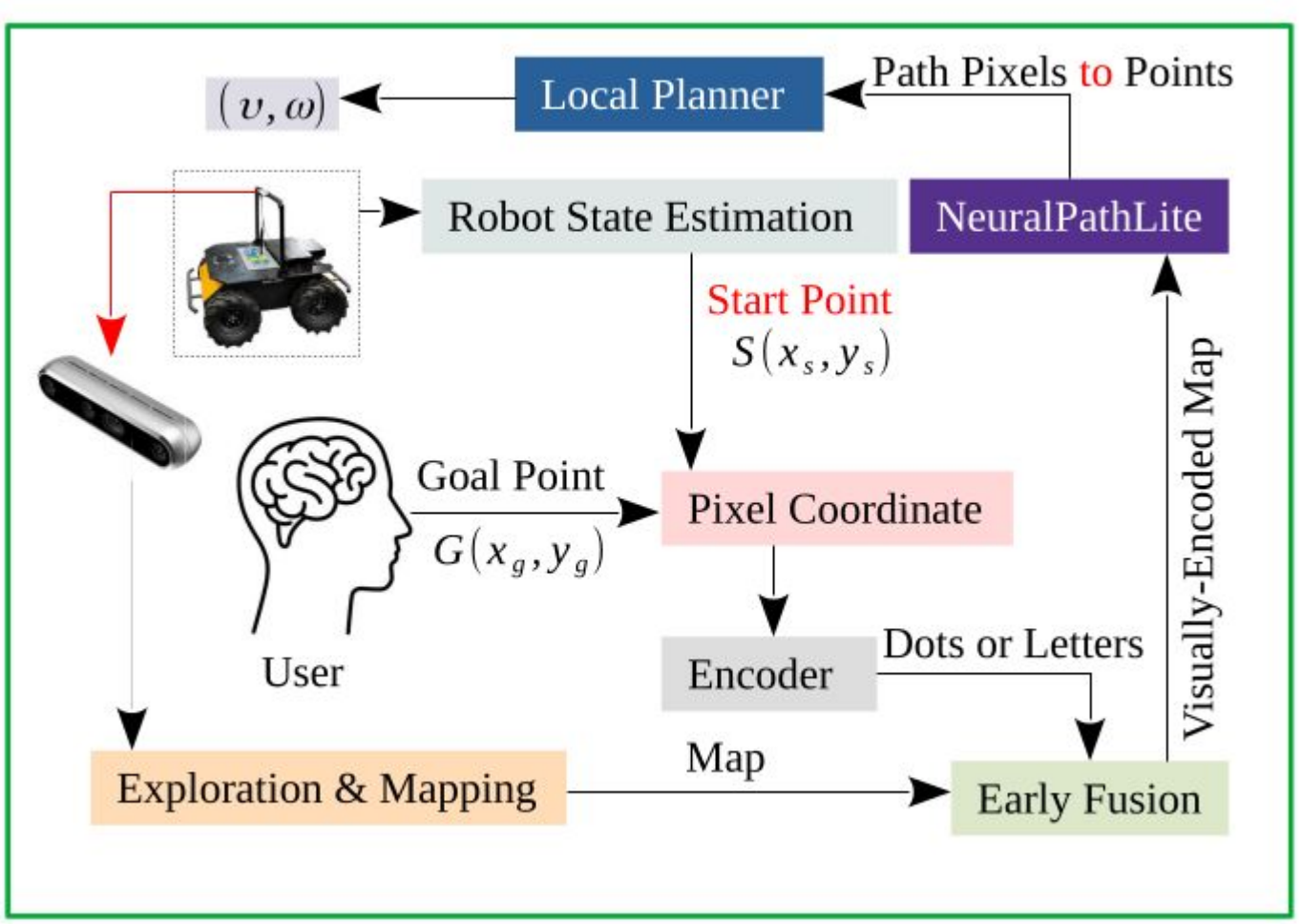
Performance boundary analysis of NeuralPathLite on diverse navigation scenarios. (a) Successful path generation on OOD environments, achieving a success rate of 92%. Obstacles are represented with multiple colors, and start/goal locations are indicated by “+” markers (green: start, blue: goal). (b) Robust evaluation under varying obstacle colors and shapes, simulating real-world variability, with a success rate of 81%. (c) Representative failure cases where the model is unable to connect the start and goal points.

Navigation Pipeline with NeuralPathLite



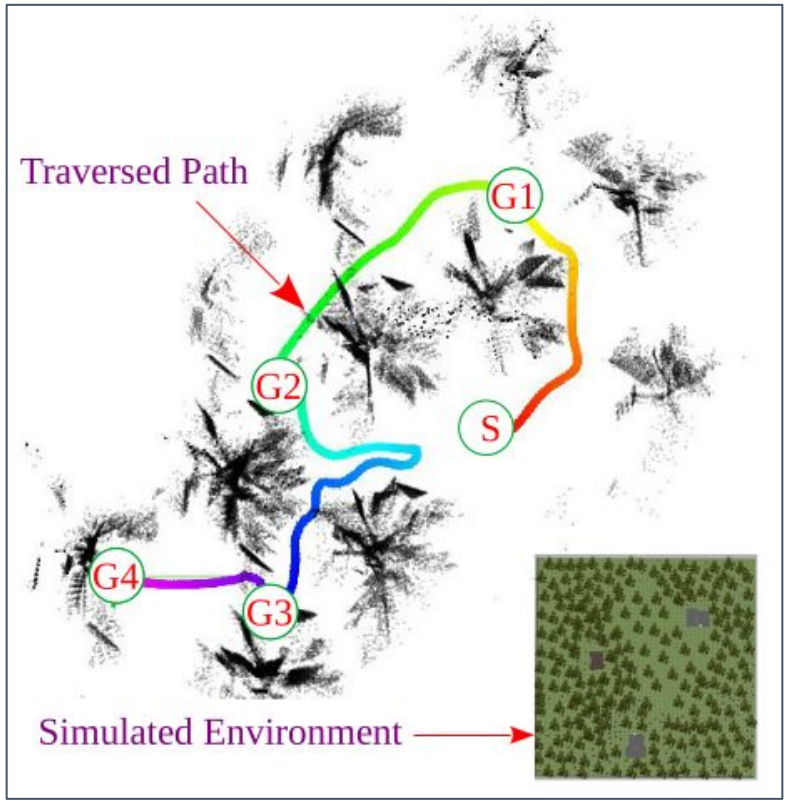
Schematic overview of the autonomous navigation framework.

Continue…

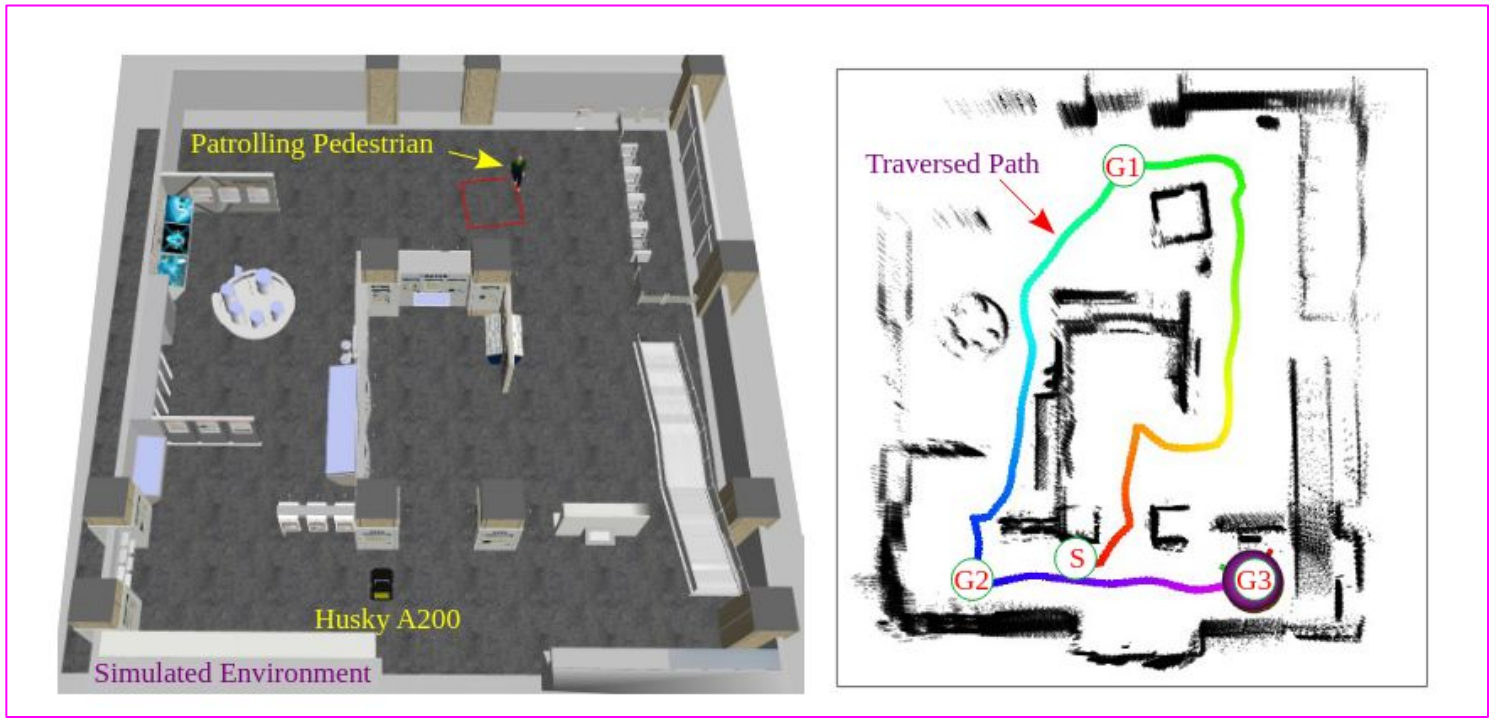


Proposed navigation pipeline from start–goal encoding to trajectory generation and control.

Navigation Results (Simulation)



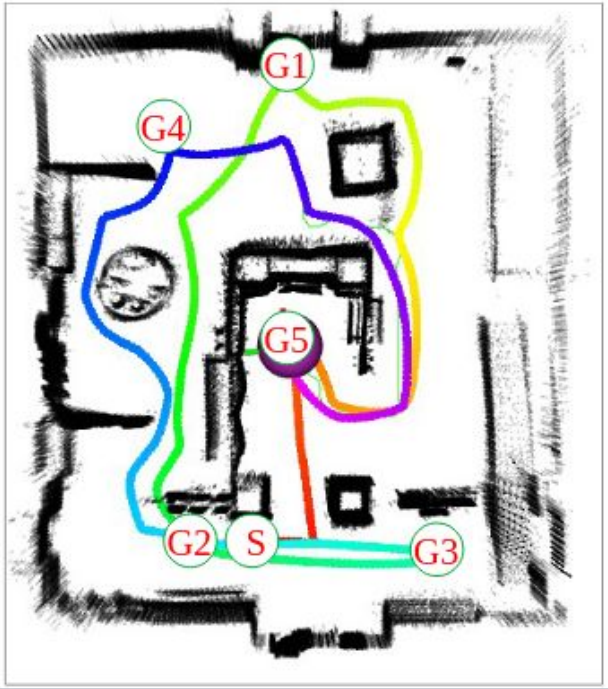
Simulated forest navigation showing trajectory S → G1 → G2 → G3→G4; the bottom-right inset shows the environment.

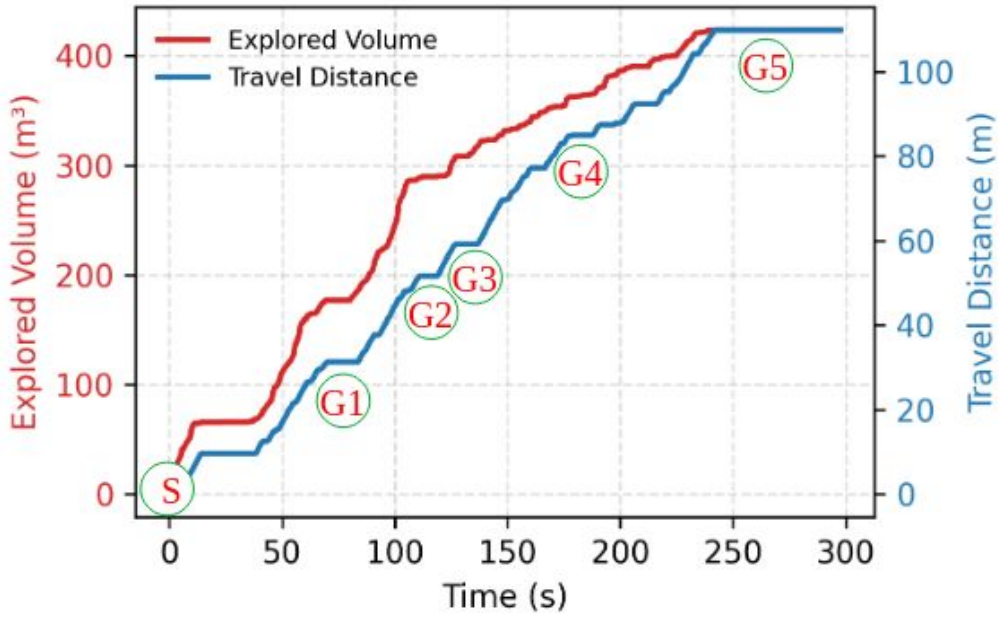


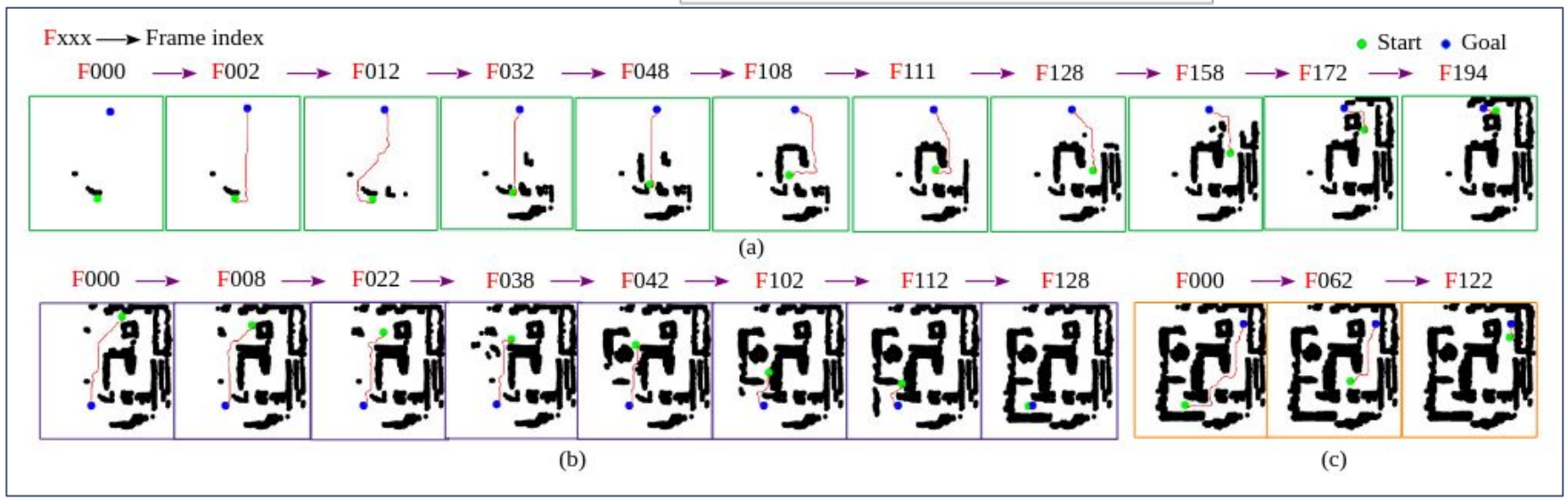
Performance verification with dynamic and spatial constraints. (Left) The simulation environment with the Husky A200 navigating alongside a patrolling pedestrian and a red-marked prohibited region.

Continue…

Trajectory visualization and quantitative performance analysis.







Validation of the proposed framework in a dense simulated environment across three independent test cases.

Continue…

QUANTITATIVE RESULTS OF NEURALPATHLITE . WE EVALUATE THE PLANNER IN THE DYNAMIC SIMULATED ENVIRONMENT.

Scenario #	**Task Difficulty** $\|\vec{S} - \vec{G}\|_2$ (m)	**Success Rate** (Success / Total)	**Primary Failure Mode**	**Path Quality Mean Length (m)**
1	16.88	33.3% (1/3)	Infeasible Path	18.79
2	17.19	100.0% (1/1)	None	22.52
3	10.53	100.0% (1/1)	None	13.31
4	10.56	50.0% (1/2)	Collision	5.56
5	5.41	14.3% (1/7)	Infeasible Path	31.72
6	5.09	100.0% (1/1)	None	18.86
7	14.64	33.3% (1/3)	Infeasible Path	18.46
8	16.11	100.0% (1/1)	None	26.84
9	15.43	100.0% (1/1)	None	21.32
10	15.00	50.0% (1/2)	Infeasible Path	18.37
11	21.25	50.0% (1/2)	Collision	19.75
12	16.56	100.0% (1/1)	None	16.56
13	11.74	50.0% (1/2)	Infeasible Path	21.66

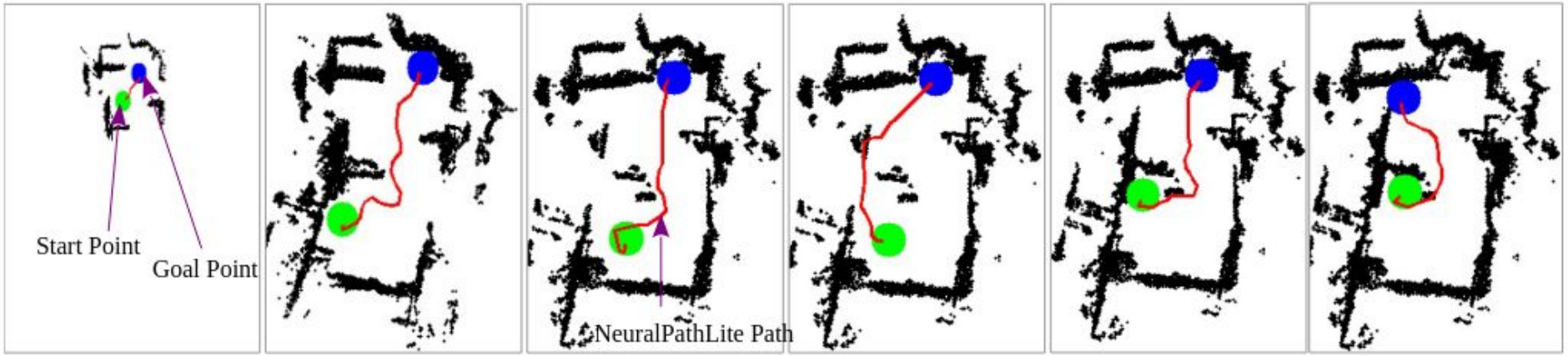
Navigation Results (Real-World)



(a) Planning Performance and Limitations. The figure illustrates the planner's ability to solve long-horizon navigation tasks between distant start–goal pairs (S1, G1) and (S2, G2). The region marked with a cross (×) highlights a failure case where the model struggles to converge to a valid path without multiple re-inference attempts, indicating a limitation to be addressed in future work. and (b) Real-world experimental setup.

Continue…

Online exploration and mapping of an unknown environment during real-world deployment.



Scenario #	Task Difficulty $\|\vec{S}-\vec{G}\|_2$ (m)	Success Rate (Success / Total)	Primary Failure Mode	Path Quality Mean Length (m)
1	04.28	33.3% (1/3)	Infeasible Path	08.25
2	09.32	100.0% (1/1)	None	12.25
3	02.43	50.0% (1/2)	None	3.32
4	08.60	25.0% (1/4)	Collision	12.42
5	11.62	20.0% (1/5)	Infeasible Path	13.14
6	10.76	100.0% (1/1)	None	15.68
7	06.82	50.0% (1/2)	Infeasible Path	10.26
8	09.21	50.0% (1/2)	None	11.84
9	05.92	100.0% (1/1)	None	10.52
10	03.20	50.0% (1/2)	Infeasible Path	08.40

Failure Analysis and Quantitative Performance in Real-World Deployment.

Source Code and Demonstration Videos

Source Code Link



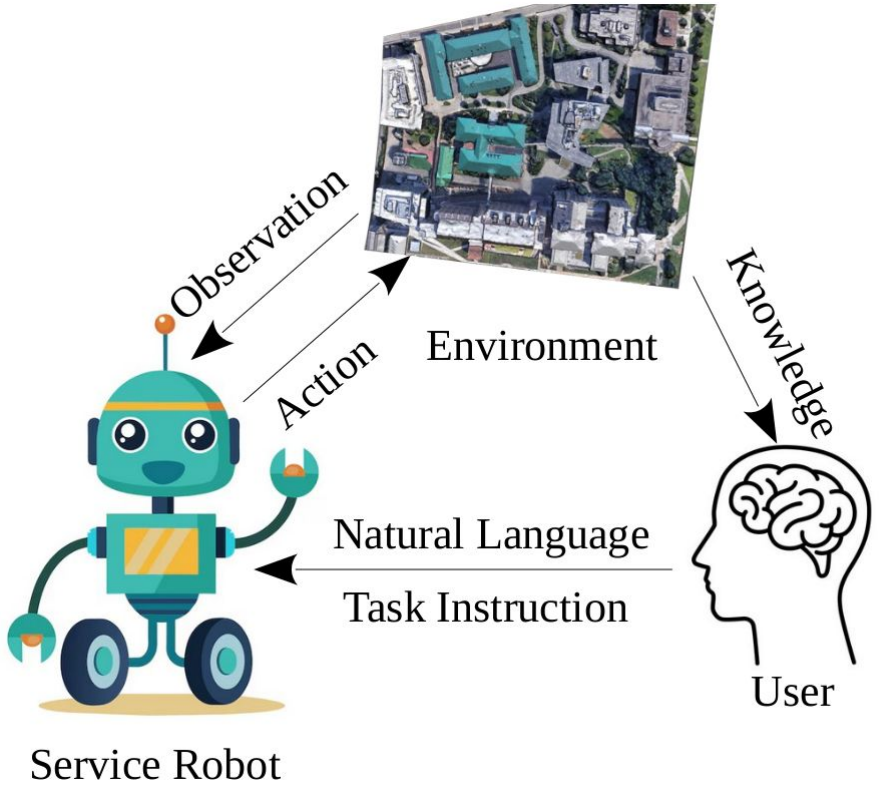
- P. Kumar, P. Gupta and T. Sandhan, “NeuralPathLite: Fast and Robust Diffusion-Based Path Planning for Autonomous Navigation,” in IEEE Transactions on Automation Science and Engineering, vol. 23, pp. 12471-12485, 2026, doi: 10.1109/TASE.2026.3710089. (Published)

Mapless Interactive Vision-Language Navigation with Drift Resets

(Drift-Free VLN): Can multi-task visual perception networks combined with Large Language Models (LLMs) enable robots to interpret human language commands, execute vision-grounded action plans, and eliminate long-horizon odometry drift through target-anchored visual landmark resets?

Mapless Interactive Vision-Language Navigation with Drift Resets

1. Eliminate Reliance on Global Maps.
2. Natural Language-Driven Navigation (LLM Integration)
3. Egocentric Action Planning
4. Minimum Accumulation Drift Strategy
5. Mapless Safe Navigation in Open Worlds
6. Human-in-loop Feedback Navigation.



Overview of the proposed VLN scenario.

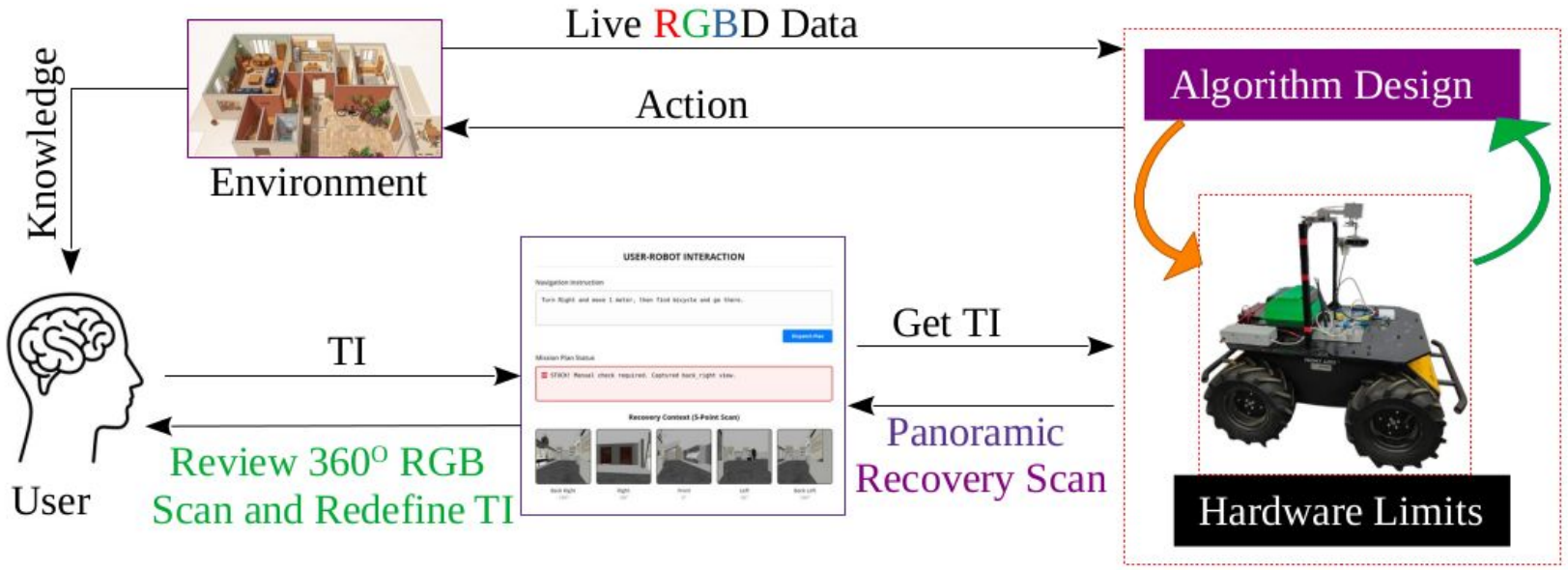
Literature Survey & SOTA Comparison

TABLE I

STATE-OF-THE-ART (SOTA) COMPARISON OF MODERN NAVIGATION PARADIGMS AGAINST THE PROPOSED MULTI-TASK VISUAL PERCEPTION NETWORK WITH LLM CONDITIONING.

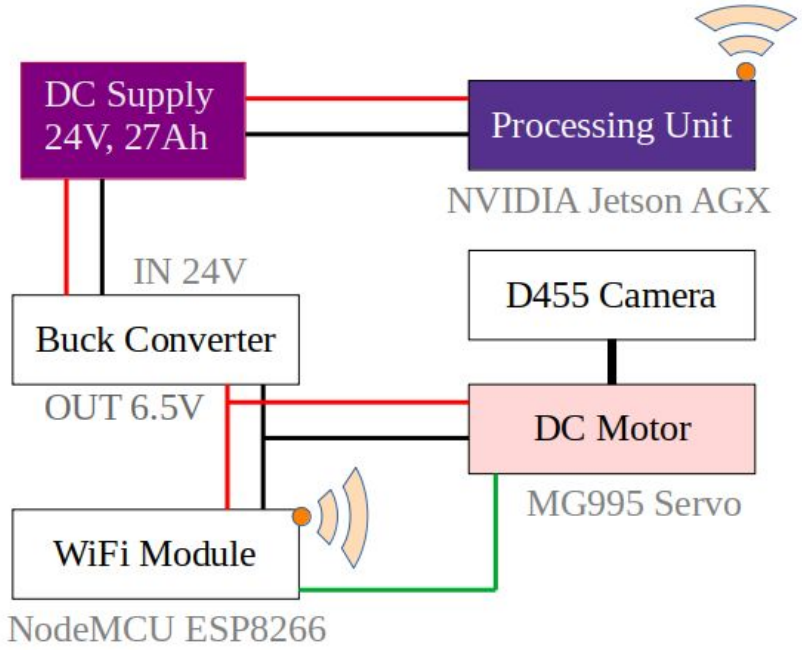
Method / Paradigm	**Global Map Required**	**Drift Sensitivity**	**Long-Term Nav.**	**Unknown Env.**	**Semantic Understand.**	**Context Awareness**	**User Interaction**	**Replanning Strategy**
Classical A^* / RRT*	Yes	High	Low	No	No	No	No	Limited
DiPPeR (ICRA 2024)	Yes	High	Low	No	No	No	No	Limited
ViT-A^* (ICML 2021)	Yes	High	Low	Partial	Partial	No	No	Limited
Vision-Aided A^*	Yes	High	Low	Partial	No	Partial	No	Online
SemExp (Semantic Exploration)	No	Partial	Partial	Yes	Yes	Partial	No	Moderate
CARe (Context-Aware Replanning)	No	Partial	Partial	Yes	Yes	Yes	No	Advanced
SEMNAV (Semantic Segmentation)	No	Low	Yes	Yes	Yes	Yes	No	Advanced
Multi-Agent Semantic Nav	No	Low	Yes	Yes	Yes	Yes	No	Advanced
Proposed Framework (Ours)	**No**	**Low (Reset)**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes (LLM)**	**Online (343 Primitives)**

Continue…

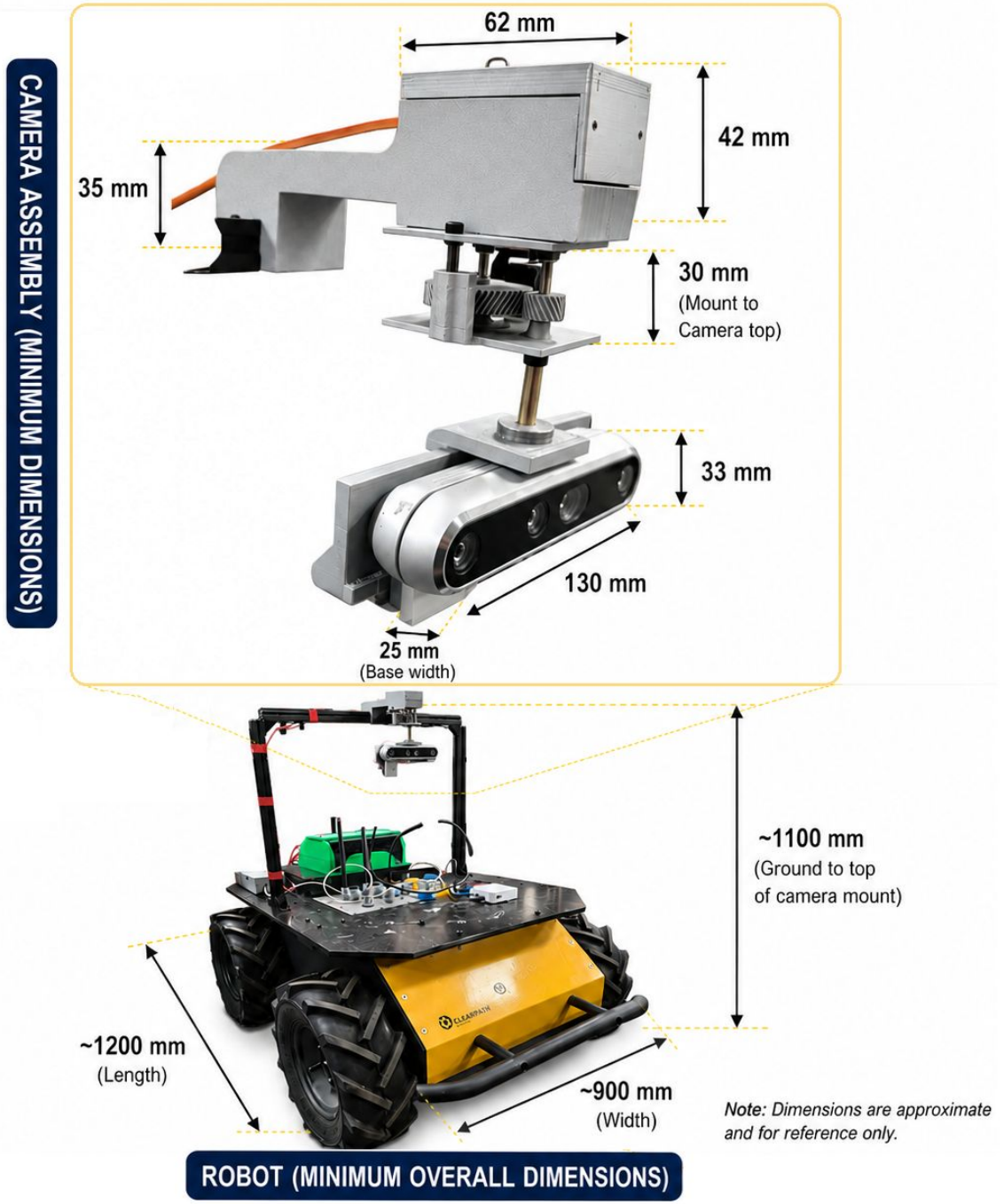


Human-in-the-Loop Feedback Pipeline for Vision-Language Navigation.

Continue…



Pan Camera Communication Architecture.



The Latency Challenge

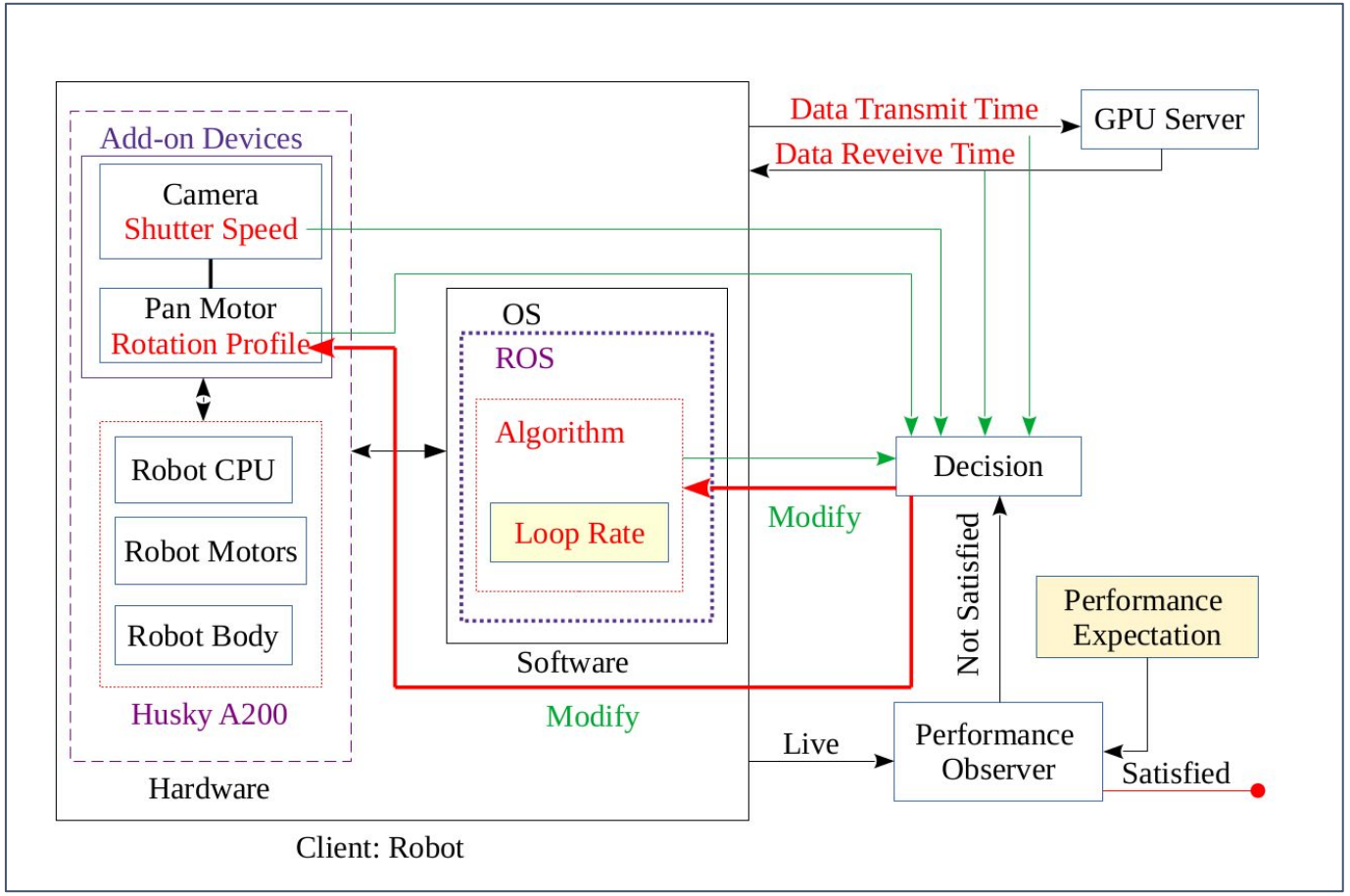
Why Perfect Path Planners Still Fail in Dynamic Environments.

- High algorithmic accuracy does **not** guarantee real-world safety if temporal delays are unmanaged.
- Collisions occur due to **Spatial-Temporal Mismatch** —executing valid commands on *stale environment states*.

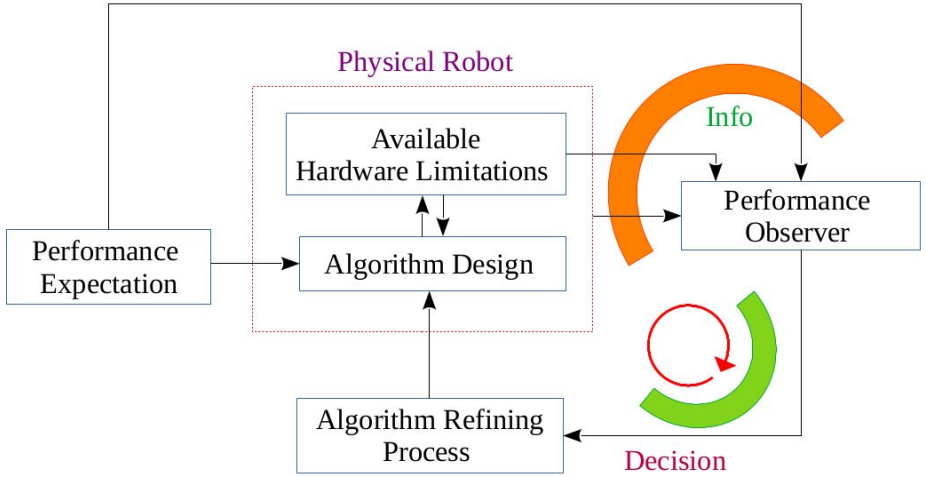
1. **End-to-End (Sensor-to-Actuator) Latency**
2. **Temporal Asynchrony (Mismatched clocks and update frequencies across sensors, ROS nodes, and microcontrollers).**
3. **Perception Overhead (Heavy deep learning inference time (e.g., semantic segmentation/depth processing).**
4. **Actuation Delay (Intrinsic hardware latency in motor controllers generating mechanical torque).**

- **Hardware Acquisition** (Δt_{sens})**:** Frame exposure, capture, and data transfer over USB/Ethernet.
- **Software Processing** (Δt_{proc})**:** ROS message deserialization, deep model inference, and costmap updating.
- **Algorithmic Rate Mismatch** (Δt_{plan})**:** Discrepancy between high-frequency sensors and lower-frequency motion planners.
- **Control & Actuation** (Δt_{act})**:** Command transmission to microcontrollers and physical motor torque response.

Methodology



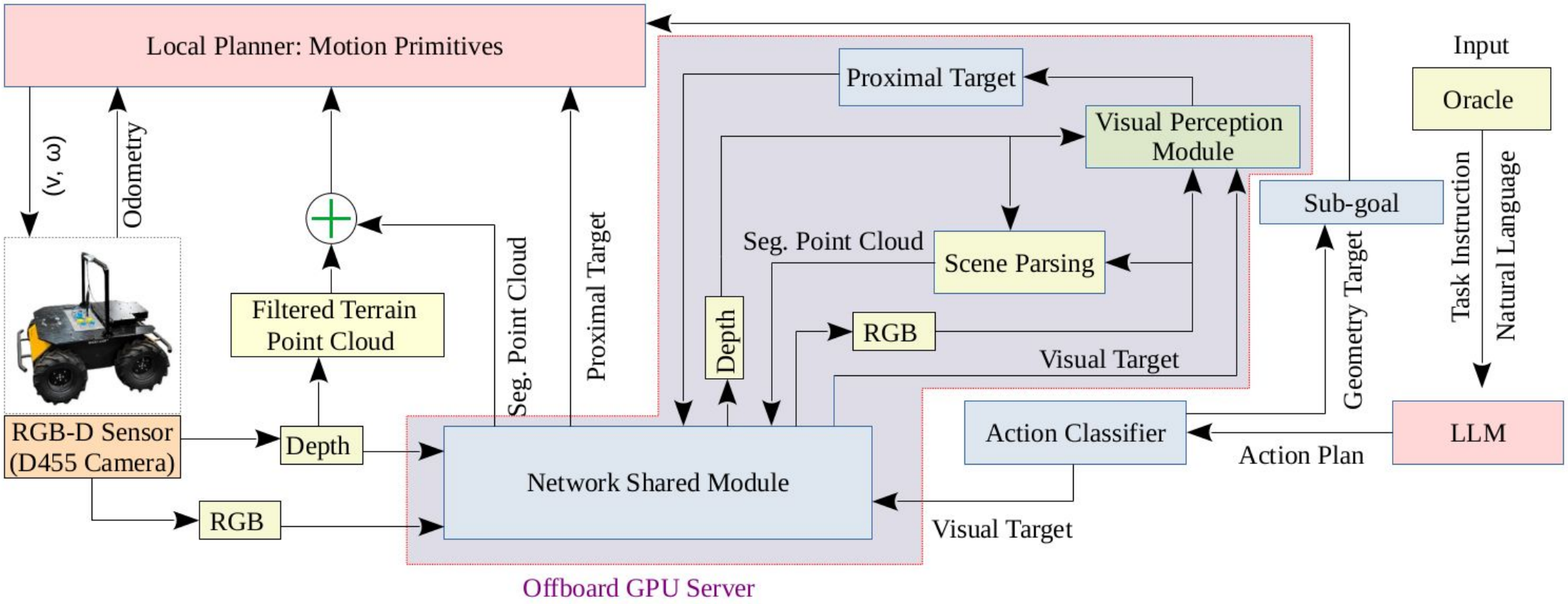
Latency Diagnosis and Performance Optimization Framework.



Closed-Loop Algorithm Refinement Framework.

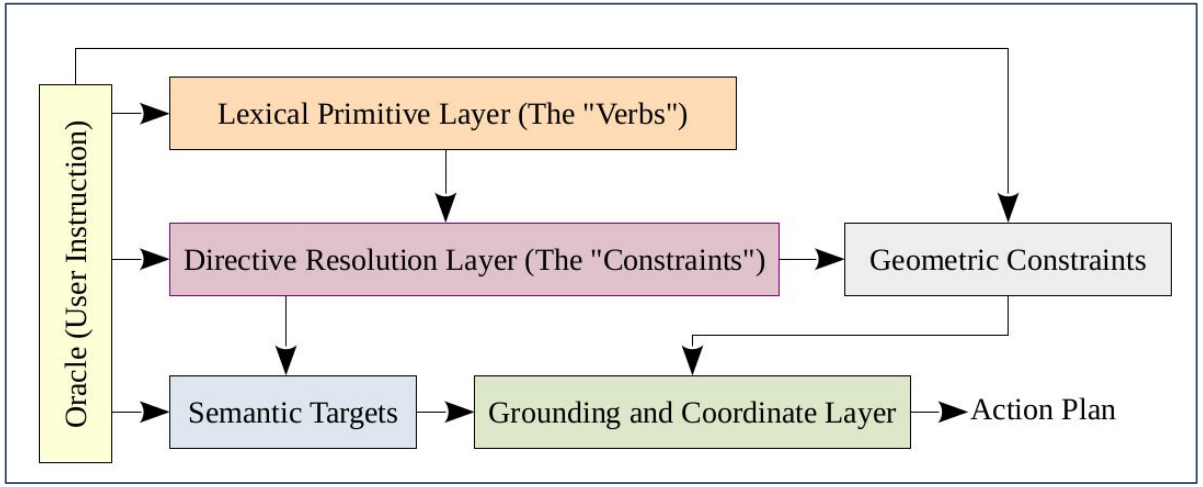
- **Timestamp Alignment:** Synchronizing heterogeneous sensor streams at the hardware/software level.
- **Deterministic Loops:** Enforcing constant-frequency control loops across system nodes.
- **Latency Compensation:** Predicting obstacle trajectories forward by Δt_{total} prior to actuation.

Continue…

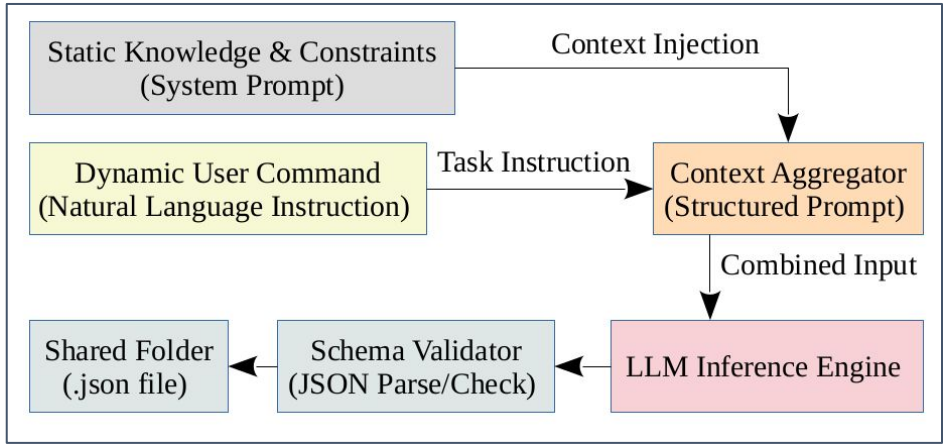


Overview of the proposed vision-language navigation framework.

Continue…

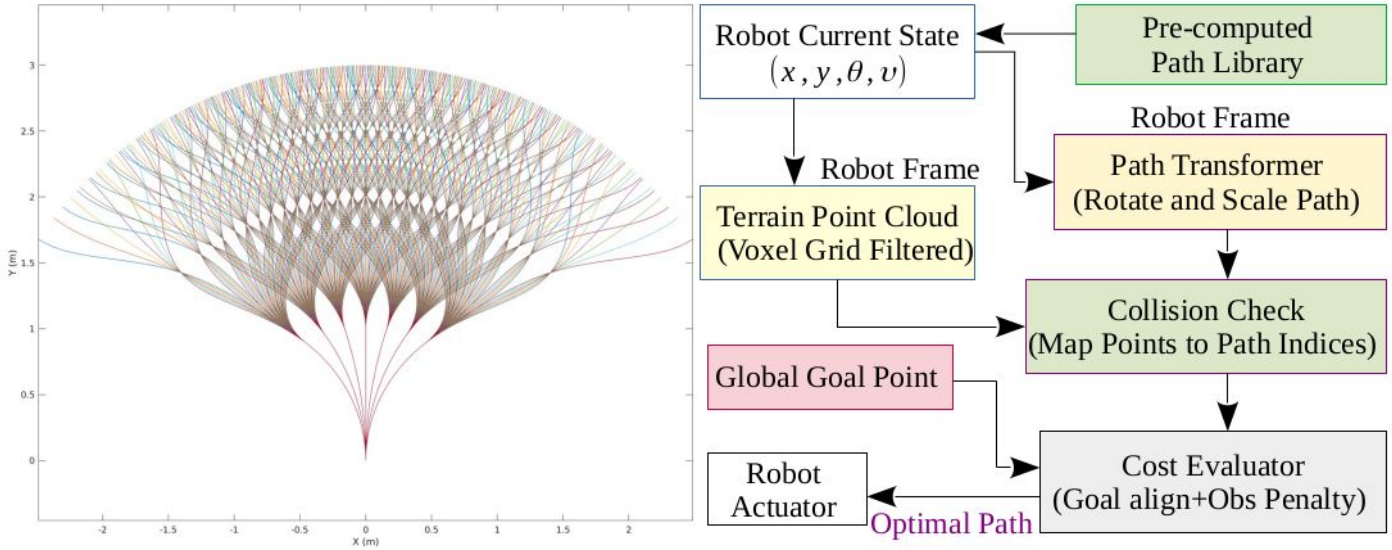


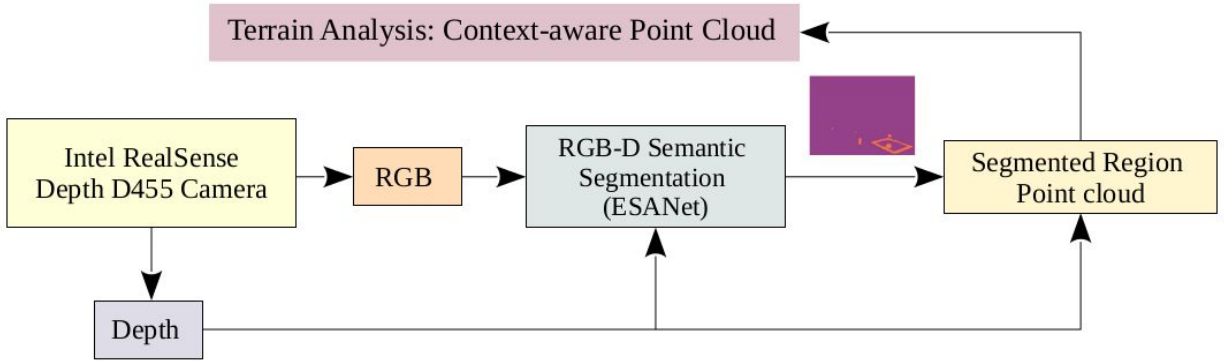
The multi-layered architecture for converting user task instructions into robot action plan. The system resolves lexical ambiguity and distinguishes between geometric goal and semantic target constraints before grounding them into a unified robot system for execution.



Structured prompt is created via injecting static system constraints into the context window alongside the user task instruction. LLM translates the natural language request into a machine-readable format.

Continue…

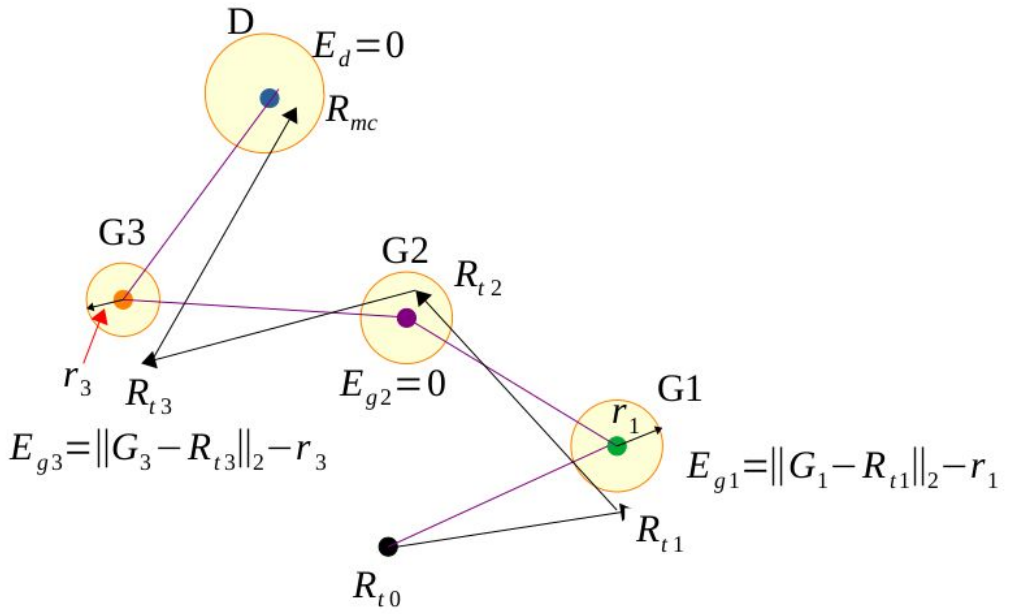




Visual perception to point cloud generation pipeline.

- Set of candidate motion trajectories produced using the motion primitive framework.
- Data flow of the local planner.

Continue…



1. Optimal Path Selection (τ^*)

$$\tau^* = \arg\min_{\tau\in\Omega_{\text{feasible}}} \left[w_g J_{\text{goal}}(\tau)+w_o J_{\text{obs}}(\tau)+w_s J_{\text{semantic}}(\tau)\right]$$

2. Physical & Semantic Barrier Penalties

$$J_{\text{obs}}(\tau) = \exp\left(-\alpha \min_{t\in\tau} d_{\text{phy}}(\mathbf{p}(t))\right), \quad J_{\text{semantic}}(\tau) = \exp\left(-\beta \min_{t\in\tau} d_{\text{sem}}(\mathbf{p}(t))\right)$$

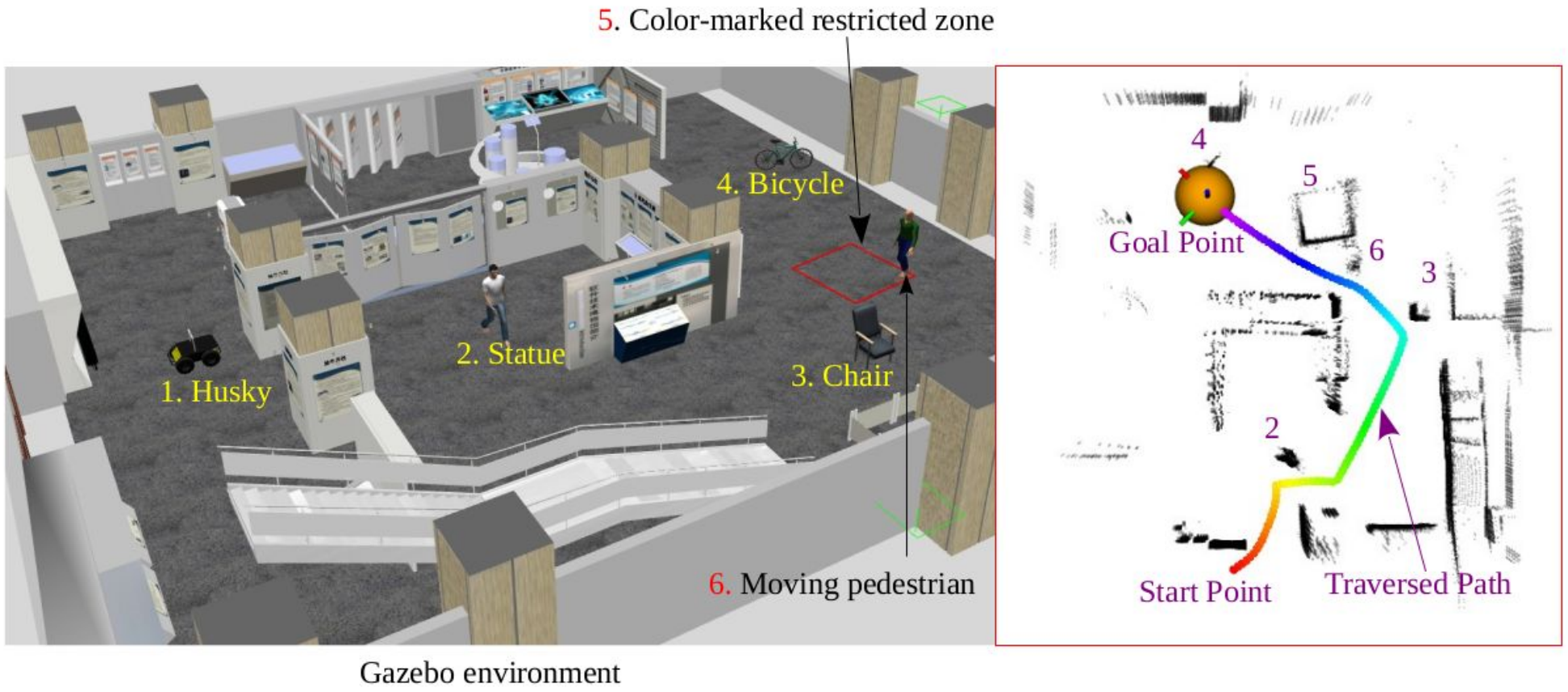
3. Subgoal Odometry Drift Formulation (E_{g_k})

$$E_{g_k} = \|G_k - R_{t_k}\|_2 - r_k, \qquad E_d = \|D - R_{t_{\text{final}}}\|_2 - r_d \approx 0$$

Target Reset Strategy: $E_{\text{net}} = \max_k E_{g_k} \ll E_{\text{unbounded_SLAM}}$

Illustration of odometry drift estimation during navigation. The robot moves through intermediate subgoals G 1 –G 3 toward the final destination D, and the drift is computed based on the deviation between the robot pose and the corresponding target.

Results



Gazebo environment

Experimental validation in the Gazebo simulation environment.

Continue…

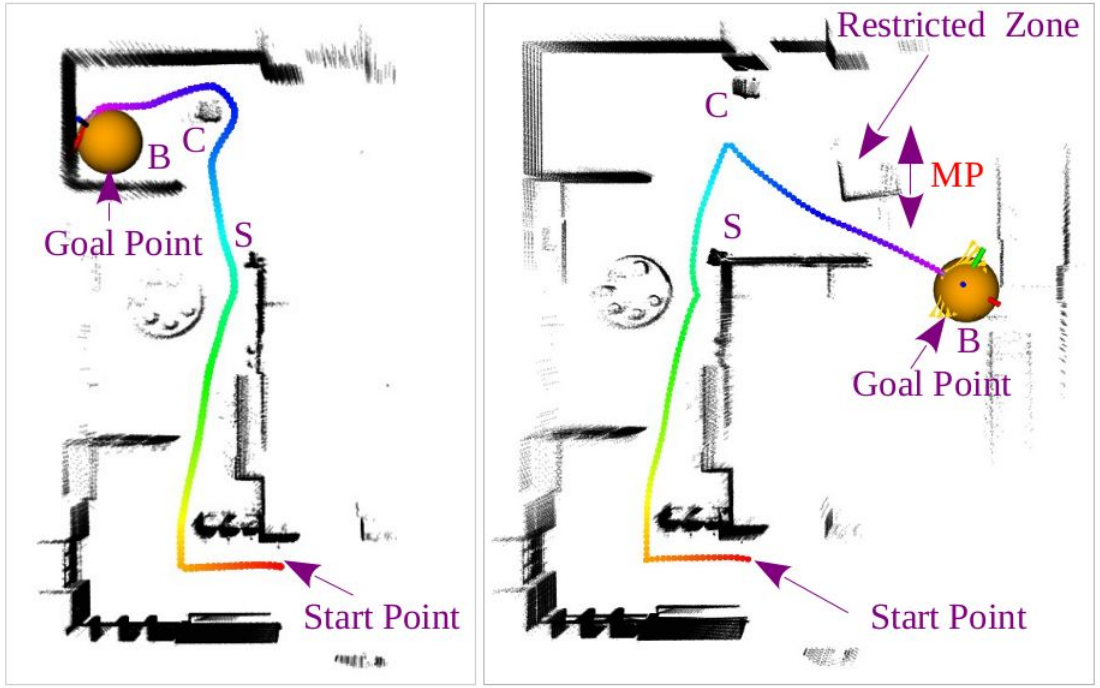
TABLE II
QUALITATIVE EVALUATION OF LLM-GUIDED INSTRUCTION PARSING.

Case 1: Complex Navigation
User Input: *"Buddy go to the statue (human), take a right turn and move a little ahead, then find the visitor chair to reach close to it, and on the right you may see a bicycle."*

LLM Output:
`[{"action": "FIND", "target": "statue (human)", "id": 0},`
`{"action": "MOVE", "desc": "turn right...", "f_dist": 2.5, "yaw": -1.57},`
`{"action": "FIND", "target": "visitor chair", "id": 56},`
`{"action": "FIND", "target": "bicycle", "id": 1}]`
Status: Success

Case 2: Metric Constraints
User Input: *"Buddy take left turn, move 4 m ahead, then find the standing person to reach close to it, In your right you may see the visitor chair follow it. Finally go at spot where bicycle is standing."*

LLM Output:
`[{"action": "MOVE", "desc": "left turn 4m", "f_dist": 4.0, "yaw": 1.57},`
`{"action": "FIND", "target": "standing person", "id": 0},`
`{"action": "FIND", "target": "visitor chair", "id": 56},`
`{"action": "FIND", "target": "bicycle", "id": 1}]`
Status: Success



Explored environment and traversed paths for two distinct navigation tasks.

Continue…

TABLE I
COMPARISON OF MODERN NAVIGATION PARADIGMS (EXTENDED). ✓ SUPPORTED, ✗ NOT SUPPORTED, △ PARTIAL.

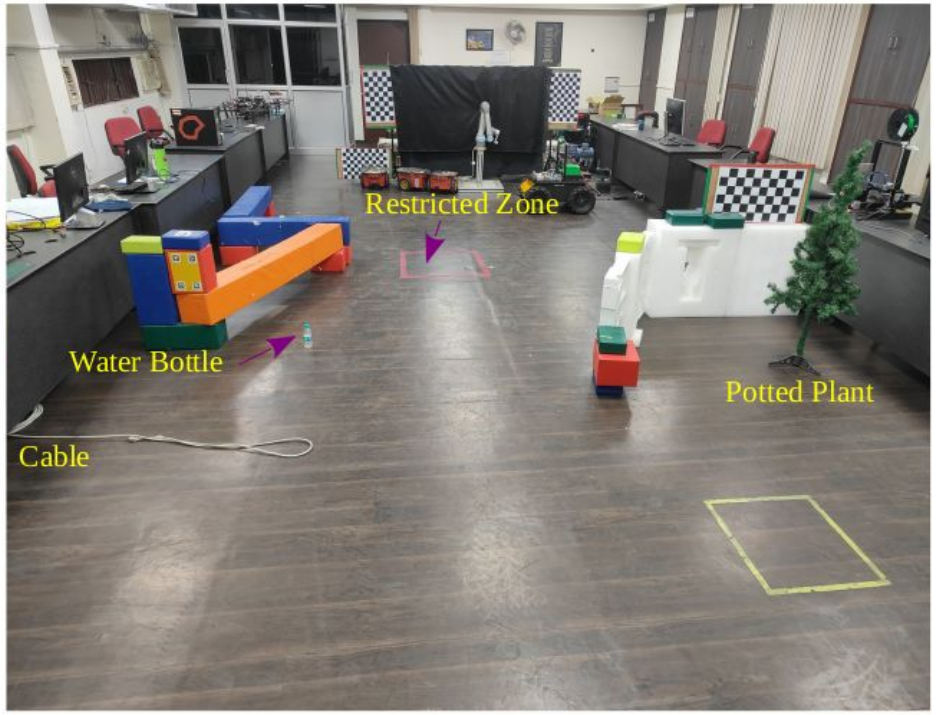
Method	Global Map	Drift Sensitivity	Long-Term Nav.	Unknown Env.	Semantic Understanding	Context Awareness	User Interaction	Replanning
A* / RRT*	✓	✓	✗	✗	✗	✗	✗	Limited
DiPPer	✓	✓	△	✗	✗	✗	✗	Limited
ViT-A*	✓	✓	△	△	△	✗	✗	Limited
Vision-Aided A* [20]	✓	✓	✗	✓	✗	✓	✗	Online
SemExp (Semantic Exploration) [21]	✗	△	△	✓	✓	✓	✗	Moderate
CARe (Context-Aware Replanning) [22]	✗	△	✓	✓	✓	✓	✗	Advanced
SEMNAV (Semantic Segmentation) [23]	✗	✗	✓	✓	✓	✓	✗	Advanced
Multi-Agent Semantic Nav [24]	✗	✗	✓	✓	✓	✓	✗	Advanced
Ours	✗	✗	✓	✓	✓	✓	✓	**Online**

Environment	Success Rate (%)	Soft Collision (%)	Hard Collision (%)	Avg. Path Length (m)	Drift Error (m)
Simulation					
Indoor	100	10.0	0.0	12.4	0.03 ± 0.01
Garage	100	8.5	0.0	20.8	0.05 ± 0.03
Campus	98	10.0	0.0	30.5	0.05 ± 0.02
Forest	95	15.2	0.3	15.5	0.08 ± 0.01
Tunnel	98	14.0	0.0	25.2	0.05 ± 0.02
Real-World Experiments (Husky A200 Platform)					
Indoor Lab (Fig. 10)	95	15.0	0.5	12.4	0.1 ± 0.02
Outdoor Area	98	5.6	0.0	124.6	0.05 ± 0.01

PERFORMANCE EVALUATION RESULTS OVER 50 TRIALS PER SCENARIO .

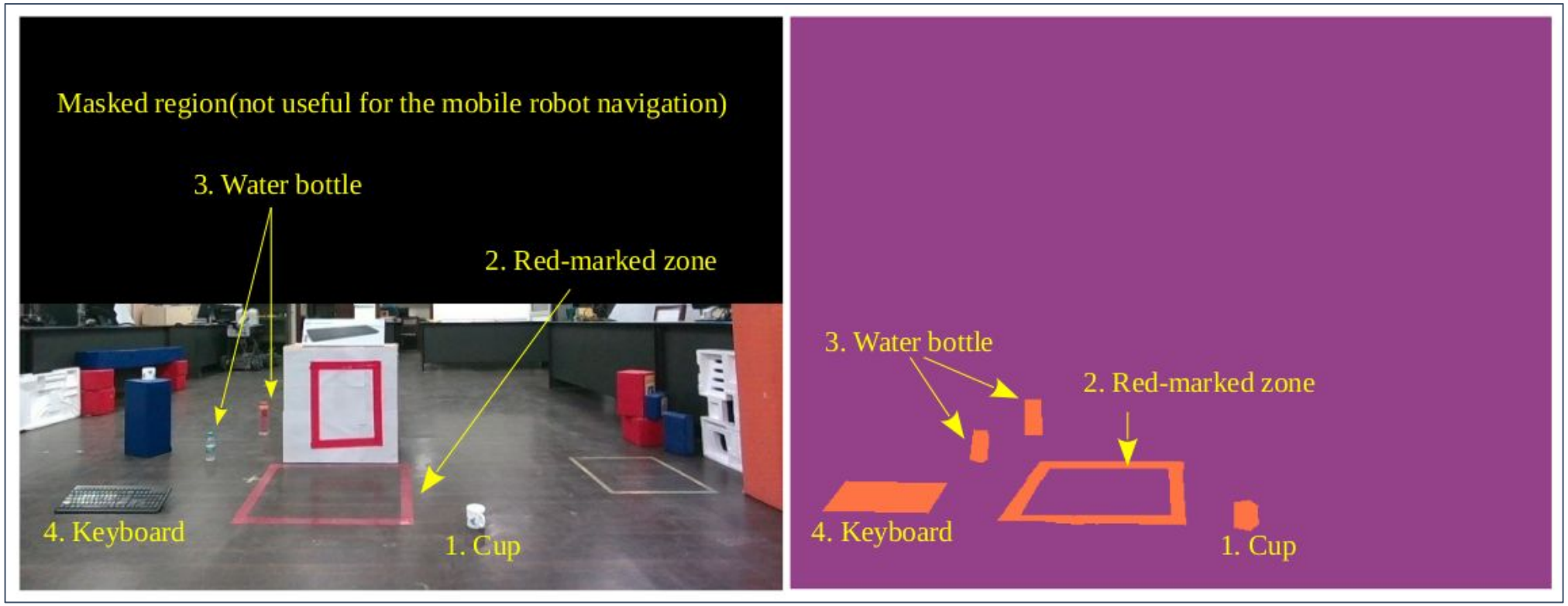
Results(Real-world)





Indoor experimental environment with the Husky A200 robot and onboard Intel RealSense D455 camera.

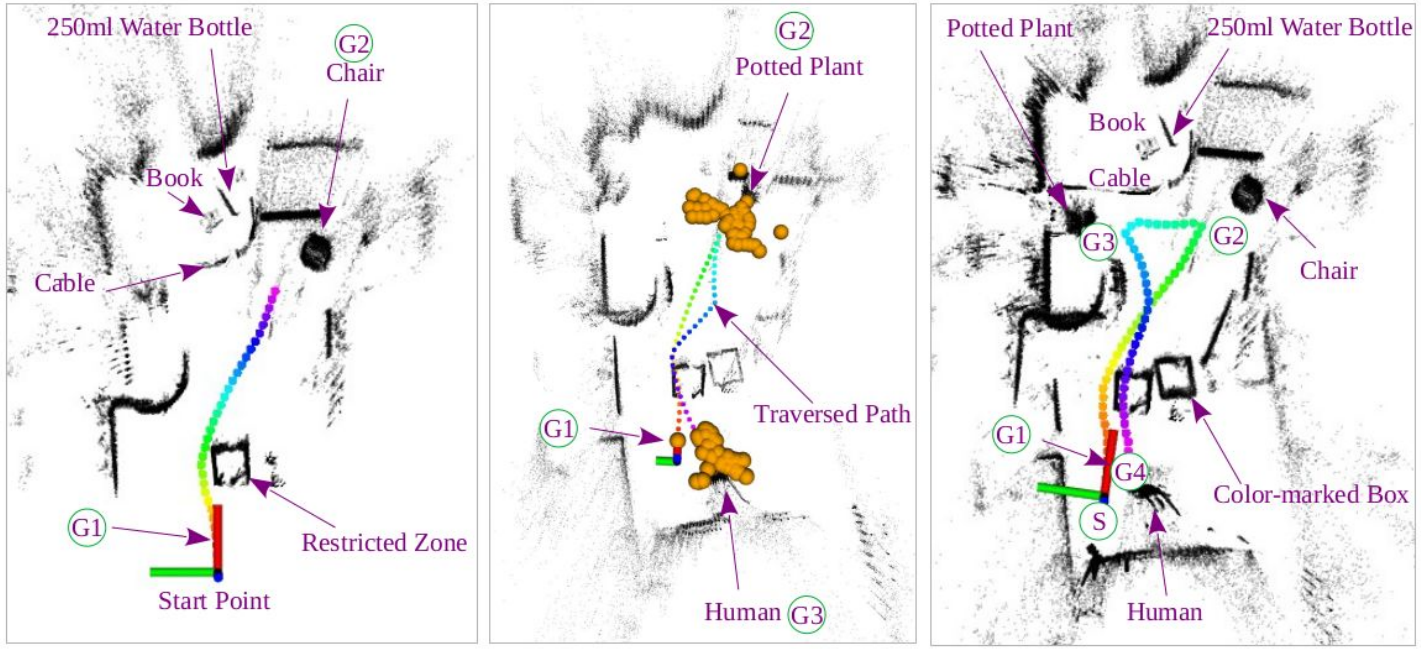
Continue…

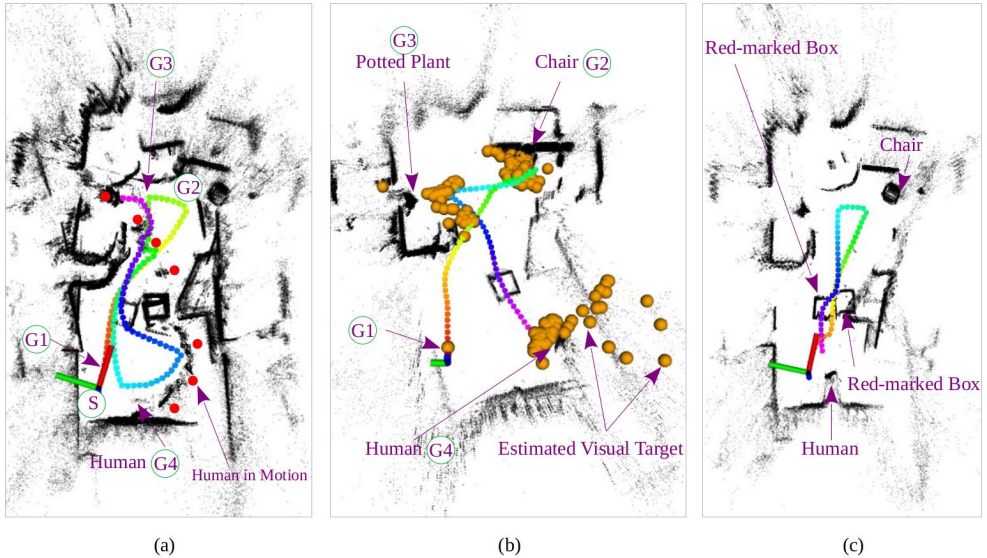


ESANet model input and output

Continue…

Real-world navigation results across multiple trials.





Experimental results of the visual navigation system. (a) The wheeled robot successfully tracking and following a dynamic human target. (b) Estimation of the fixed visual target point; note that despite the spatial spread of the estimated points, the robot successfully navigated to the visual destination. (c) The view of the red-marked box target, showing lateral shifts caused by the robot's to-and-fro motion during navigation.

Continue…

TABLE IV
PERFORMANCE COMPARISON OF NAVIGATION MODALITIES UNDER DIFFERENT ENVIRONMENTAL CONSTRAINTS

Scenario	Constraint	Geometric	Visual	Hybrid (Ours)
Featureless Corridor	No visual landmarks	92.0	15.0	**95.0**
Symmetric Room	Geometric ambiguity	40.0	94.0	**96.0**
Occluded Target	Target not visible	85.0	30.0	**91.0**
Precise Positioning	Exact stopping required	96.0	65.0	**98.0**
Complex Mission	Sequential task execution	10.0	25.0	**88.0**

TABLE III
INFERENCE TIME PERFORMANCE ACROSS COMPUTING PLATFORMS

Module	Task	Latency (ms)	Execution Mode
Server Platform (NVIDIA RTX A4000)			
RedNet	RGB-D Segmentation	29.2 ± 1.8	Continuous
ESANet	RGB-D Segmentation	34.8 ± 0.7	Continuous
OSNet	Person Re-ID	50 ± 5	On-demand
YOLOv8	Object Detection	12.3 ± 1.4	On-demand
LLM (Gemini Pro)	Instruction Parsing	150 ± 30	Per command
Onboard Platform (Husky A200 – Jetson AGX Xavier)			
RedNet	RGB-D Segmentation	87 ± 7	Continuous
ESANet	RGB-D Segmentation	108 ± 14	Continuous
OSNet	Person Re-ID	80 ± 10	On-demand
YOLOv8	Object Detection	38 ± 8	On-demand
LLM (Gemini Pro)	Instruction Parsing	190 ± 35	Per command

Source Code and Demonstration Videos

Source Code Link

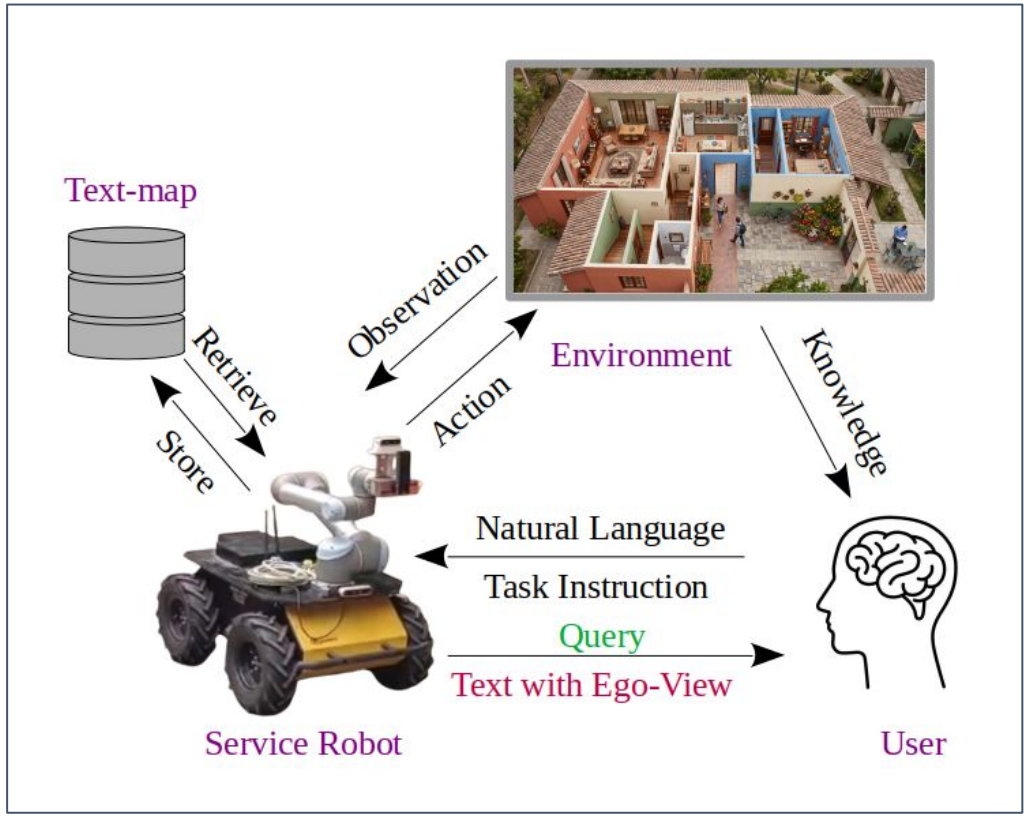


- P. Kumar, K. R. Guruprasad, and T. Sandhan, “Multi-Task Visual Perception Network with LLM Conditioning for Autonomous Navigation,” in Proceedings of the 2026 International Joint Conference on Neural Networks (IJCNN), presented at the 2026 IEEE World Congress on Computational Intelligence (WCCI), Maastricht, The Netherlands, June 21–26, 2026. **(Published)**

(Vision-Text Memory & Text-Space Navigation): Is it possible to eliminate Cartesian coordinate maps (x, y, θ) by representing environment memory as a Vision-Text Memory Map (VTMMap), and how can text-space path planning combined with interactive user communication progressively enhance a service robot's commandability and operational efficiency?

VTMMap: Geometry-Free Vision-to-Text World Models & Text Paths

- Visual-Text Environment Representation.
  - Convert visual scene graphs into geometry-free, text-based semantic maps.
  - Encode spatial relations directly as natural language records, replacing 2D/3D grids.
- Textual Path Finding (LLM-Driven Reasoning).
  - Execute dual-level LLM reasoning across global routes and local sub-goal actions.
  - Translate abstract human commands into sequential action primitives without (x,y) targets.
- Incremental Exploration Learning
  - Dynamically update textual topological graph memory during continuous exploration.
  - Mitigate odometry drift and adapt to dynamic human-centric environmental shifts.



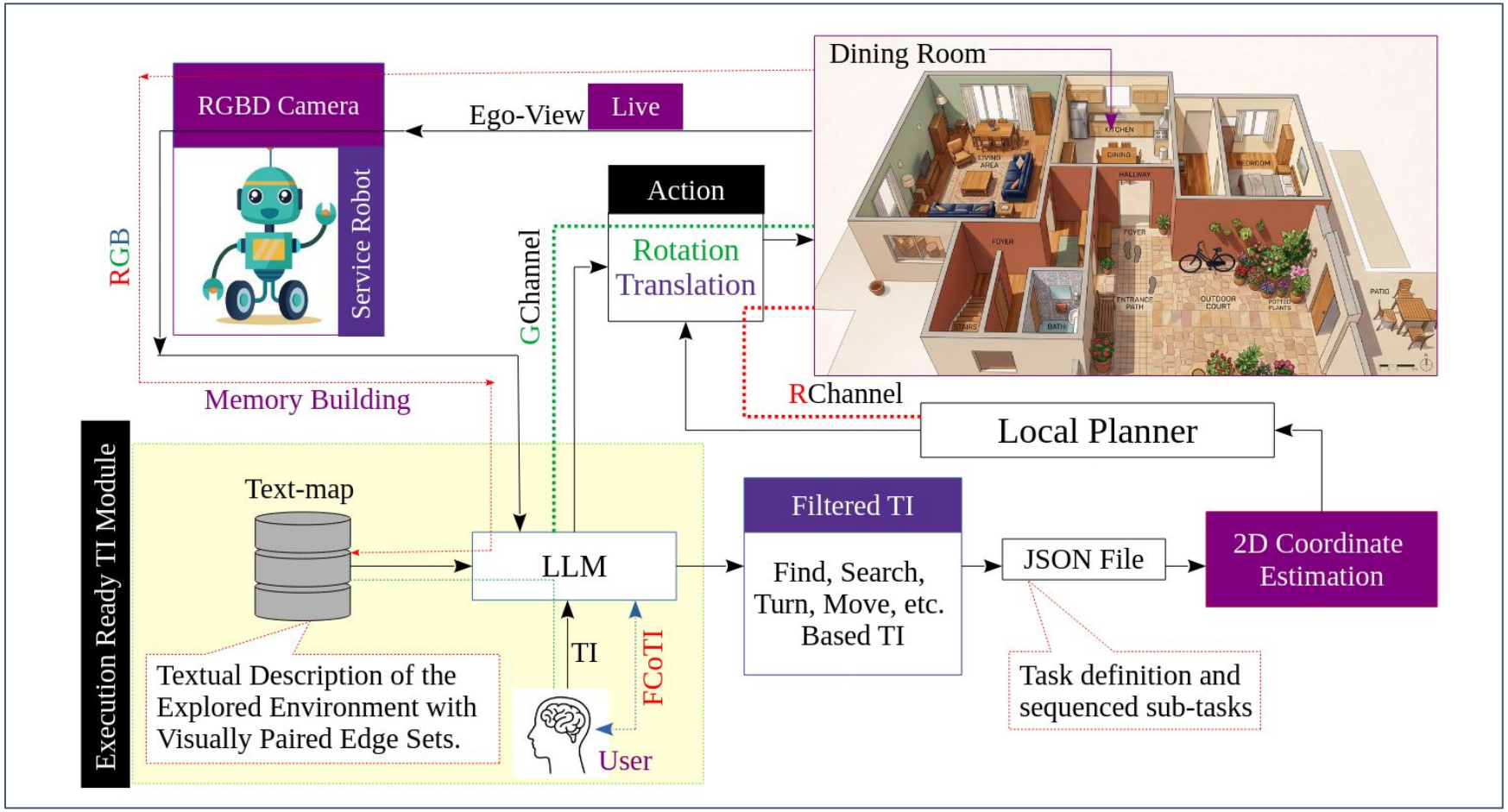
VTMMap-Driven Vision-Language Navigation Framework.

Literature Survey & SOTA Comparison

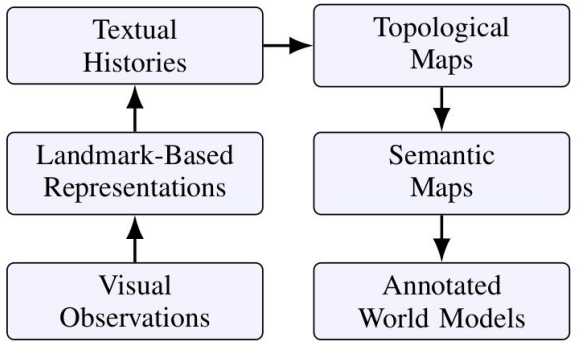
1. **Dense Metric & Geometric SLAM**
   - **Baselines:** Occupancy Grids, 3D Voxel Maps, ORB-SLAM2.
   - **Bottlenecks:** High memory footprint; sensitive to cumulative odometry drift and wheel slip; cannot parse natural language instructions directly.
2. **Dense Embedding-Based Maps**
   - **Baselines:** VLMaps (Voxel + CLIP), OpenMap, CLIP-Nav.
   - **Bottlenecks:** Require heavy GPU-bound feature extraction, continuous metric point cloud updating, and high-latency vector similarity searches during runtime.
3. **Graph & Scene Graph Approaches**
   - **Baselines:** SayNav, ConceptGraphs, TopologicalMap.
   - **Bottlenecks:** Require offline map construction; struggle with online incremental updates in unseen spaces; lack deterministic path guarantees.

Method	Map Type	Natural Language Query	Path Planning	Online Build	GPU-Free
TopologicalMap	Topological Graph	✓	Transformer	Partial	No
VLMaps	Voxel + CLIP	✓ (LLM)	Partial	Partial	No
SayNav	Scene Graph	✓ (LLM)	✓	Partial	No
ConceptGraphs	3D Scene Graph	✓	Partial	Partial	No
CLIP-Nav	Embedding	Partial	×	Partial	No
OpenMap	Open-Vocabulary VL Map	✓ (LLM)	Partial	Partial	No
VTMap (Ours)	**Semantic Graph (JSON)**	**✓ (Keyword + LLM)**	**✓ (BFS)**	**✓ (Incremental)**	**✓ (CPU 4ms)**

Methodology

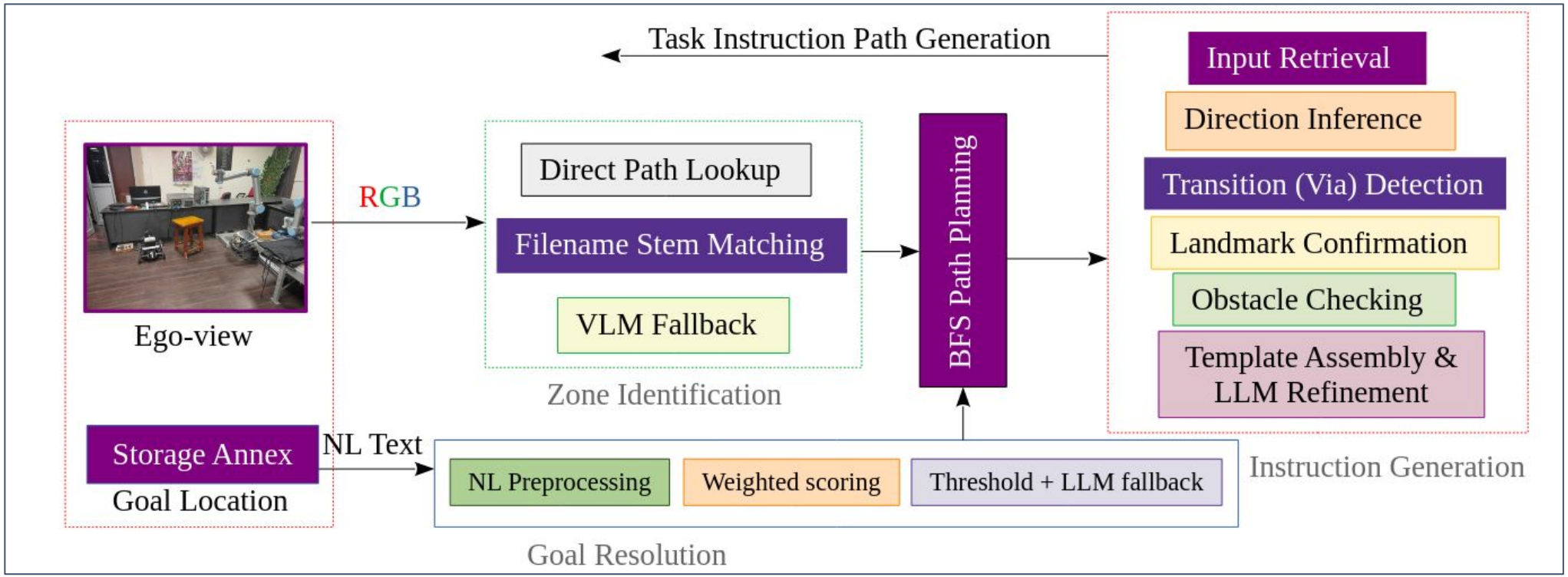


Overview of the proposed text-map-assisted vision-language navigation framework.



Evolution of memory representations in vision-language navigation.

Continue…



VTMMap (VTM) navigation pipeline. Given an egocentric RGB observation and a natural language destination query, the pipeline first localizes the robot using direct image correspondence, metadata-based semantic retrieval, and VLM-based localization.

Continue…

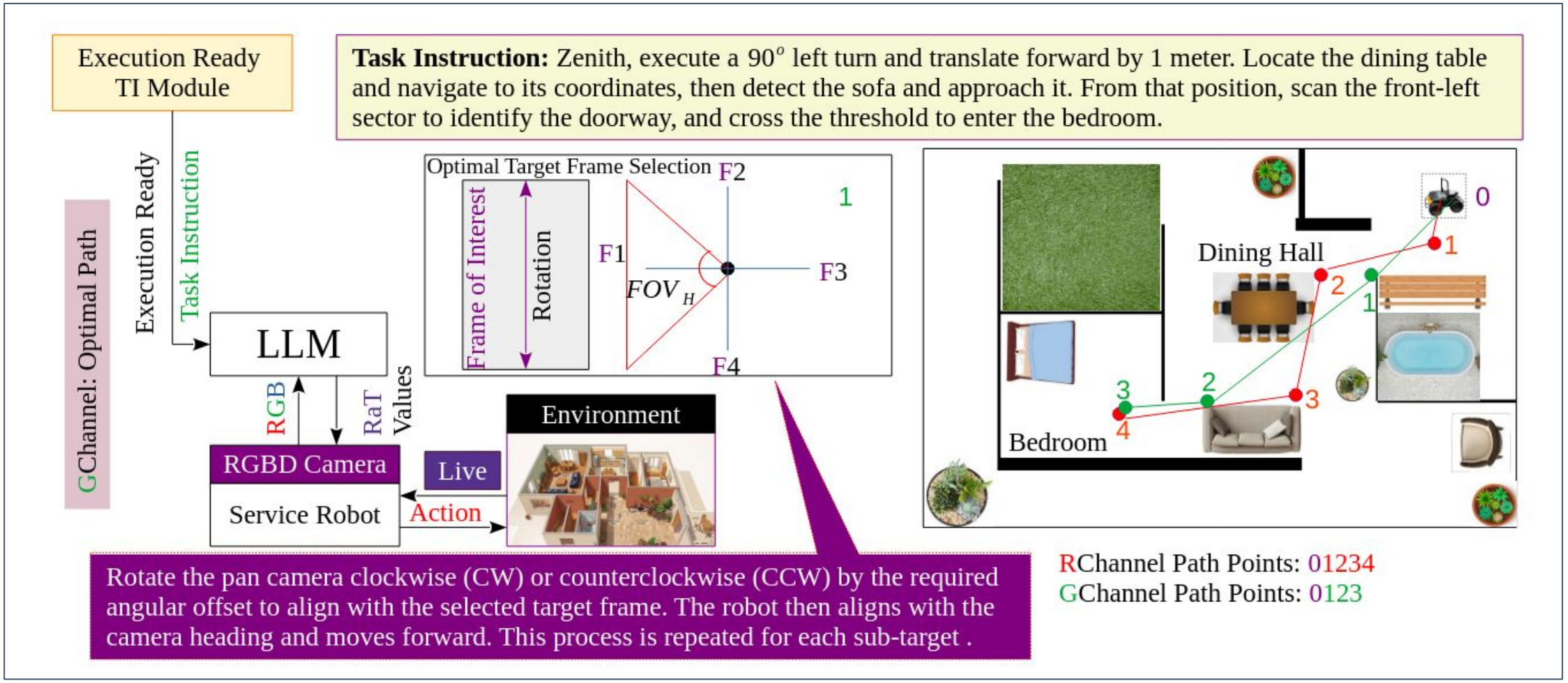


Illustration of the proposed LLM-guided egocentric navigation workflow. The user provides a natural-language task instruction, which is interpreted by the LLM after the execution-ready module validates system readiness.

Continue…

World Model & Goal Resolution Details

8-Tuple Semantic Node Definition: Lightweight Topological World Model

Topological Graph & Node Formulation

$$G = (V, E), \quad v = \langle L_v, T_v, D_v, M_v, N_v, G_v, O_v, I_v \rangle$$

Graph Components:

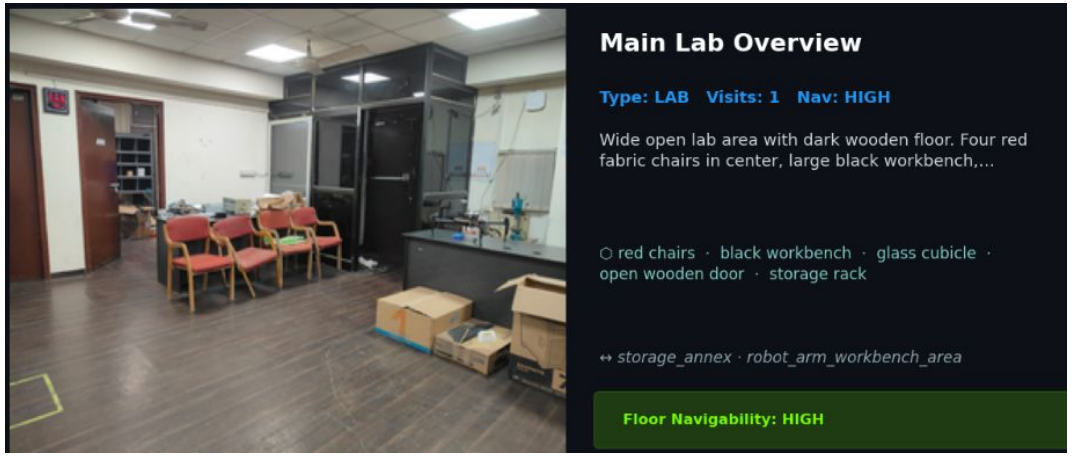
- G: Topological knowledge graph.
- V: Set of semantic nodes ($V = \{v_1, \ldots, v_n\}$).
- E: Set of navigable topological edges.

Node Metadata:

- L_v (**Zone Label**): Identifier tokens (e.g., *"workbench area"*).
- T_v (**Zone Type**): Structural category (*"corridor"*, *"office"*).
- D_v (**Description**): Natural language scene tokens.

Spatial & Perception Attributes:

- M_v (**Landmarks**): Visual objects set ($M_v = \{m_1, m_2, \ldots\}$).
- N_v (**Directional Map**): Navigable headings $\{N_{\text{front}}, N_{\text{left}}, N_{\text{right}}, N_{\text{back}}\}$.
- G_v (**Semantic Tags**): Keywords for LLM query retrieval.
- O_v (**Floor Obstacles**): Ground hazards and Beware Items.
- I_v (**Image Paths**): File paths to registered RGB-D frames.



Example of a Semantic Node encoding egocentric visual observation, zone metadata, semantic landmarks, topological connections, and floor navigability status.

Continue…

2. Multi-Field Goal Resolution Score

$$\text{Score}(v, Q) = w_l|Q \cap L_v| + w_{tp}\mathbb{I}(T_v \in Q) + w_m \sum_{m \in M_v} \mathbb{I}(q \subset m) + w_g|Q \cap G_v| + w_d|Q \cap D_v|$$

Explanation: Resolves free-form user text commands into target node v_{target} via weighted token intersection ($w_l = 0.5, w_m = 0.35$) in **0.388 ms**.

Path Planning & Direction Inference

1. Guaranteed Shortest-Path BFS Planning

$$P = (v_0, v_1, \ldots, v_k) = \text{BFS}(v_{\text{start}}, v_{\text{target}}) \quad \text{Complexity: } \mathcal{O}(|V| + |E|)$$

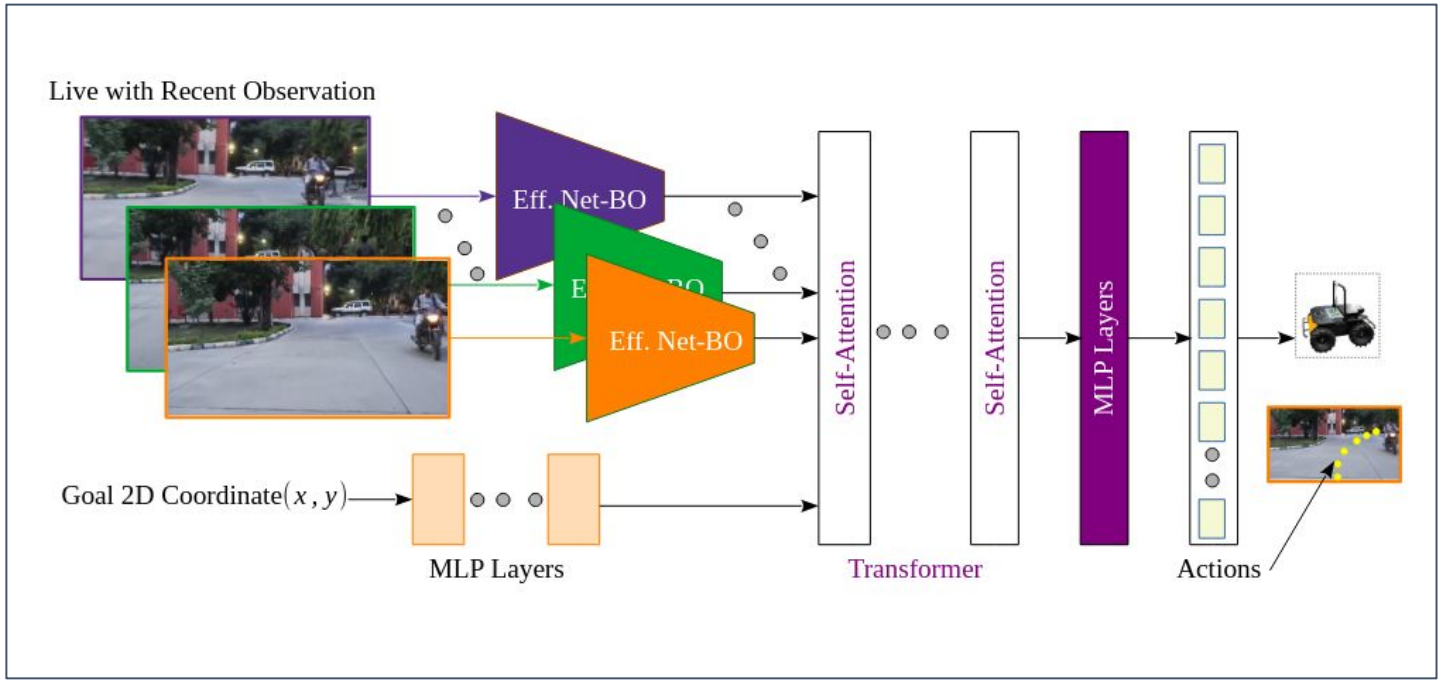
Explanation: Guarantees minimum-hop, minimum-turn navigation without requiring heuristic design or metric map optimization (**0.019 ms execution**).

Continue…

2. Cardinal Direction Alignment (d^*)

$$d^* = \arg \max_{d \in \{\text{front, left, right, back}\}} \sum_{t \in L_{v_{i+1}}} \mathbb{I}\Big(t \in N_{v_i}[d]\Big)$$

Explanation: Maps topological node transitions directly to actionable cardinal turn directives for the local robot controller.



Overview of the LogoNav local motion controller [12] employed for local navigation.

Results

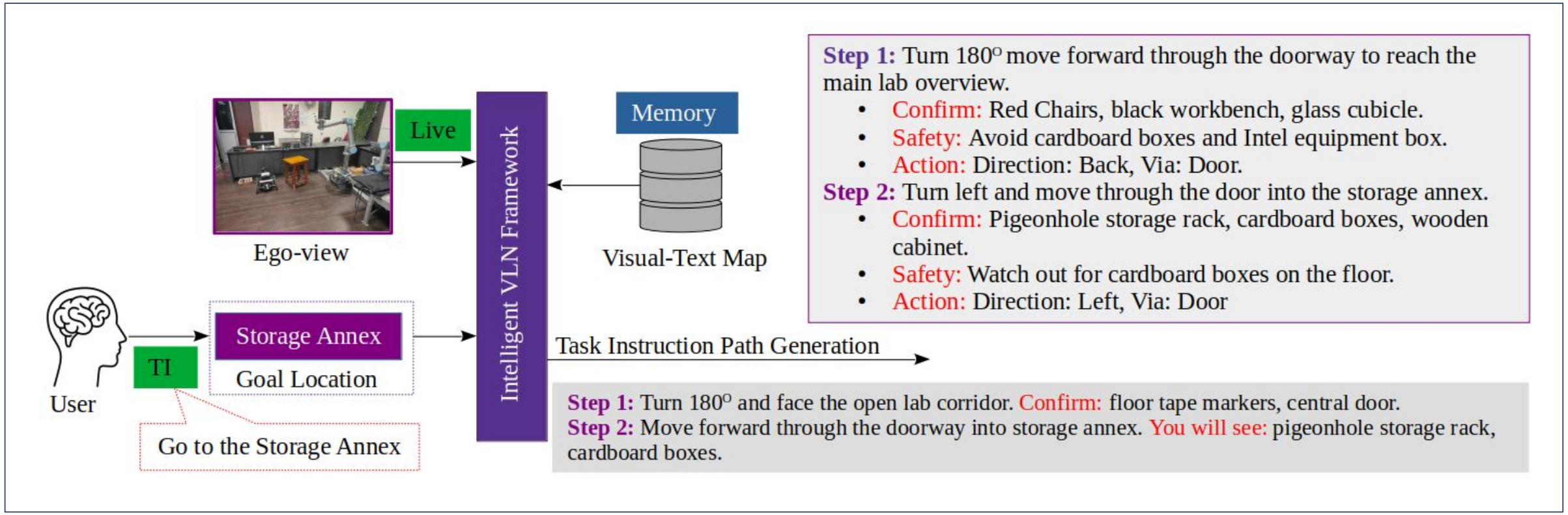


Illustration of the instruction generation sequence in VTM. The process begins with an egocentric RGB image captured by the robot and a user-provided natural language task (e.g., “Go to the Storage Annex”).

Continue…

TABLE I
REPRESENTATIVE NAVIGATION PATHS GENERATED BY THE BFS PLANNER.

Start → Goal	BFS Path	Hops	Shortest
Corridor → Storage Annex	Corridor → Storage Annex	1	Yes
Entrance → Glass Cubicle	Entrance → Seating → Glass Cubicle	2	Yes
Office Desk → Storage	Office Desk → Lab Overview → Storage	2	Yes
Robot Arm → Seating	Robot Arm → Lab Overview → Seating	2	Yes†

†One of multiple shortest paths with the same hop count.

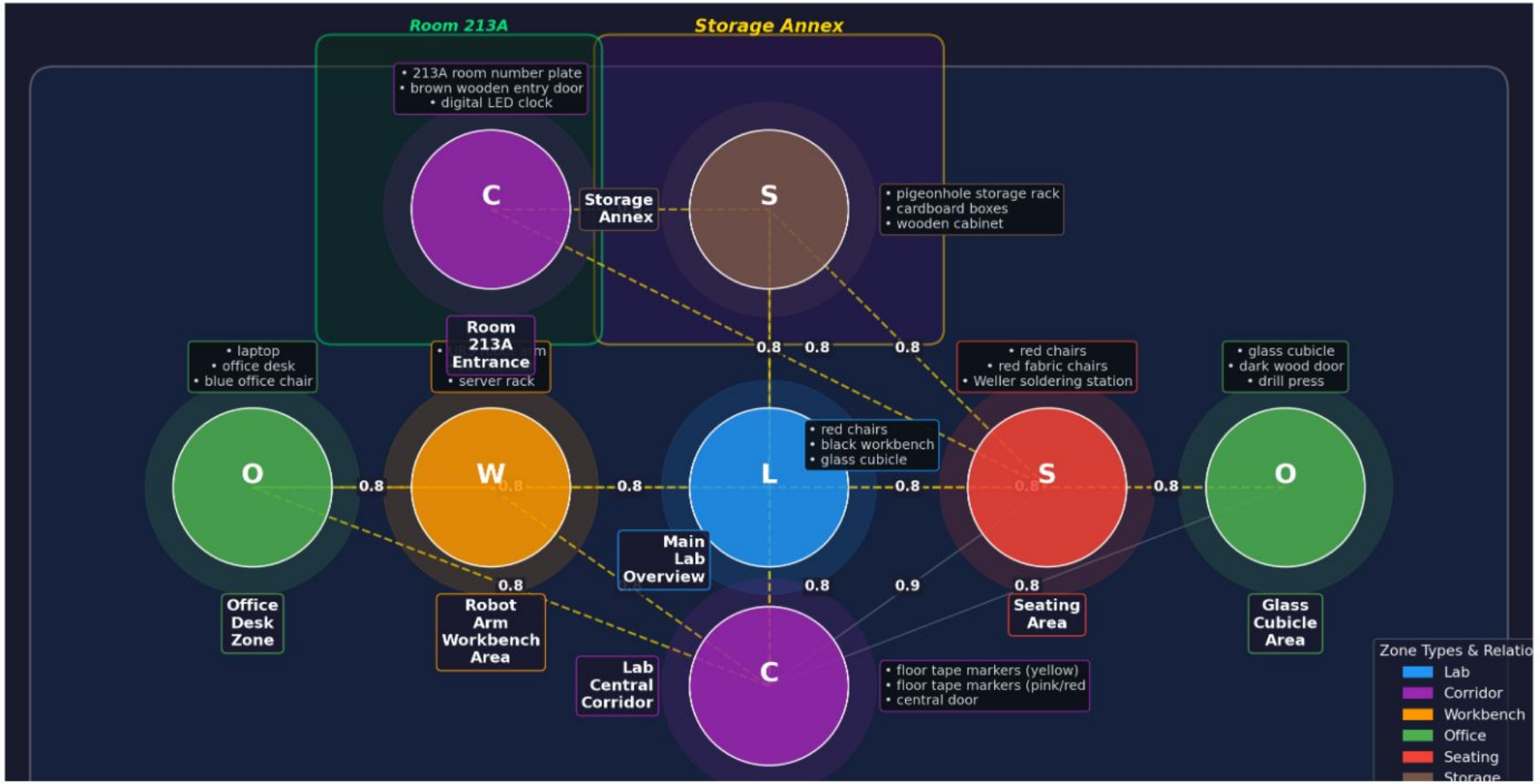
TABLE II
COMPUTATIONAL LATENCY OF THE PROPOSED VTMMAP PIPELINE.

Stage	Method	Latency	Hardware
Frame description (VLM)	Gemini Flash 2.0 API	∼1–2 s/frame	Cloud API
Graph construction	Jaccard merge	2.0 ms	CPU
NL query resolution	Keyword scoring	0.388 ms	CPU
BFS path planning	Graph traversal	0.019 ms	CPU
Offline pipeline	Build + Query + BFS	4.0 ms	CPU

TABLE III
EVALUATION OF VTMMAP ON ENVIRONMENTS OF VARYING COMPLEXITY.

Environment	Frames	Semantic Zones	Zone Accuracy	Avg. Landmarks	Avg. Edges	Merge Accuracy	Graph Connectivity
Indoor (Hall+2 Rooms)	37	8	100%	7.0	9.0	0.60	100%
Indoor + Outdoor	52	11	95%	6.8	8.6	0.60	100%
Office	65	14	98.5%	7.5	10.1	0.70	100%
Corridor	81	17	90%	6.9	8.8	0.60	100%

Continue…



Example of a VTMMap (VTM) generated from a single exploration run in an indoor laboratory environment. E

Continue…

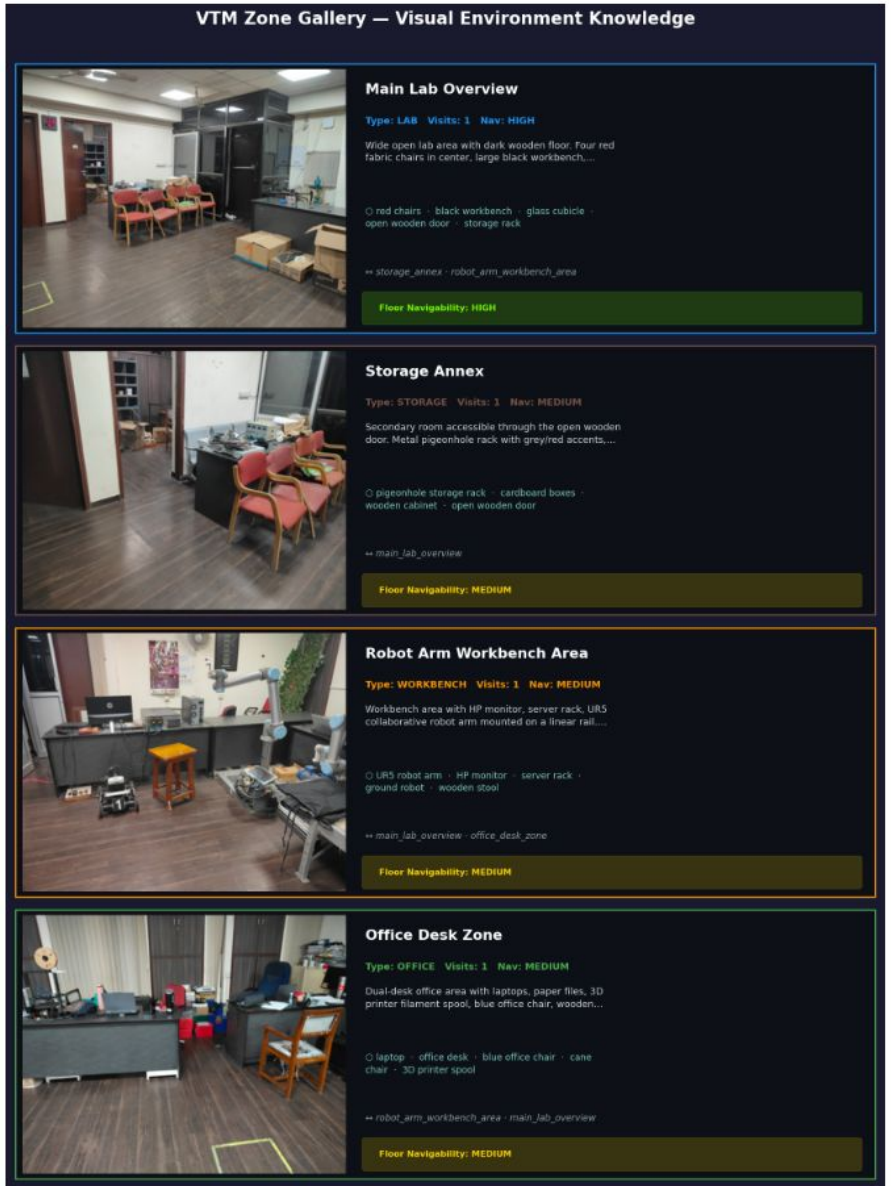
TABLE IV
COMPARISON OF VTMMAP DESIGN CHOICES WITH COMMONLY USED ALTERNATIVES.

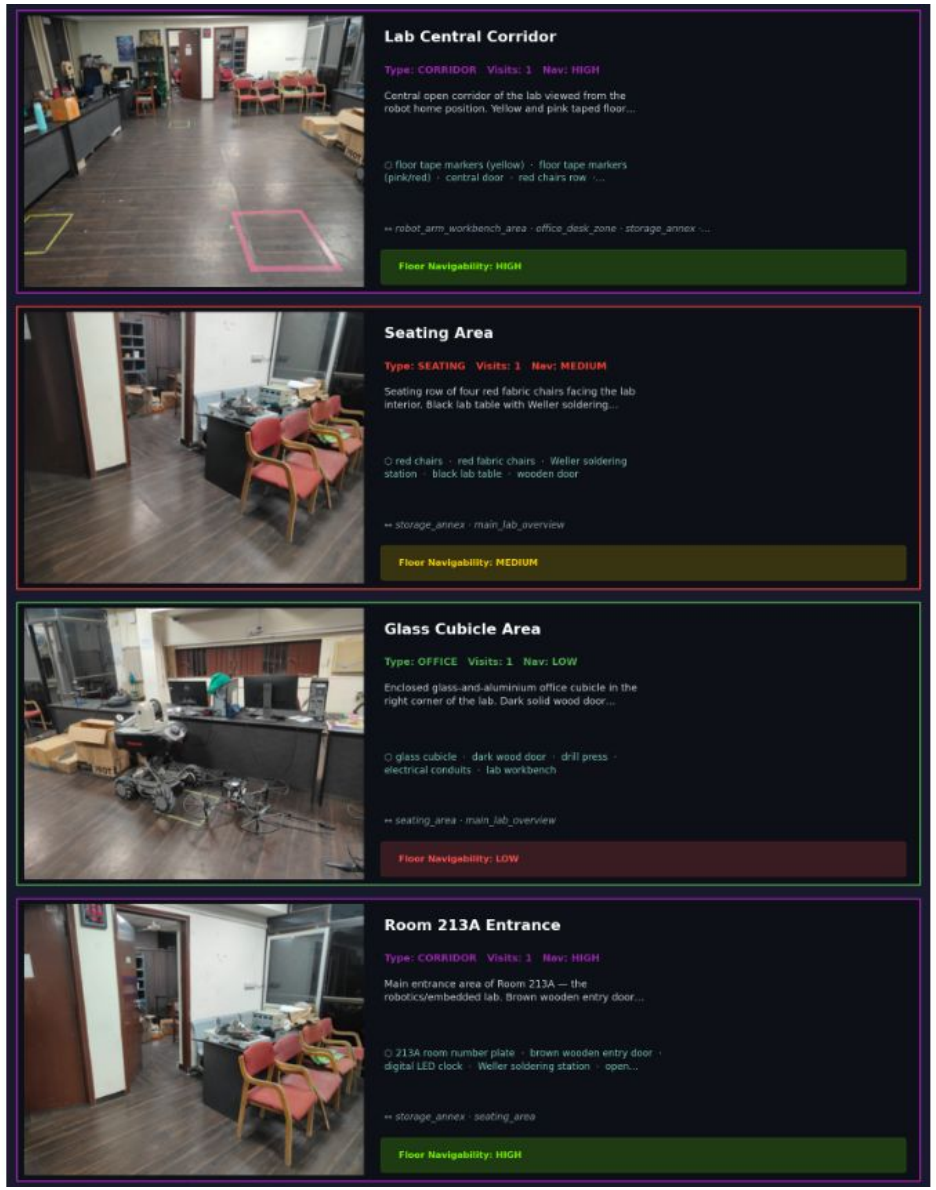
Component	VTMMap Choice	Alternative	Rationale
Map representation	Semantic graph (JSON)	Occupancy grid / Voxel map	Lightweight, interpretable, and directly supports natural-language queries
Zone matching	Jaccard similarity on semantic tags	CLIP embeddings	Computationally efficient; no GPU required
Natural language query	Keyword scoring + LLM	Pure LLM / CLIP retrieval	Supports fast offline execution with LLM fallback
Path planning	Breadth-First Search (BFS)	Dijkstra / A*	Simple, interpretable, and efficient for unweighted semantic graphs
Scene description	Vision-Language Model (Gemini/GPT-4o)	Manual annotation	Scalable to unseen environments with minimal human effort
Knowledge storage	JSON-based graph	SQL / Vector database	Portable, lightweight, and server-free deployment

TABLE V
COMPARISON OF VTMMAP WITH REPRESENTATIVE ENVIRONMENT MAPPING AND VISION-LANGUAGE NAVIGATION APPROACHES.

Method	Map Type	NL Query	Path Planning	Online Build	GPU-free	Landmark Grounding
VTMMap (Ours)	Semantic Graph	✓ Keyword+LLM	✓ BFS	✓	✓	✓
TopologicalMap [13]	Topological Graph	✓	Transformer	×	×	×
VLMaps [14]	Voxel + CLIP	✓ LLM	Partial	✓	×	✓
SayNav	Scene Graph	✓ LLM	✓	×	×	✓
ConceptGraphs [15]	3D Scene Graph	✓	Partial	✓	Partial	✓
CLIP-Nav [16]	Embedding	Partial	×	×	×	Partial
OpenMap [17]	Open-Vocabulary VL Map	✓ LLM	Partial	✓	×	✓

Continue…


VTM Zone Gallery — Visual Environment Knowledge
Main Lab Overview
Type: LAB Visits: 1 Nav: HIGH
Wide open lab area with dark wooden floor. Four red fabric chairs in center, large black workbench,...
red chairs · black workbench · glass cubicle · open wooden door · storage rack
storage_annex · robot_arm_workbench_area
Floor Navigability: HIGH
Storage Annex
Type: STORAGE Visits: 1 Nav: MEDIUM
Secondary room accessible through the open wooden door. Metal pigeonhole rack with grey/red accents,...
pigeonhole storage rack · cardboard boxes · wooden cabinet · open wooden door
main_lab_overview
Floor Navigability: MEDIUM
Robot Arm Workbench Area
Type: WORKBENCH Visits: 1 Nav: MEDIUM
Workbench area with HP monitor, server rack, UR5 collaborative robot arm mounted on a linear rail,...
UR5 robot arm · HP monitor · server rack · ground robot · wooden stool
main_lab_overview · office_desk_zone
Floor Navigability: MEDIUM
Office Desk Zone
Type: OFFICE Visits: 1 Nav: MEDIUM
Dual-desk office area with laptops, paper files, 3D printer filament spool, blue office chair, wooden...
laptop · office desk · blue office chair · cane chair · 3D printer spool
robot_arm_workbench_area · main_lab_overview
Floor Navigability: MEDIUM


Lab Central Corridor
Type: CORRIDOR Visits: 1 Nav: HIGH
Central open corridor of the lab viewed from the robot home position. Yellow and pink taped floor...
floor tape markers (yellow) · floor tape markers (pink/red) · central door · red chairs row · ...
robot_arm_workbench_area · office_desk_zone · storage_annex · ...
Floor Navigability: HIGH
Seating Area
Type: SEATING Visits: 1 Nav: MEDIUM
Seating row of four red fabric chairs facing the lab interior. Black lab table with Weller soldering...
red chairs · red fabric chairs · Weller soldering station · black lab table · wooden door
storage_annex · main_lab_overview
Floor Navigability: MEDIUM
Glass Cubicle Area
Type: OFFICE Visits: 1 Nav: LOW
Enclosed glass-and-aluminium office cubicle in the right corner of the lab. Dark solid wood door...
glass cubicle · dark wood door · drill press · electrical conduits · lab workbench
seating_area · main_lab_overview
Floor Navigability: LOW
Room 213A Entrance
Type: CORRIDOR Visits: 1 Nav: HIGH
Main entrance area of Room 213A — the robotics/embedded lab. Brown wooden entry door...
213A room number plate · brown wooden entry door · digital LED clock · Weller soldering station · open...
storage_annex · seating_area
Floor Navigability: HIGH

List of Publications

1. P. Kumar, P. Gupta and T. Sandhan, “NeuralPathLite: Fast and Robust Diffusion-Based Path Planning for Autonomous Navigation,” in IEEE Transactions on Automation Science and Engineering, vol. 23, pp. 12471-12485, 2026, doi: 10.1109/TASE.2026.3710089. (Published)

2. P. Kumar and T. Sandhan, “Vision-Aided Online A* Path Planning for Efficient and Safe Navigation of Service Robots,” International Journal of Intelligent Robotics and Applications (Springer), 2026, doi: 10.1007/s41315-026-00540-z. **(Published)**

3. P. Kumar, K. R. Guruprasad, and T. Sandhan, “Multi-Task Visual Perception Network with LLM Conditioning for Autonomous Navigation,” in Proceedings of the 2026 International Joint Conference on Neural Networks (IJCNN), presented at the 2026 IEEE World Congress on Computational Intelligence (WCCI), Maastricht, The Netherlands, June 21–26, 2026. **(Published)**

4. P. Kumar and T. Sandhan, “Path-Tracking Control of the 4WIS4WID Electric Vehicle by Direct Inverse Control using Artificial Neural Network,” 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-8, doi: 10.1109/ICCCNT56998.2023.10306595. **(Published)**

5. P. Kumar, R. M. Singhee, K. R. Guruprasad and T. Sandhan, “VTMMap Driven Vision-Language Navigation: From Visual Relations into Textual Paths,” in IEEE Transactions on Automation Science and Engineering (Submitted).

6. P. Kumar, R. M. Singhee, K. R. Guruprasad and T. Sandhan, “Optimizing Human-in-the-Loop Vision-Language Navigation Through Hardware-Software Co-Design,” IEEE Robotics and Automation Practice (Submitted).

This presentation is organized based on the findings reported in these publications.

Conclusion & Future Possibilities

Conclusion

- **Minimalist RGB-D Perception:** Operates using only a single RGB-D camera for 3D environment sensing, avoiding expensive sensor rigs.
- **Vision-Text Memory Mapping:** Replaces traditional grid maps with a vision-text memory map that directly aligns with user instructions.
- **Semantic Passability & Safety:** Respects user-defined safety rules (beware lists) to prevent the robot from crossing unsafe or untraversable areas.
- **Natural Language Commanding:** Allows users to give commands using plain English text or oracle instructions.
- **Egocentric Path Generation:** Converts starting camera images directly into step-by-step textual navigation paths to the goal.
- **Fast Path Planning:** Calculates optimal paths in just 0.16 seconds across diverse mazes, regardless of size, shape, color, or camera resolution.
- **Drift Resetting at Sub-goals:** Eliminates accumulating position errors by resetting odometry drift at every intermediate sub-goal.
- **Interactive Stuck Handling:** Uses a human-robot interaction module to gracefully recover if the robot ever gets stuck or deadlocked.
- **Latency Diagnosis & Optimization:** Detects and fixes system delays to guarantee fast, real-time performance.
- **Fleet-Scalable Architecture:** Designed with a modular system structure that can easily scale across multiple robots.

Continue…

Future Possibilities

1. **Multi-Floor and Uneven Terrain Navigation:** Expanding the framework beyond flat floors to navigate uneven surfaces and multi-storey buildings using elevators, ramps, and stairs.
2. **Hardware Generalization & Mobile Manipulation:** Transitioning the software from wheeled robots to legged and humanoid platforms, and mounting robotic arms for complete pick-and-place task execution.
3. **Scalable Vision-Text Memory Mapping:** Improving the speed and robustness of vision-text memory map construction for complex, large-scale real-world environments.
4. **Human-Like Semantic Reasoning:** Enhancing semantic awareness with human-like commonsense reasoning to automatically adapt safety rules and "beware lists."
5. **Proactive Failure Prevention:** Upgrading stuck-handling from reactive recovery to an intelligent system that predicts and prevents navigation freezes before they happen.
6. **Dynamic Obstacle Avoidance:** Optimizing path planning to handle unpredictably moving obstacles, such as crowds of people.

Thanks

References

[1] Bresson, Guillaume, et al. “Simultaneous localization and mapping: A survey of current trends in autonomous driving.” IEEE Transactions on Intelligent Vehicles 2.3 (2017): 194-220.

[2] Jordan, Julian, and Andreas Zell. “Real-time model based path planning for wheeled vehicles.” 2019 International Conference on Robotics and Automation (ICRA). IEEE, 2019.

[3] Pfrunder, Andreas, et al. “Real-time autonomous ground vehicle navigation in heterogeneous environments using a 3D LiDAR.” 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2017.

[4] Zang, Xiao, et al. “Robot Motion Planning as Video Prediction: A Spatio-Temporal Neural Network-based Motion Planner.” 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2022.

[5] Martínez, Eder Alejandro Rodríguez, et al. “Photometric path planning for vision-based navigation.” 2020 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2020.

[6] Thoma, Janine, et al. “Mapping, localization and path planning for image-based navigation using visual features and map.” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2019.

[7] Wang, Hengli, et al. “Learning interpretable end-to-end vision-based motion planning for autonomous driving with optical flow distillation.” 2021 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2021.

[8] Nubert, Julian, et al. “Learning-based localizability estimation for robust lidar localization.” 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2022.

[9] Ghosh, Sourish, and Joydeep Biswas. "Joint perception and planning for efficient obstacle avoidance using stereo vision." 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2017.

Continue..

[10] Deng, Xinke, et al. “Feature-constrained active visual SLAM for mobile robot navigation.” *2018 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2018.

[11] Wang, Hsueh-Cheng, et al. “Enabling independent navigation for visually impaired people through a wearable vision-based feedback system.” *2017 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2017.

[12] Sabatta, Deon, and Roland Siegwart. “Bearings-only path following with a vision-based potential field.” *2014 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2014.

[13] S. Griffith, G. Chahine, and C. Pradalier, “Symphony lake dataset,” The International Journal of Robotics Research, vol. 36, no. 11, pp. 1151–1158, 2017. [Online]. Available: https://doi.org/10.1177/0278364917730606